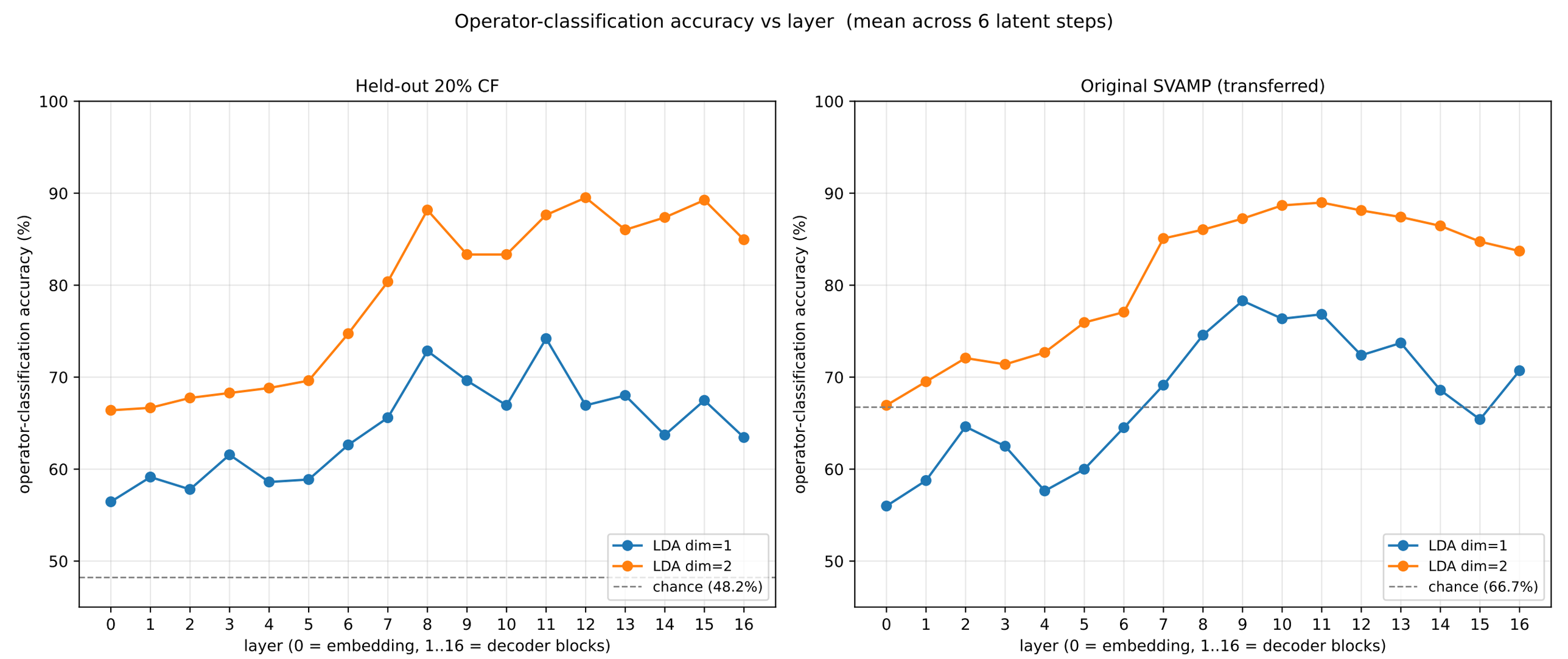

Operator-classification accuracy vs layer (mean across 6 latent steps)
Held-out 20% CF
Original SVAMP (transferred)
operator-classification accuracy (%)
layer (0 = embedding, 1..16 = decoder blocks)
LDA dim=1
LDA dim=2
chance (48.2%)
chance (66.7%)

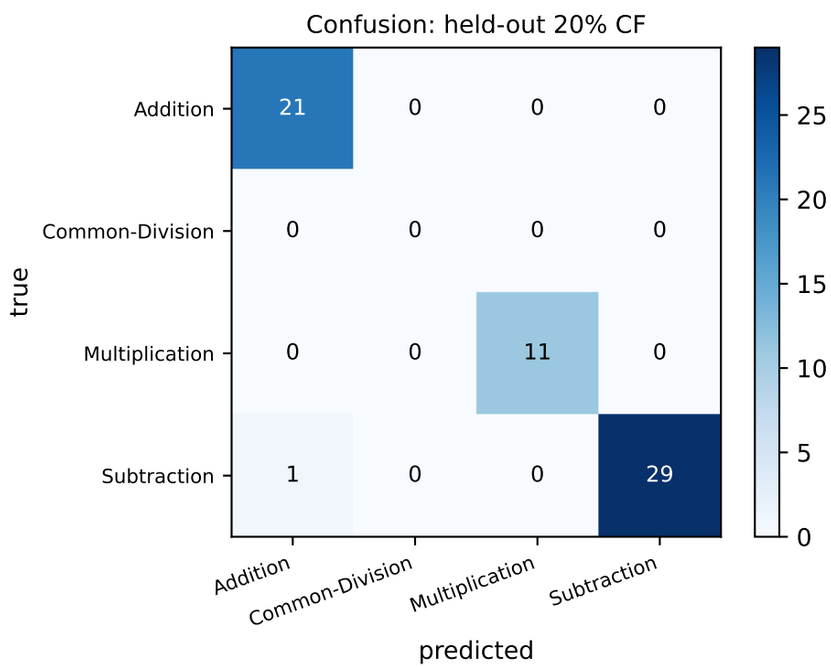
Sanity check — per-class precision / recall / F1 + confusion matrix
peak config: LDA dim=2, layer=12, latent step=5

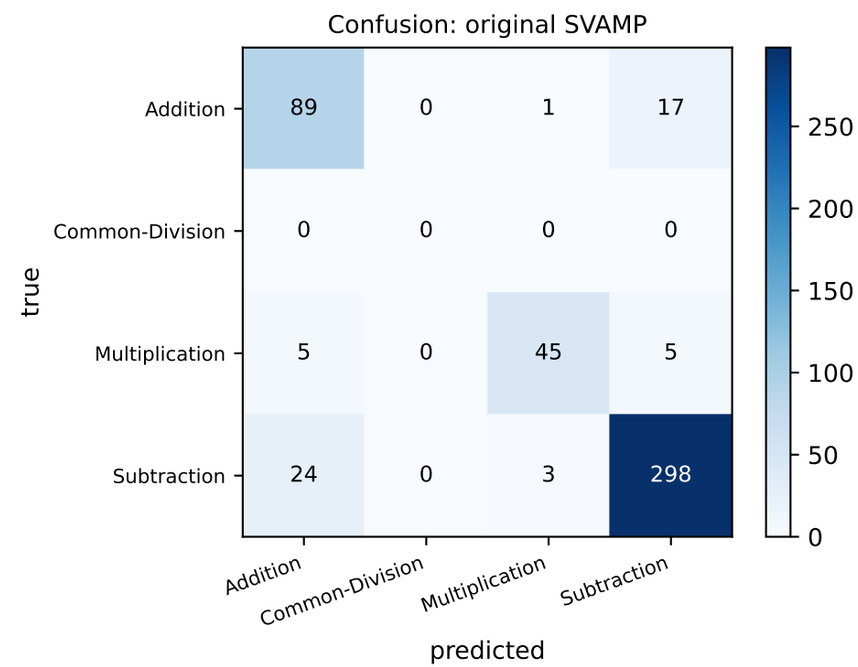
Held-out 20% CF (overall acc 0.984)

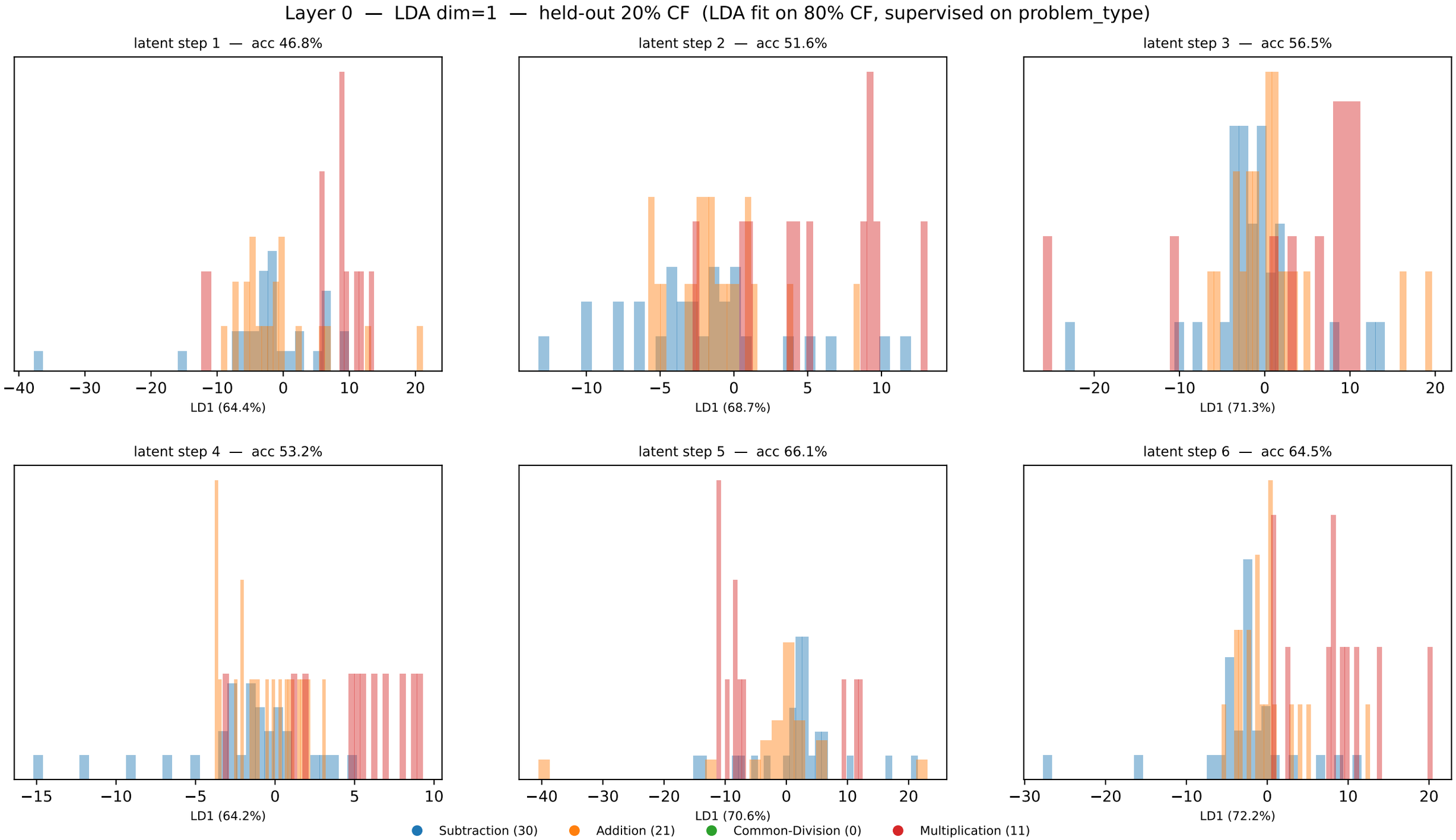
class	precision	recall	F1	n
Addition	0.955	1.000	0.977	21
Common-Divisio	0.000	0.000	0.000	0
Multiplication	1.000	1.000	1.000	11
Subtraction	1.000	0.967	0.983	30
macro avg	0.739	0.742	0.740	62
weighted avg	0.985	0.984	0.984	62

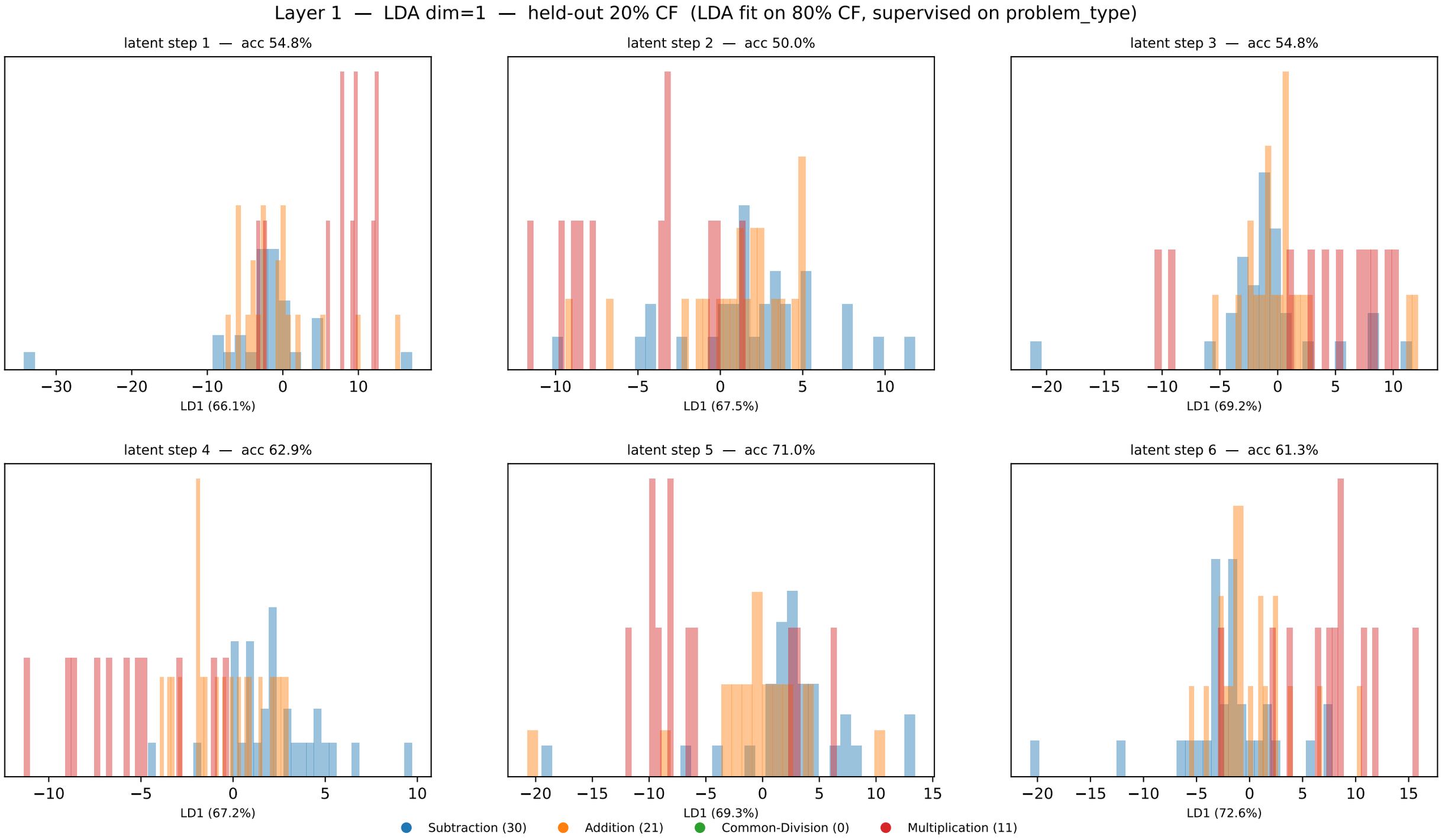
Original SVAMP (overall acc 0.887)

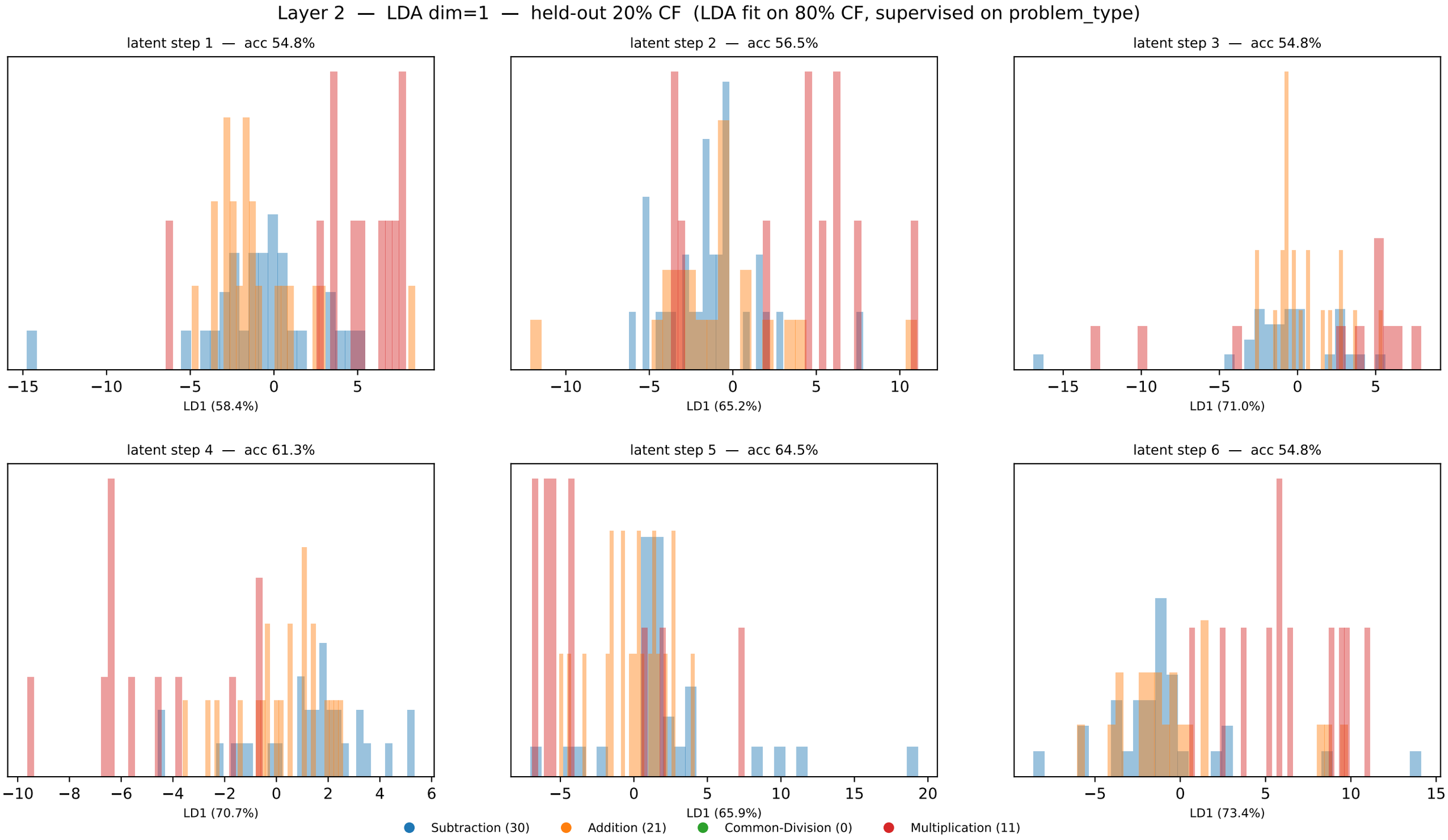
class	precision	recall	F1	n
Addition	0.754	0.832	0.791	107
Common-Divisio	0.000	0.000	0.000	0
Multiplication	0.918	0.818	0.865	55
Subtraction	0.931	0.917	0.924	325
macro avg	0.651	0.642	0.645	487
weighted avg	0.891	0.887	0.888	487

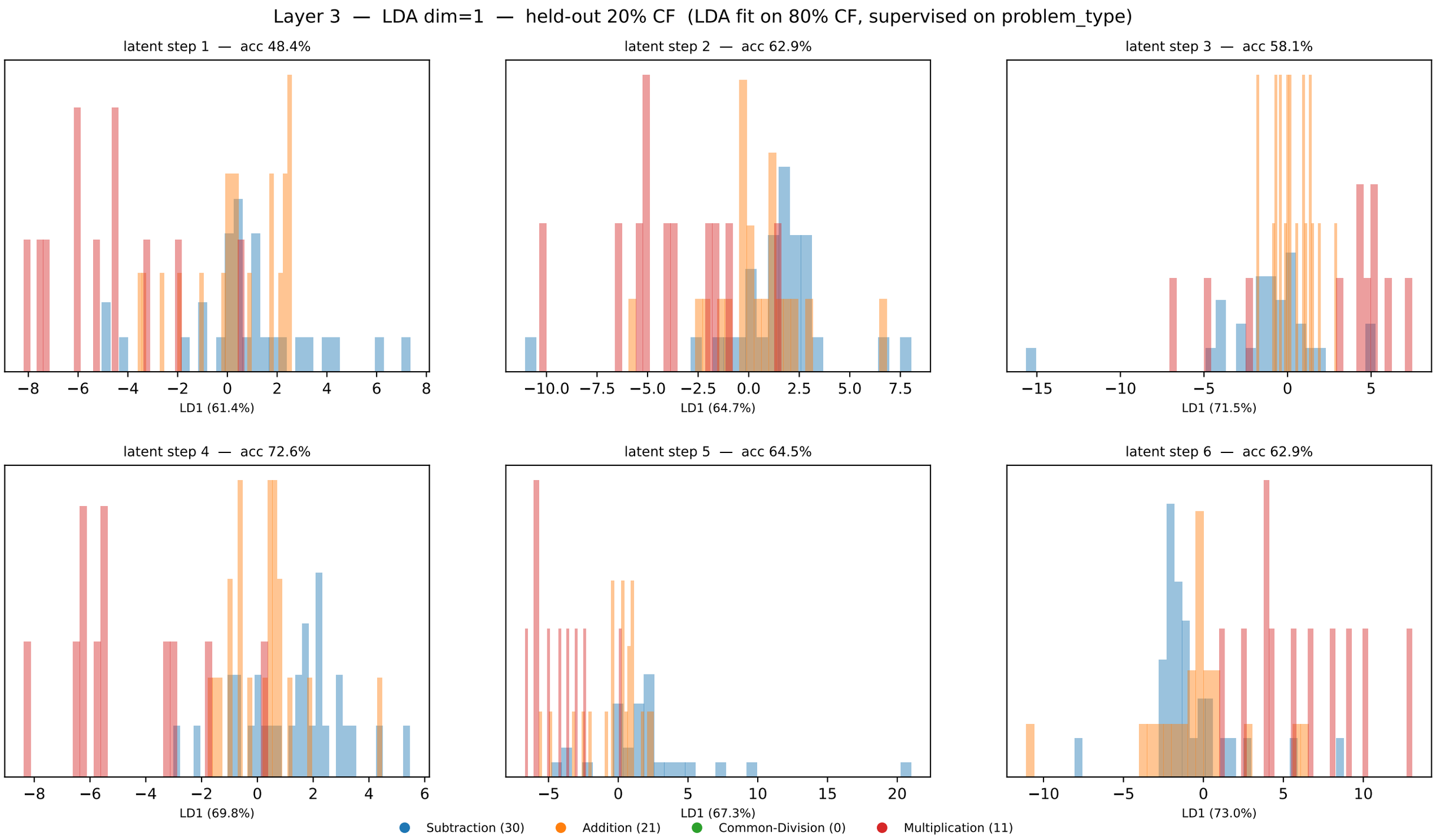


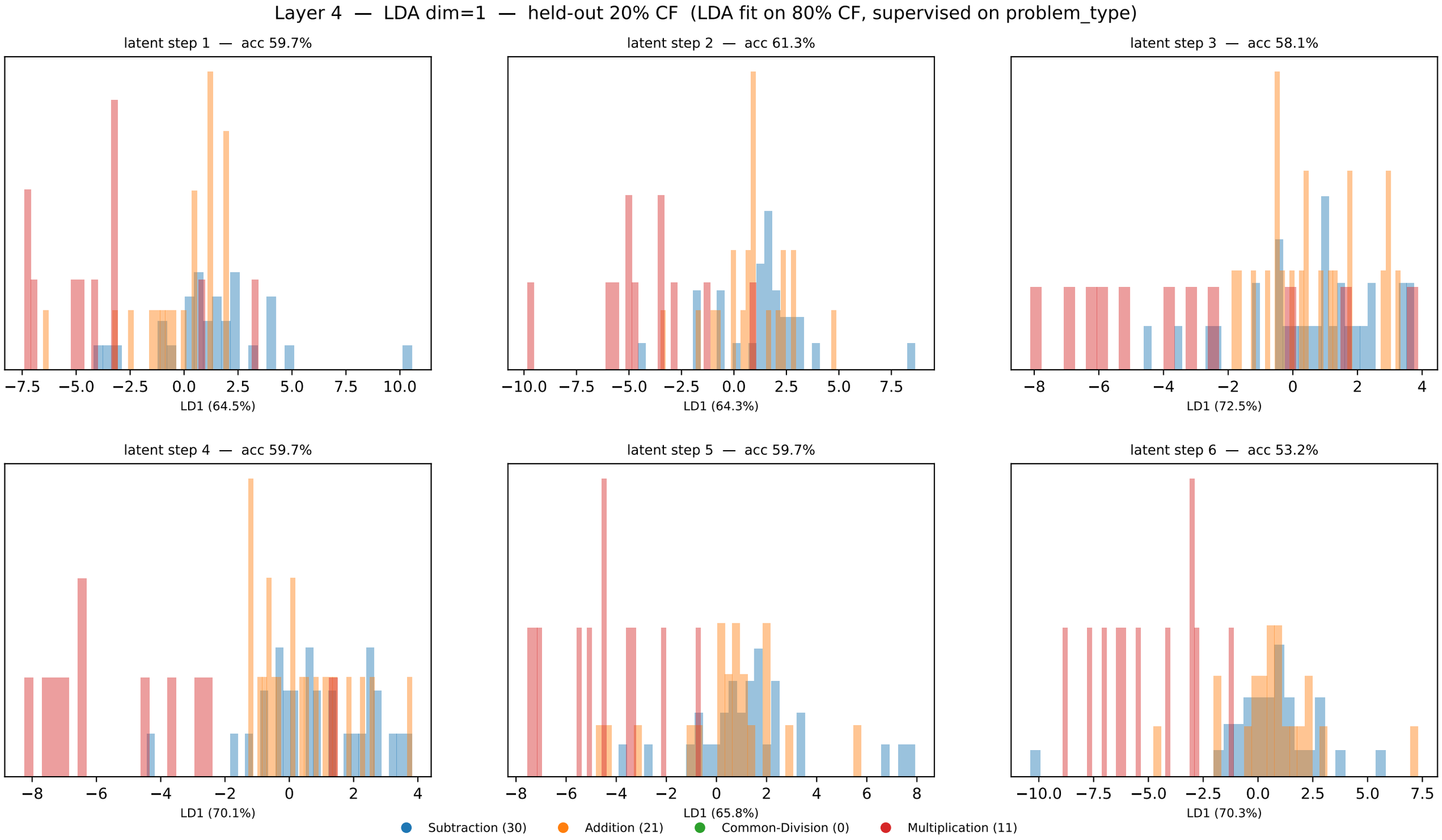


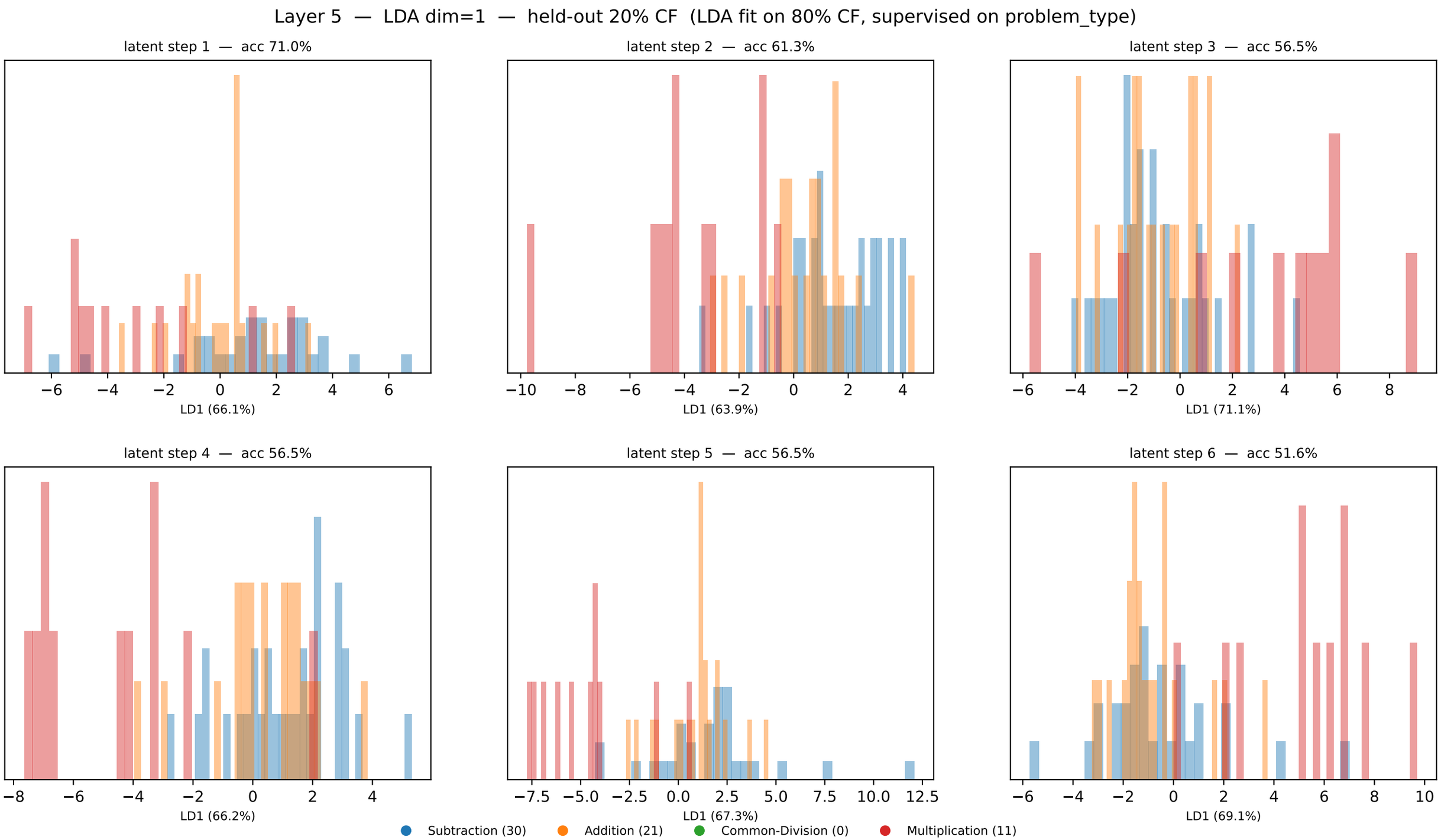

Layer 0 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 46.8%
LD1 (64.4%)
latent step 2 — acc 51.6%
LD1 (68.7%)
latent step 3 — acc 56.5%
LD1 (71.3%)
latent step 4 — acc 53.2%
LD1 (64.2%)
latent step 5 — acc 66.1%
LD1 (70.6%)
latent step 6 — acc 64.5%
LD1 (72.2%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

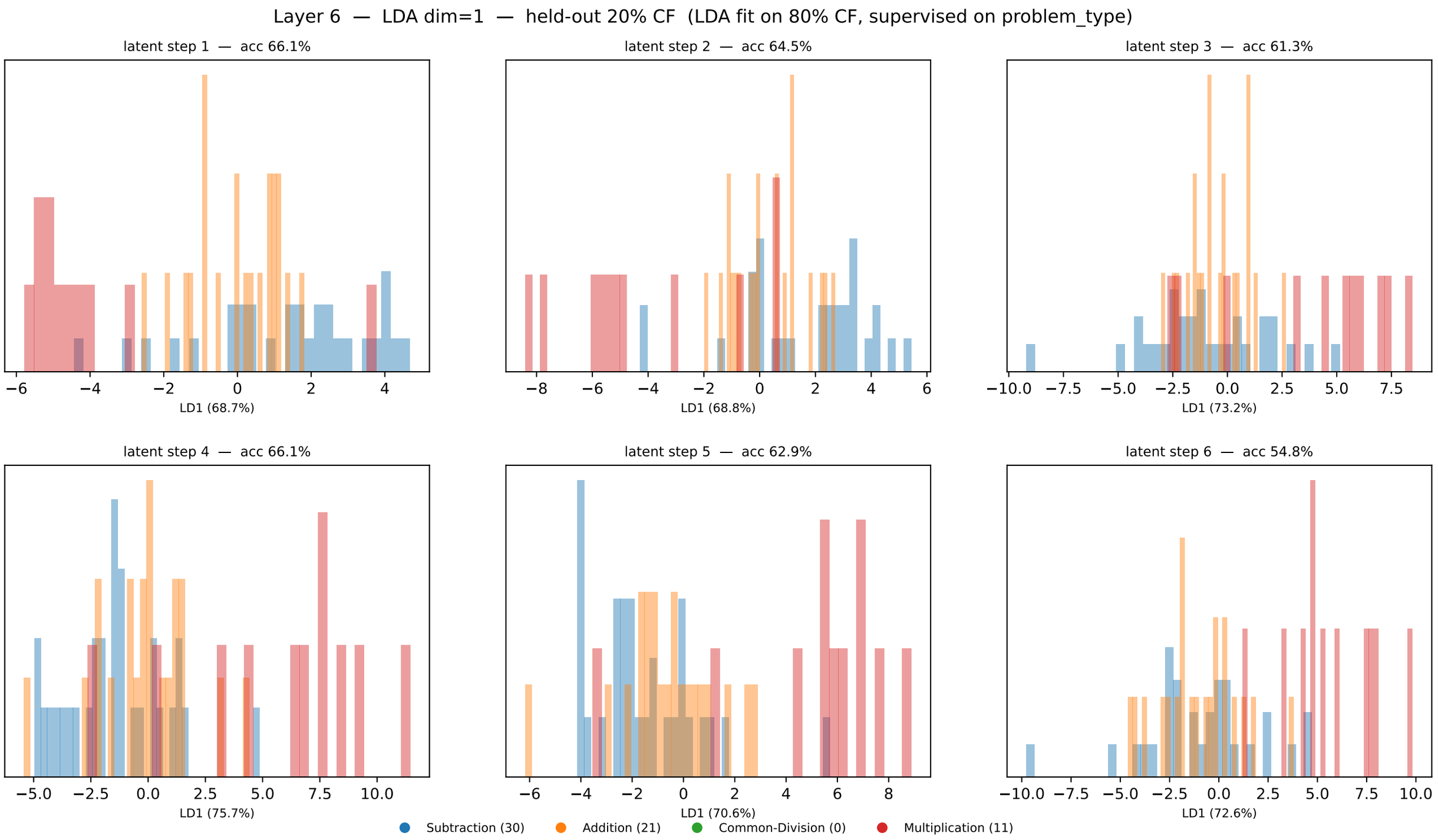

Layer 1 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 54.8%
LD1 (66.1%)
latent step 2 — acc 50.0%
LD1 (67.5%)
latent step 3 — acc 54.8%
LD1 (69.2%)
latent step 4 — acc 62.9%
LD1 (67.2%)
latent step 5 — acc 71.0%
LD1 (69.3%)
latent step 6 — acc 61.3%
LD1 (72.6%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

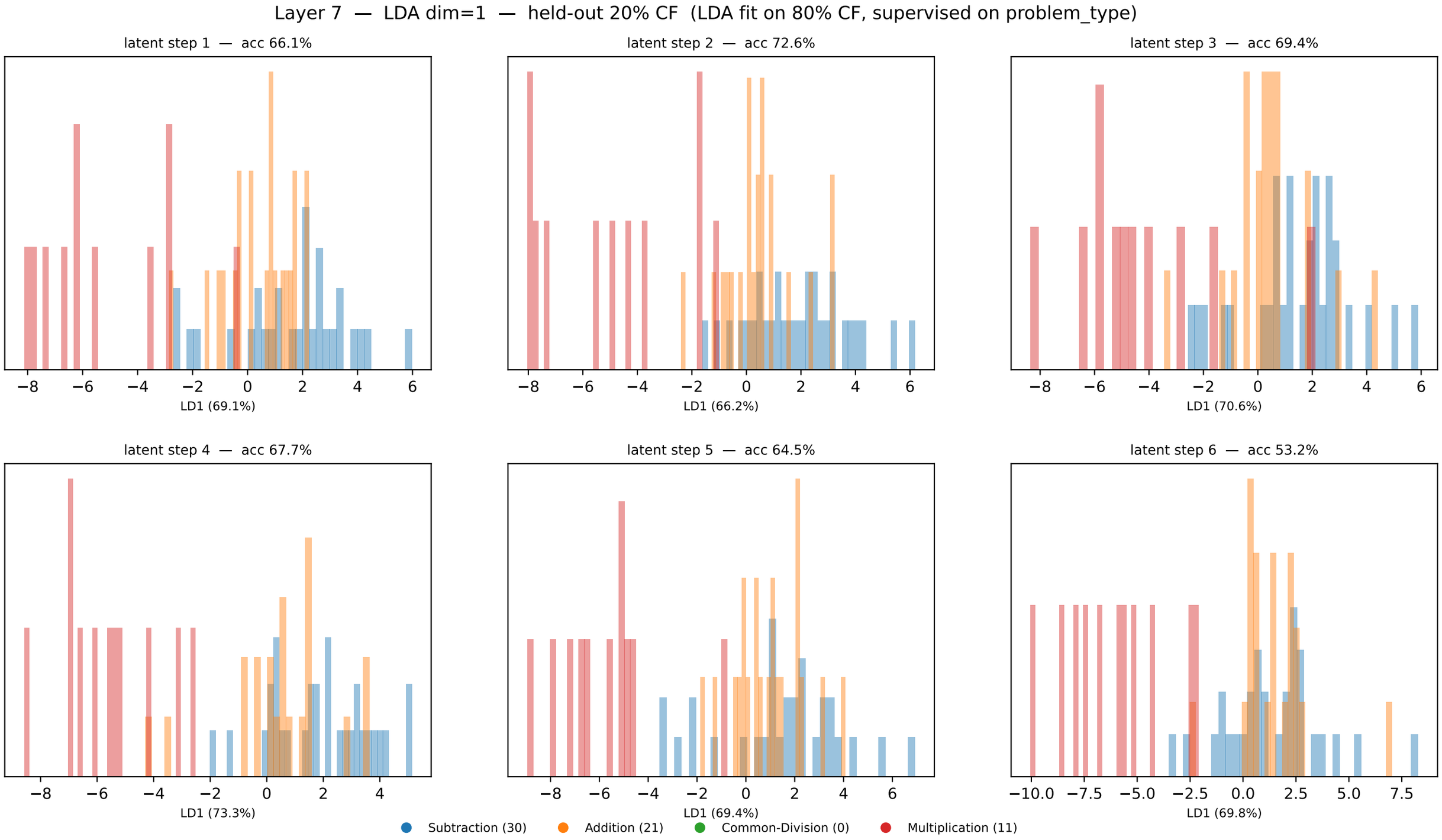

Layer 2 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 54.8%
LD1 (58.4%)
latent step 2 — acc 56.5%
LD1 (65.2%)
latent step 3 — acc 54.8%
LD1 (71.0%)
latent step 4 — acc 61.3%
LD1 (70.7%)
latent step 5 — acc 64.5%
LD1 (65.9%)
latent step 6 — acc 54.8%
LD1 (73.4%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

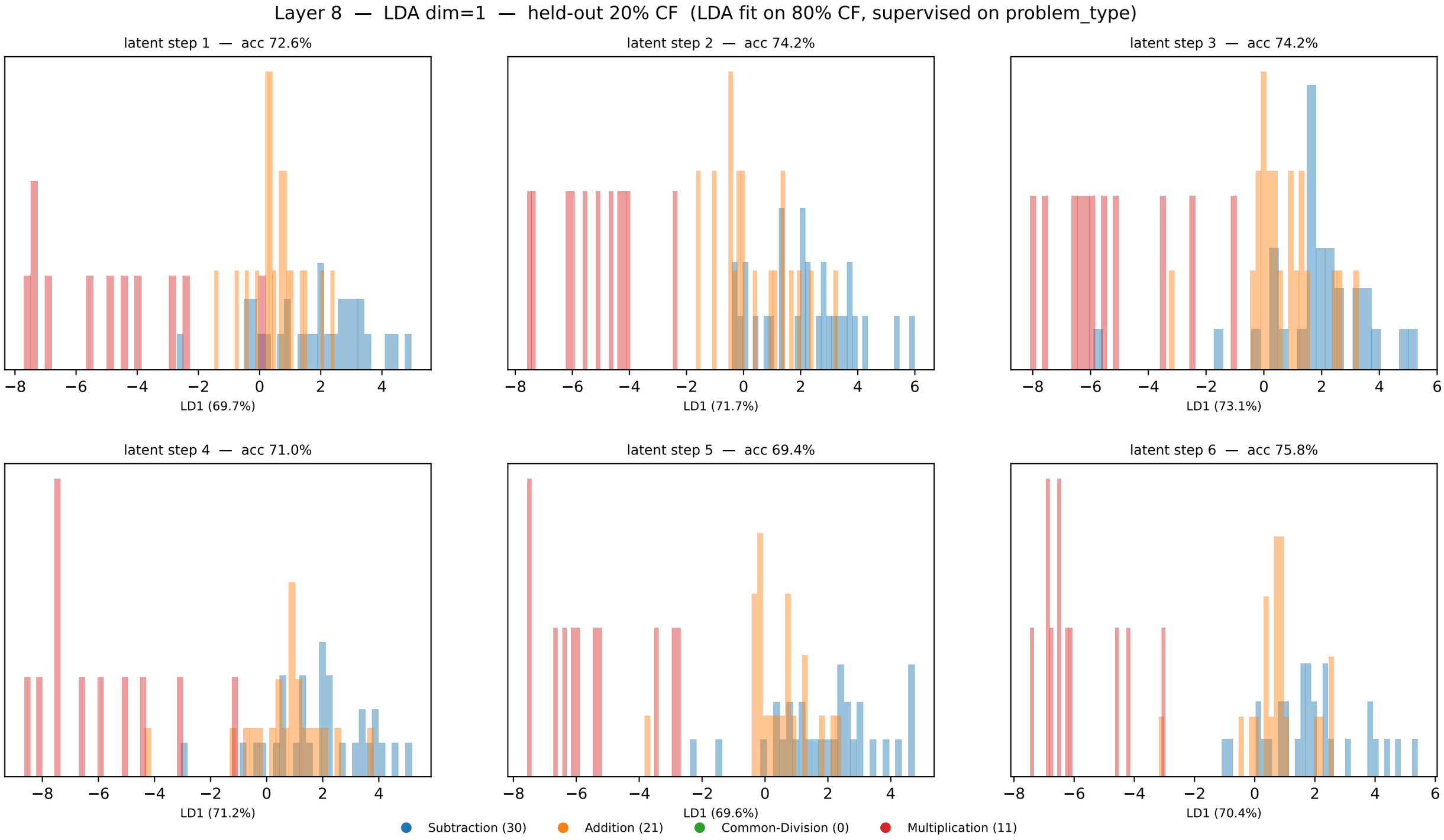

Layer 3 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 48.4%
LD1 (61.4%)
latent step 2 — acc 62.9%
LD1 (64.7%)
latent step 3 — acc 58.1%
LD1 (71.5%)
latent step 4 — acc 72.6%
LD1 (69.8%)
latent step 5 — acc 64.5%
LD1 (67.3%)
latent step 6 — acc 62.9%
LD1 (73.0%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

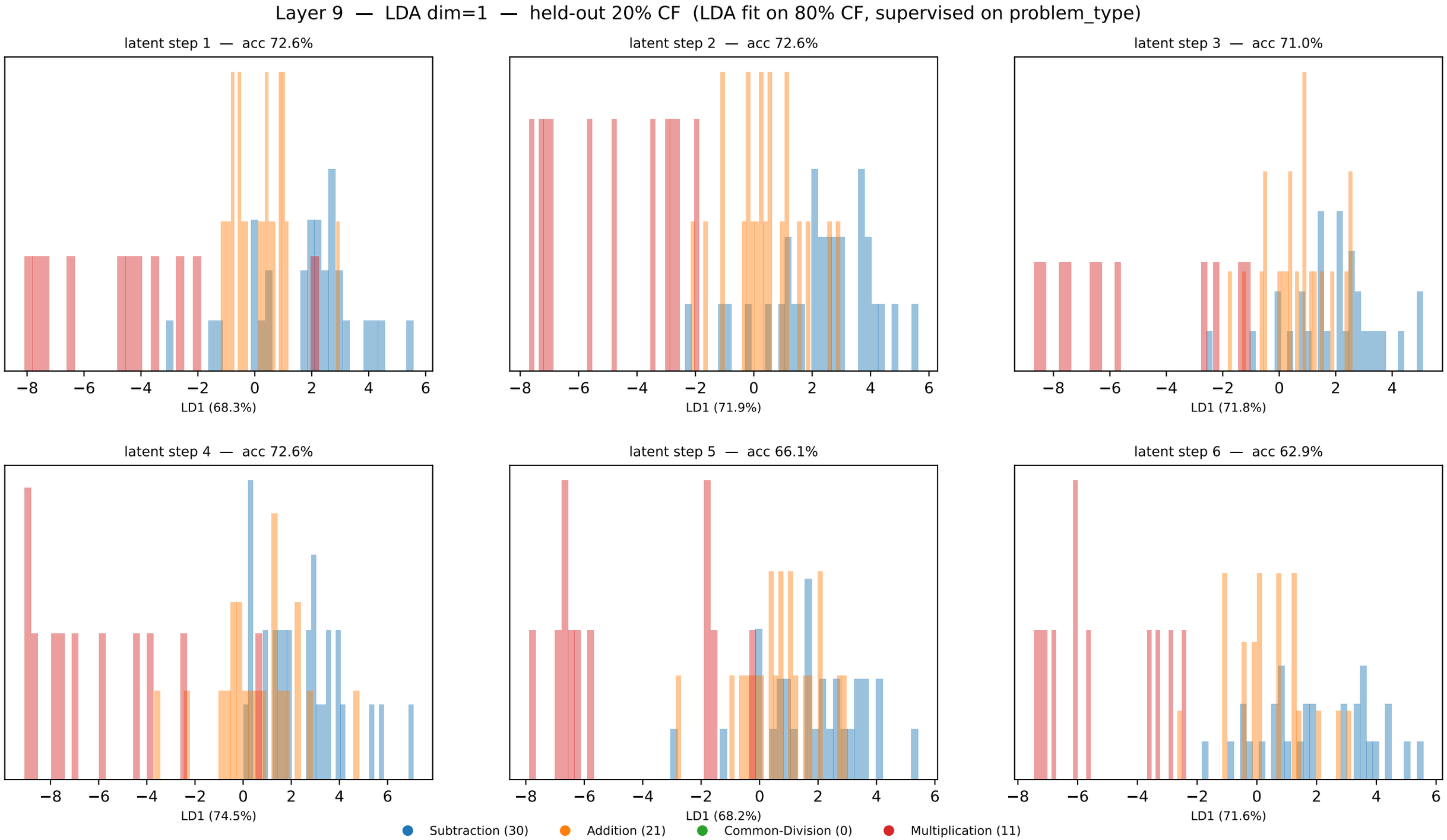

Layer 4 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 59.7%
LD1 (64.5%)
latent step 2 — acc 61.3%
LD1 (64.3%)
latent step 3 — acc 58.1%
LD1 (72.5%)
latent step 4 — acc 59.7%
LD1 (70.1%)
latent step 5 — acc 59.7%
LD1 (65.8%)
latent step 6 — acc 53.2%
LD1 (70.3%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

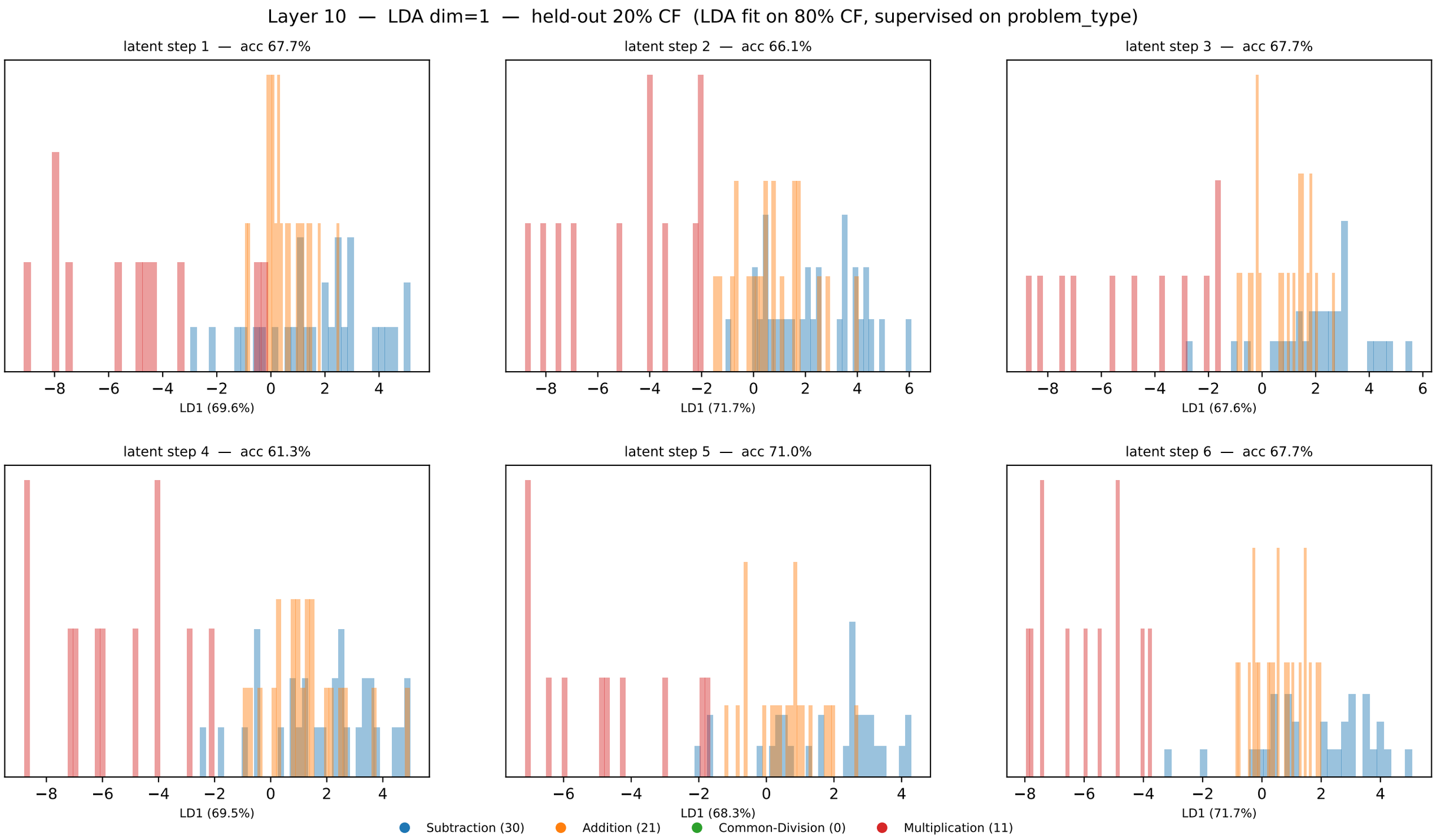

Layer 5 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 71.0%
LD1 (66.1%)
latent step 2 — acc 61.3%
LD1 (63.9%)
latent step 3 — acc 56.5%
LD1 (71.1%)
latent step 4 — acc 56.5%
LD1 (66.2%)
latent step 5 — acc 56.5%
LD1 (67.3%)
latent step 6 — acc 51.6%
LD1 (69.1%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

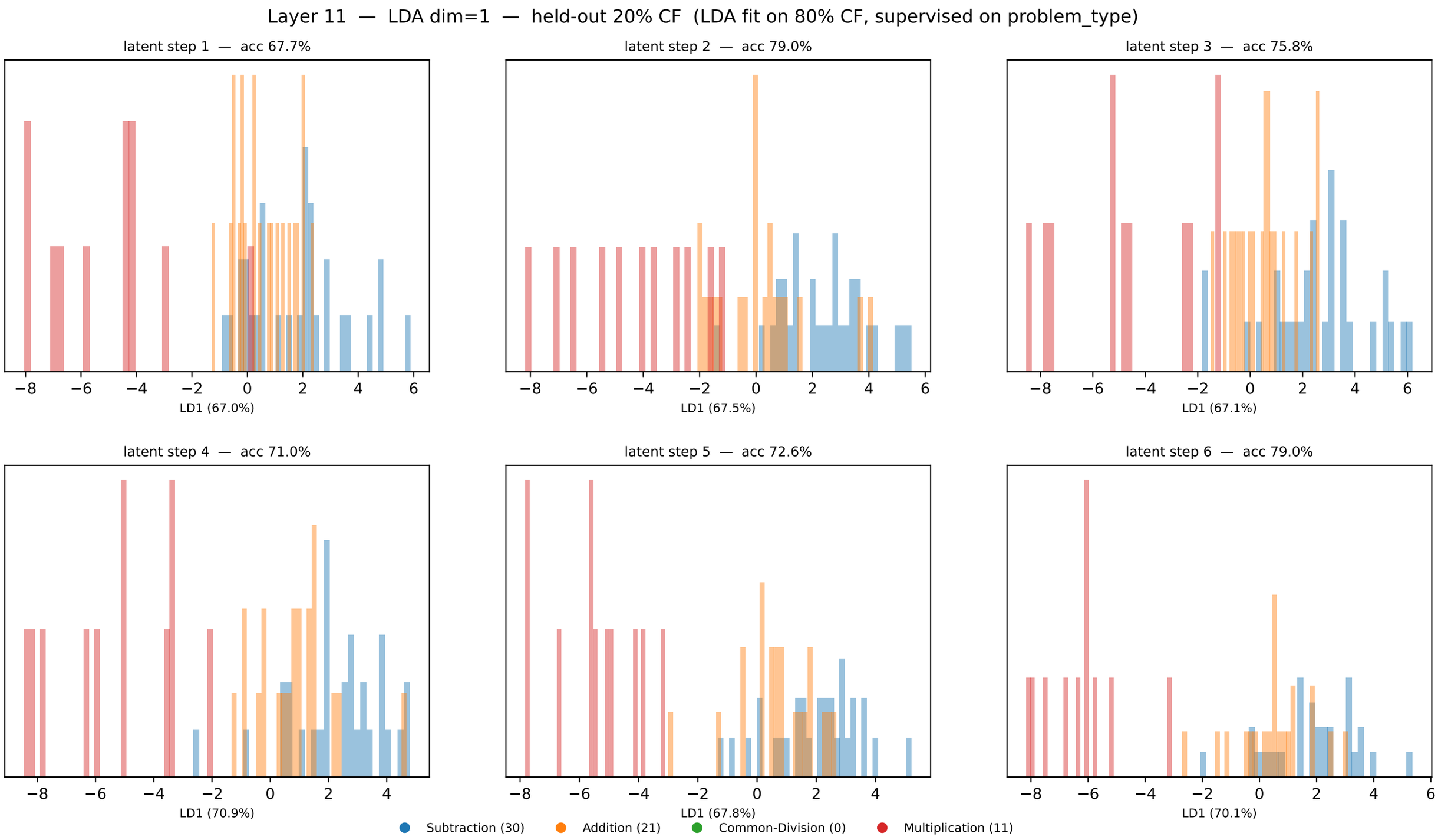

Layer 6 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 66.1%
LD1 (68.7%)
latent step 2 — acc 64.5%
LD1 (68.8%)
latent step 3 — acc 61.3%
LD1 (73.2%)
latent step 4 — acc 66.1%
LD1 (75.7%)
latent step 5 — acc 62.9%
LD1 (70.6%)
latent step 6 — acc 54.8%
LD1 (72.6%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

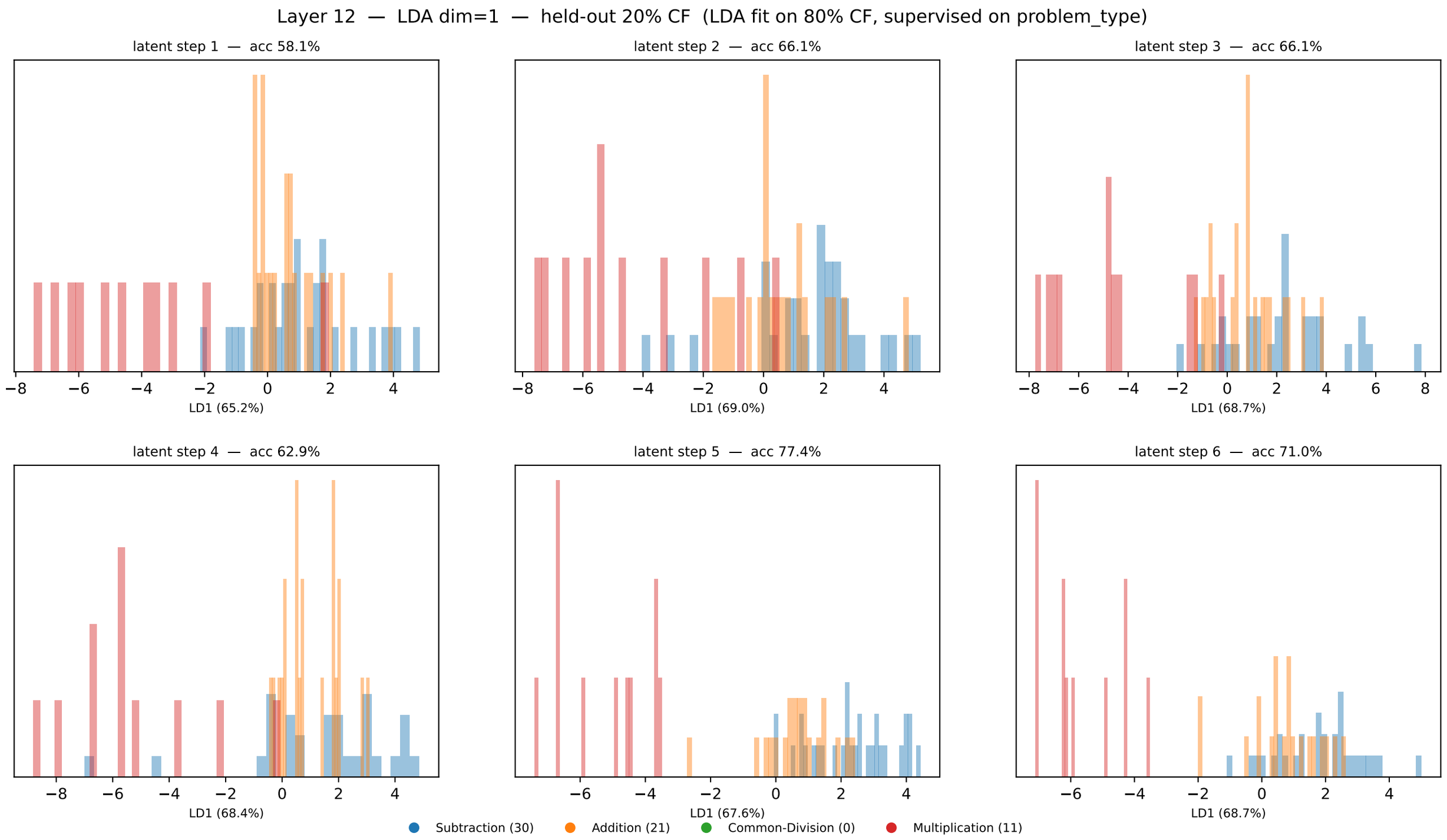

Layer 7 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 66.1%
LD1 (69.1%)
latent step 2 — acc 72.6%
LD1 (66.2%)
latent step 3 — acc 69.4%
LD1 (70.6%)
latent step 4 — acc 67.7%
LD1 (73.3%)
latent step 5 — acc 64.5%
LD1 (69.4%)
latent step 6 — acc 53.2%
LD1 (69.8%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

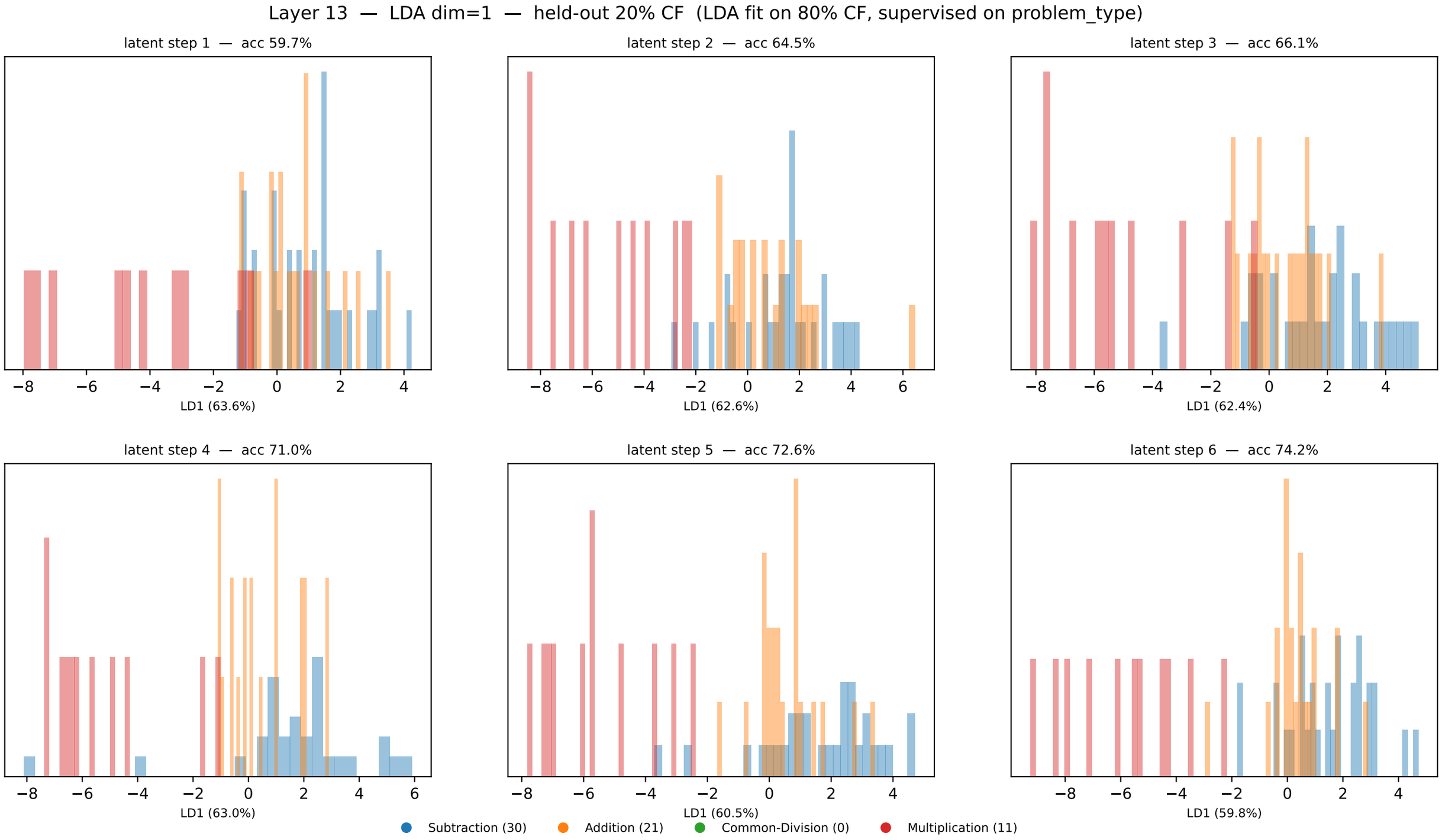

Layer 8 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 72.6%
LD1 (69.7%)
latent step 2 — acc 74.2%
LD1 (71.7%)
latent step 3 — acc 74.2%
LD1 (73.1%)
latent step 4 — acc 71.0%
LD1 (71.2%)
latent step 5 — acc 69.4%
LD1 (69.6%)
latent step 6 — acc 75.8%
LD1 (70.4%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

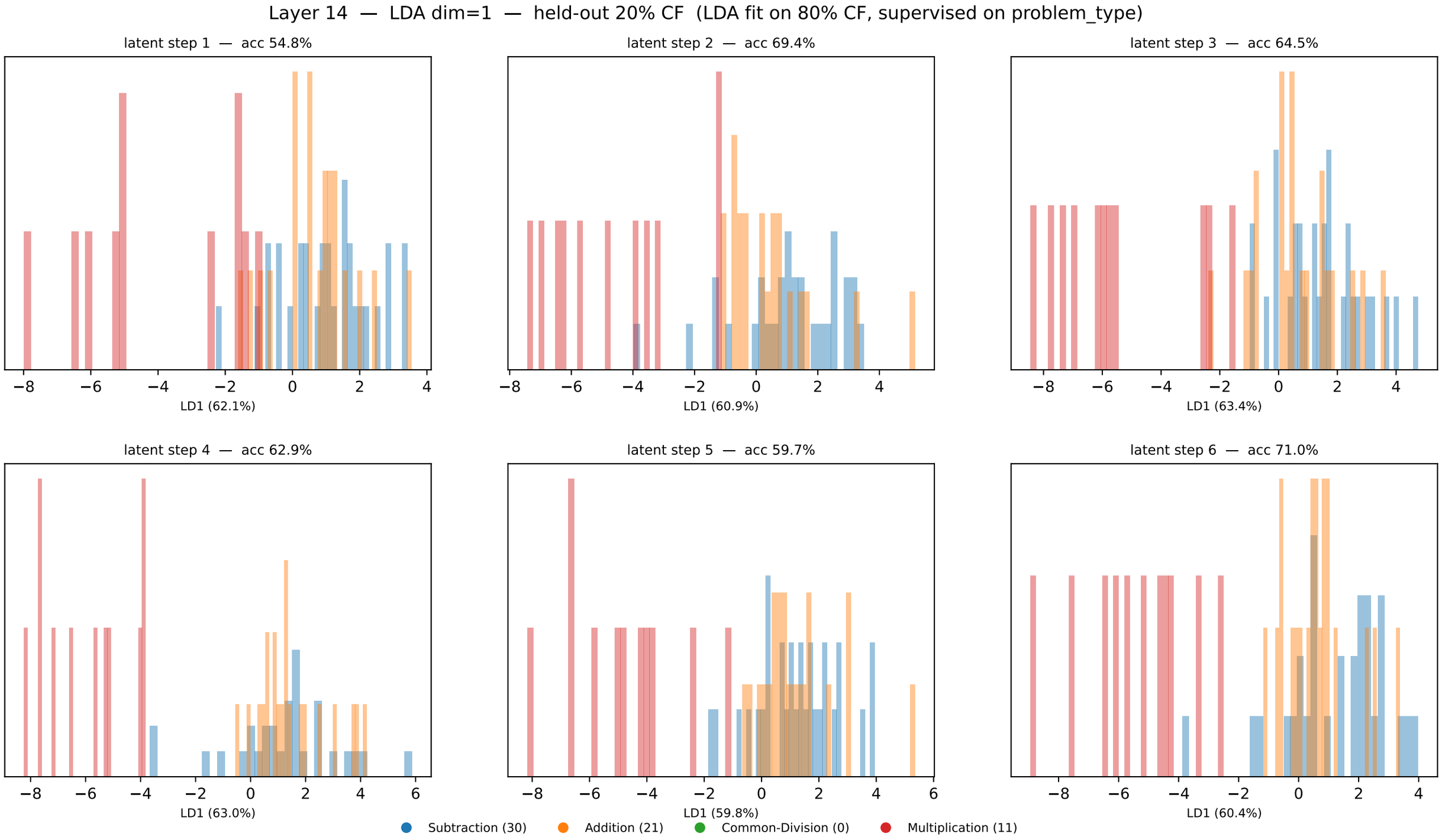

Layer 9 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 72.6%
LD1 (68.3%)
latent step 2 — acc 72.6%
LD1 (71.9%)
latent step 3 — acc 71.0%
LD1 (71.8%)
latent step 4 — acc 72.6%
LD1 (74.5%)
latent step 5 — acc 66.1%
LD1 (68.2%)
latent step 6 — acc 62.9%
LD1 (71.6%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

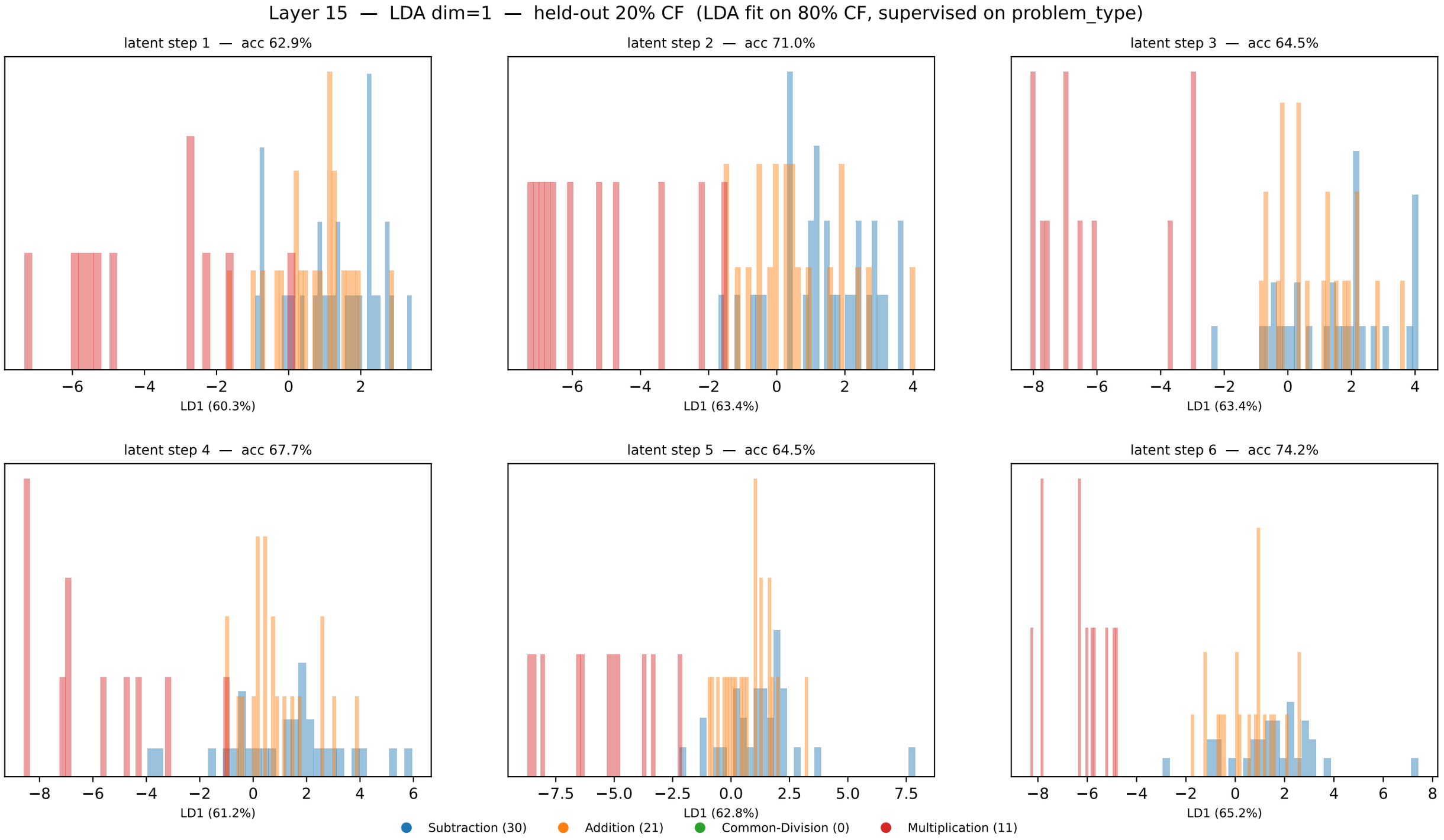

Layer 10 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 67.7%
LD1 (69.6%)
latent step 2 — acc 66.1%
LD1 (71.7%)
latent step 3 — acc 67.7%
LD1 (67.6%)
latent step 4 — acc 61.3%
LD1 (69.5%)
latent step 5 — acc 71.0%
LD1 (68.3%)
latent step 6 — acc 67.7%
LD1 (71.7%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

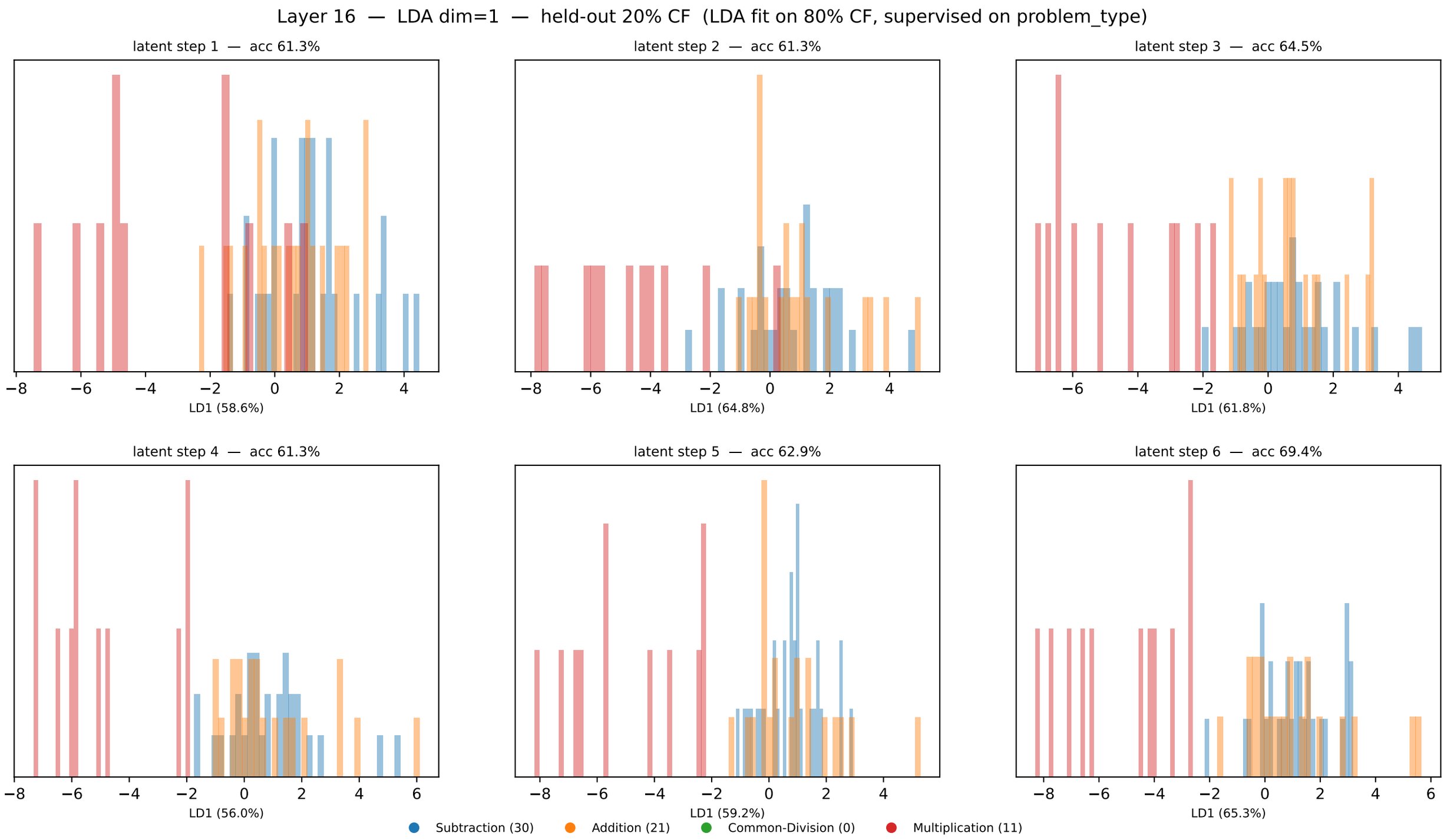

Layer 11 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 67.7%
LD1 (67.0%)
latent step 2 — acc 79.0%
LD1 (67.5%)
latent step 3 — acc 75.8%
LD1 (67.1%)
latent step 4 — acc 71.0%
LD1 (70.9%)
latent step 5 — acc 72.6%
LD1 (67.8%)
latent step 6 — acc 79.0%
LD1 (70.1%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

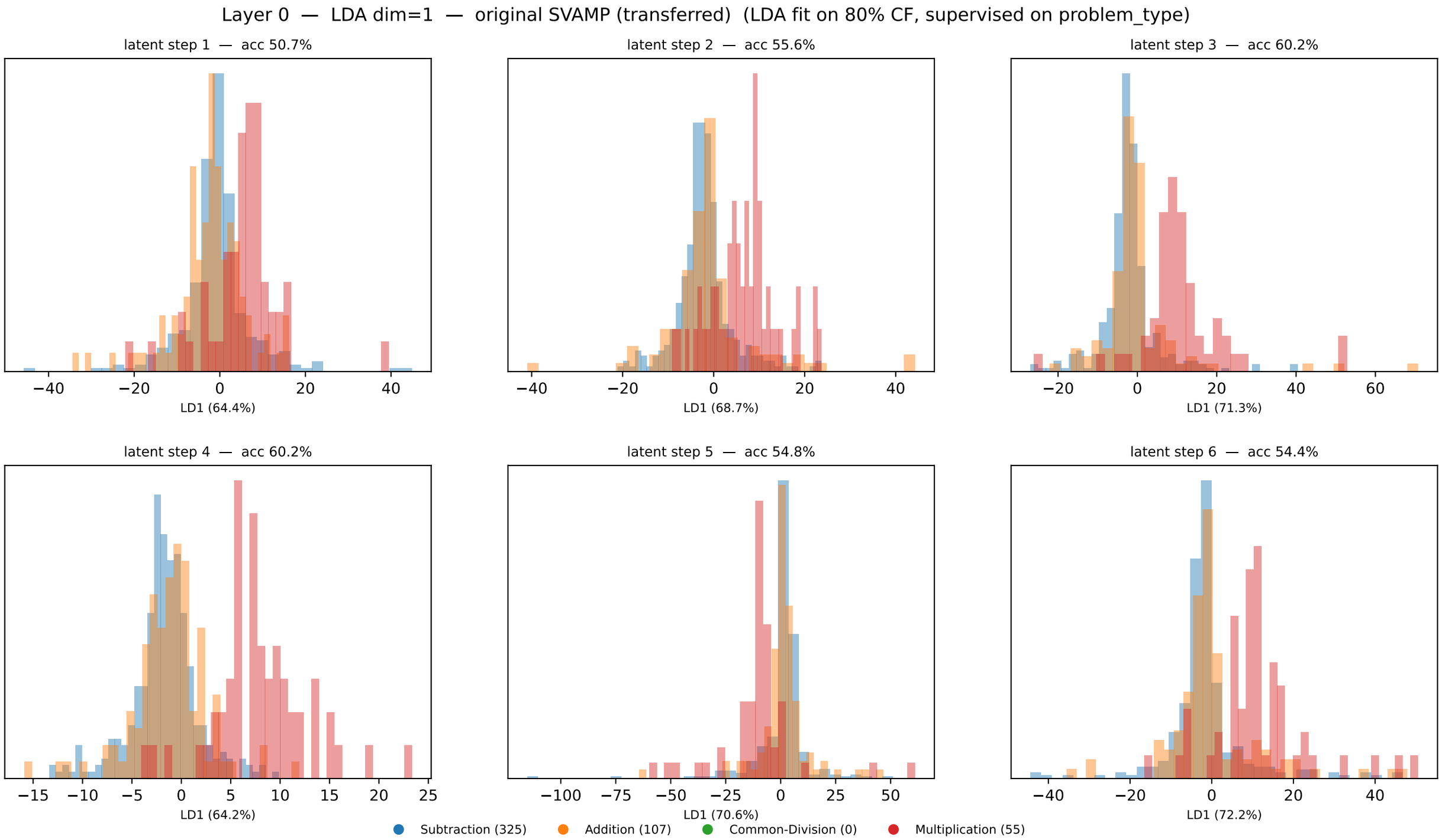

Layer 12 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 58.1%
LD1 (65.2%)
latent step 2 — acc 66.1%
LD1 (69.0%)
latent step 3 — acc 66.1%
LD1 (68.7%)
latent step 4 — acc 62.9%
LD1 (68.4%)
latent step 5 — acc 77.4%
LD1 (67.6%)
latent step 6 — acc 71.0%
LD1 (68.7%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

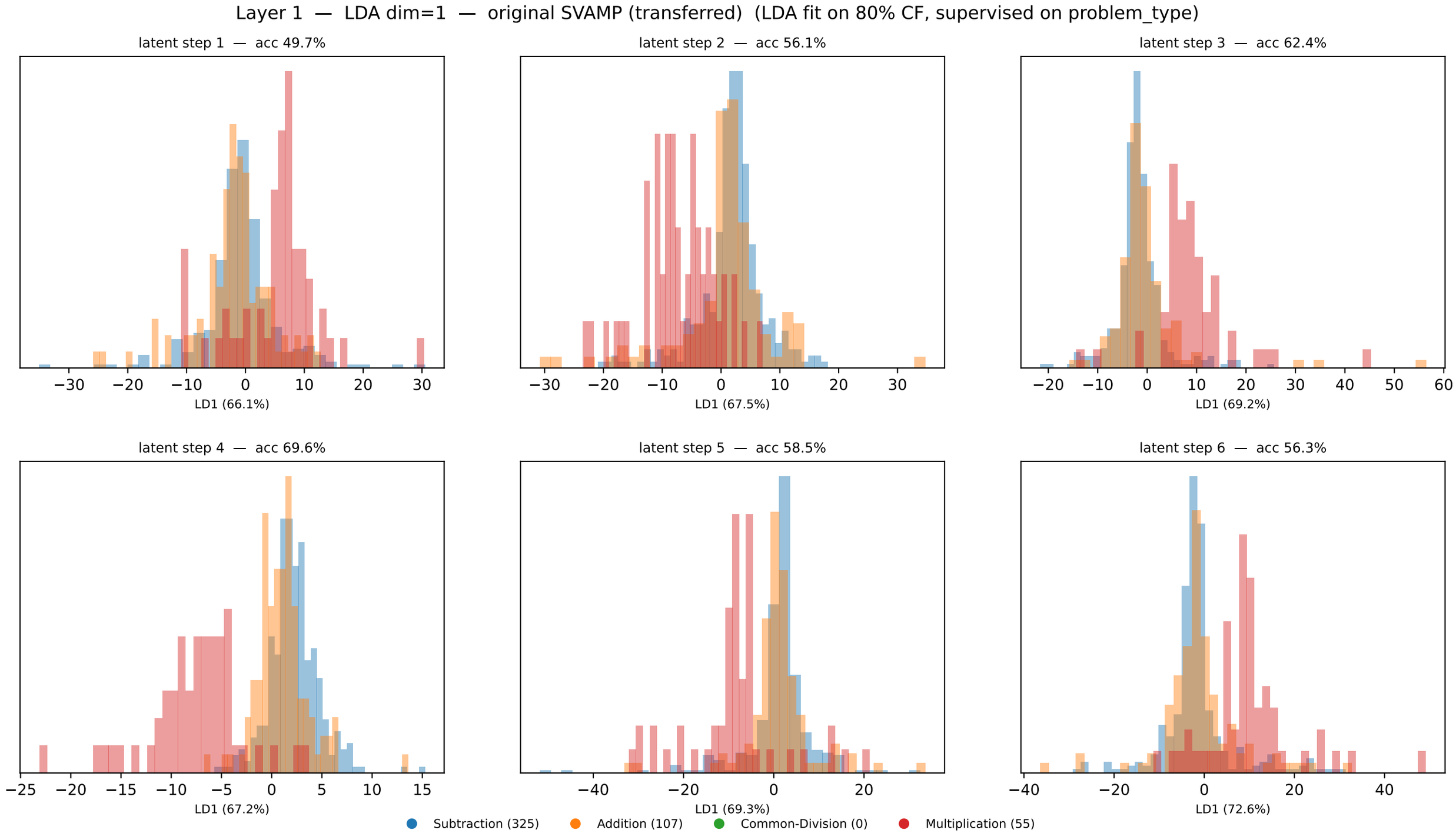

Layer 13 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 59.7%
LD1 (63.6%)
latent step 2 — acc 64.5%
LD1 (62.6%)
latent step 3 — acc 66.1%
LD1 (62.4%)
latent step 4 — acc 71.0%
LD1 (63.0%)
latent step 5 — acc 72.6%
LD1 (60.5%)
latent step 6 — acc 74.2%
LD1 (59.8%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

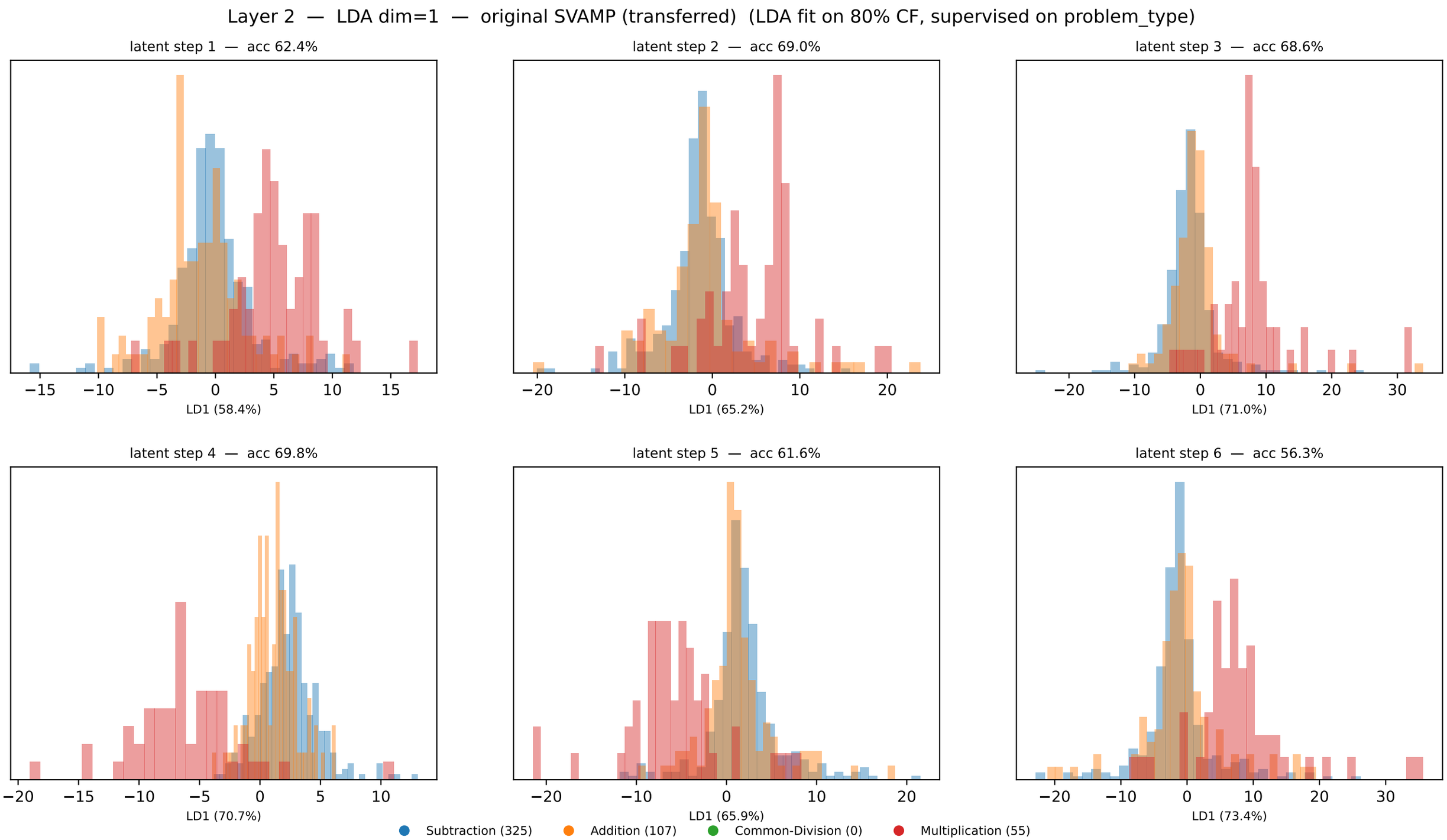

Layer 14 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 54.8%
LD1 (62.1%)
latent step 2 — acc 69.4%
LD1 (60.9%)
latent step 3 — acc 64.5%
LD1 (63.4%)
latent step 4 — acc 62.9%
LD1 (63.0%)
latent step 5 — acc 59.7%
LD1 (59.8%)
latent step 6 — acc 71.0%
LD1 (60.4%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

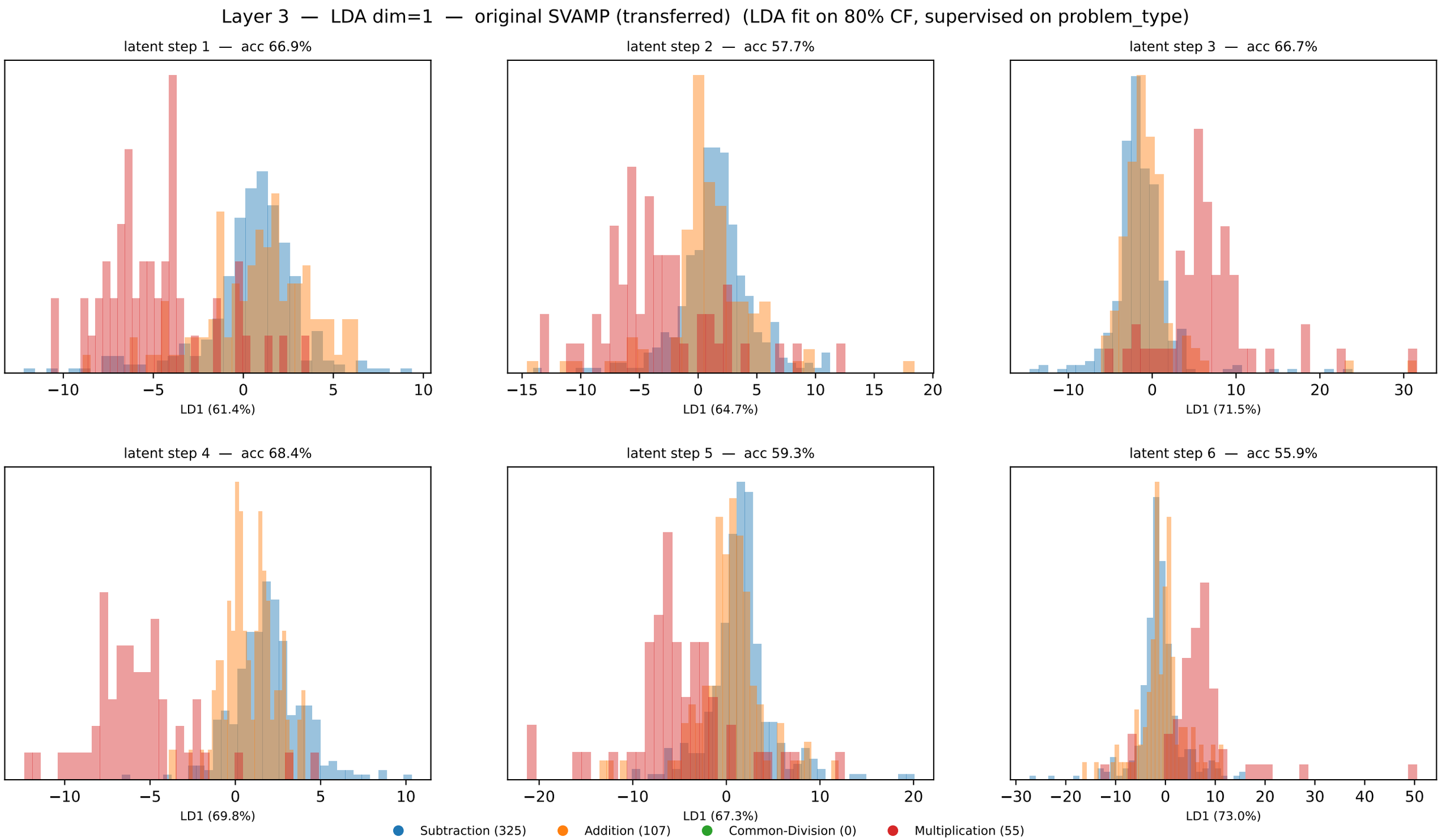

Layer 15 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 62.9%
LD1 (60.3%)
latent step 2 — acc 71.0%
LD1 (63.4%)
latent step 3 — acc 64.5%
LD1 (63.4%)
latent step 4 — acc 67.7%
LD1 (61.2%)
latent step 5 — acc 64.5%
LD1 (62.8%)
latent step 6 — acc 74.2%
LD1 (65.2%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

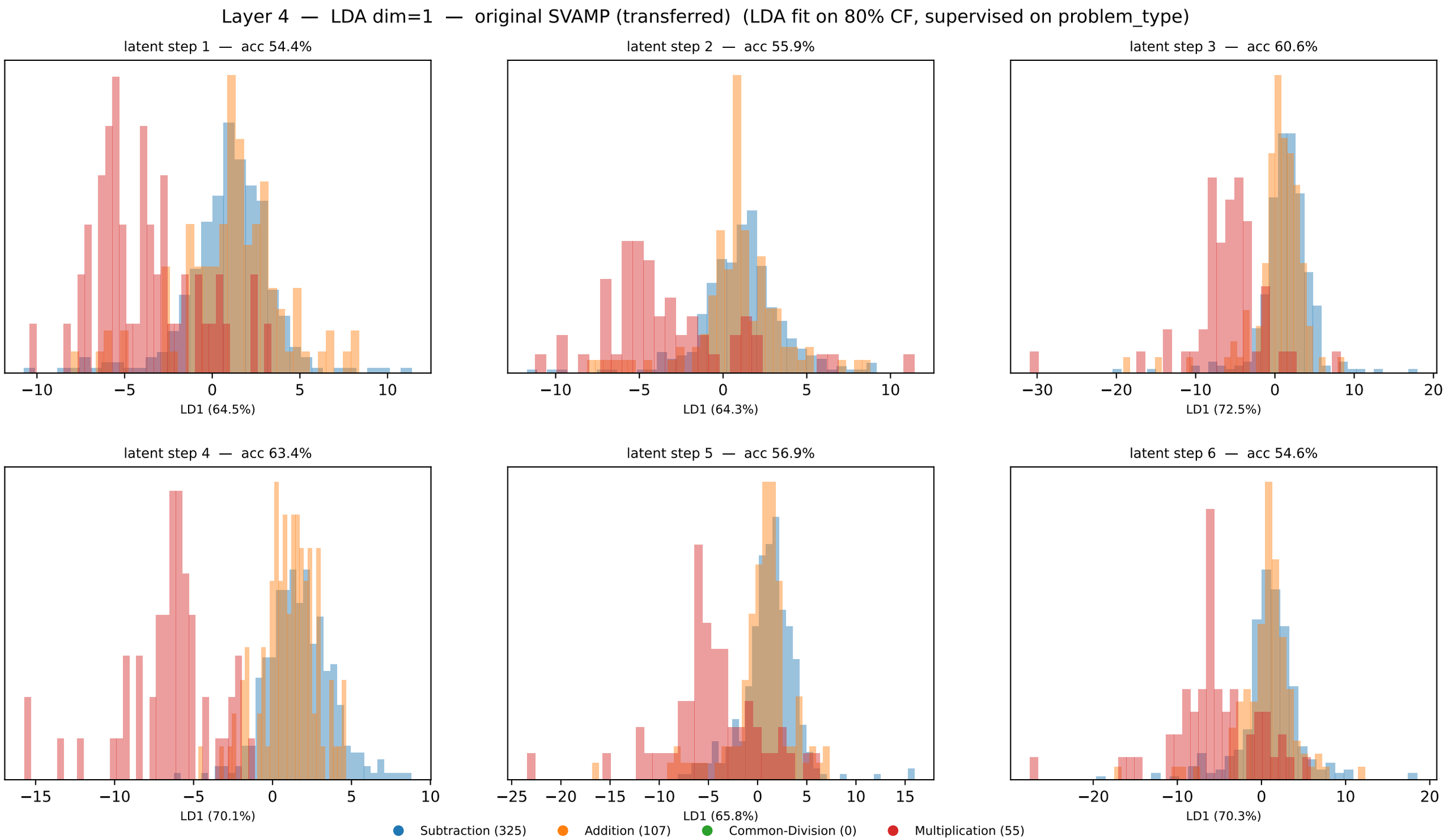

Layer 16 — LDA dim=1 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 61.3%
LD1 (58.6%)
latent step 2 — acc 61.3%
LD1 (64.8%)
latent step 3 — acc 64.5%
LD1 (61.8%)
latent step 4 — acc 61.3%
LD1 (56.0%)
latent step 5 — acc 62.9%
LD1 (59.2%)
latent step 6 — acc 69.4%
LD1 (65.3%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

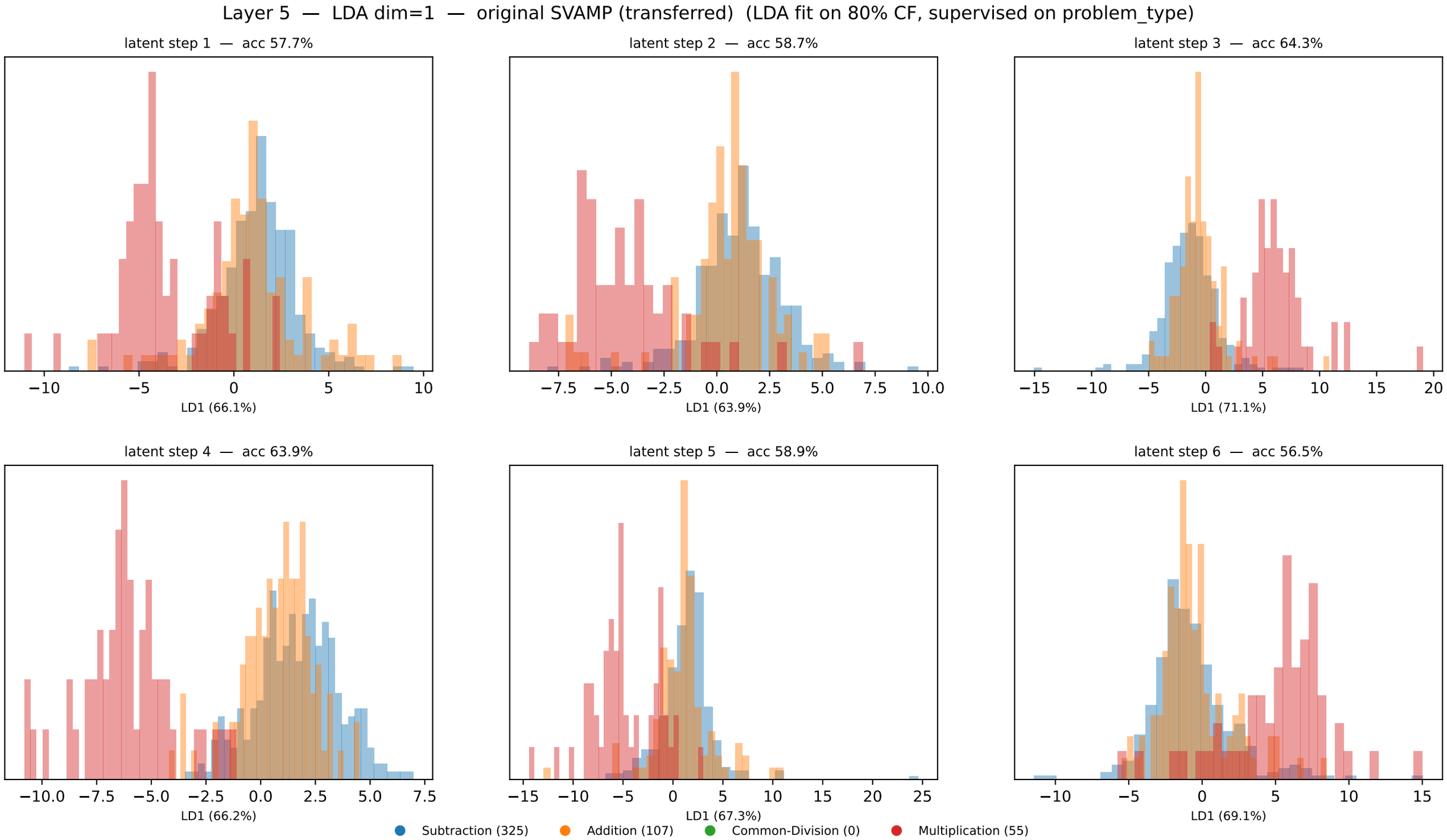

Layer 0 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 50.7%
LD1 (64.4%)
latent step 2 — acc 55.6%
LD1 (68.7%)
latent step 3 — acc 60.2%
LD1 (71.3%)
latent step 4 — acc 60.2%
LD1 (64.2%)
latent step 5 — acc 54.8%
LD1 (70.6%)
latent step 6 — acc 54.4%
LD1 (72.2%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

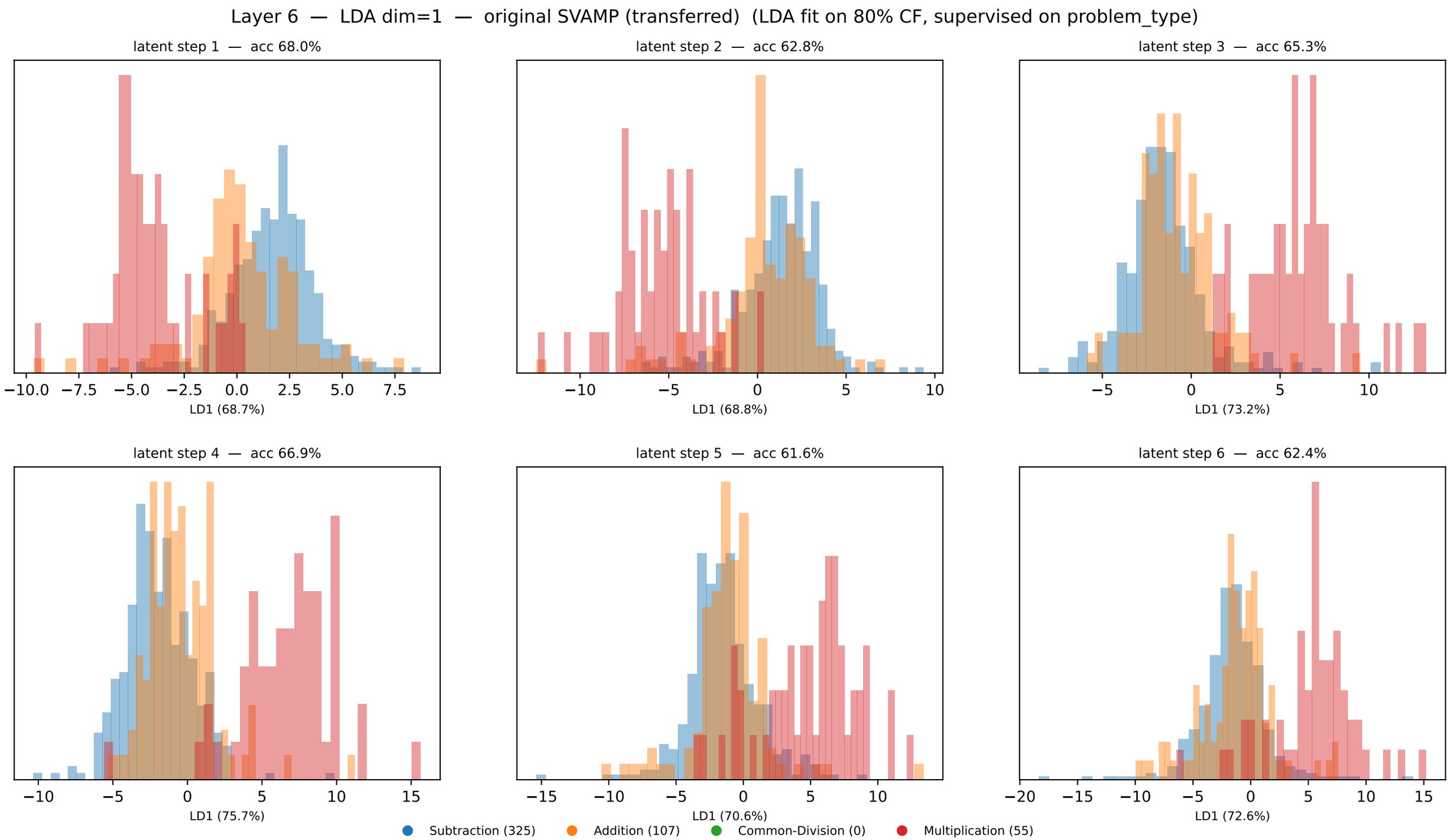

Layer 1 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 49.7%
LD1 (66.1%)
latent step 2 — acc 56.1%
LD1 (67.5%)
latent step 3 — acc 62.4%
LD1 (69.2%)
latent step 4 — acc 69.6%
LD1 (67.2%)
latent step 5 — acc 58.5%
LD1 (69.3%)
latent step 6 — acc 56.3%
LD1 (72.6%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

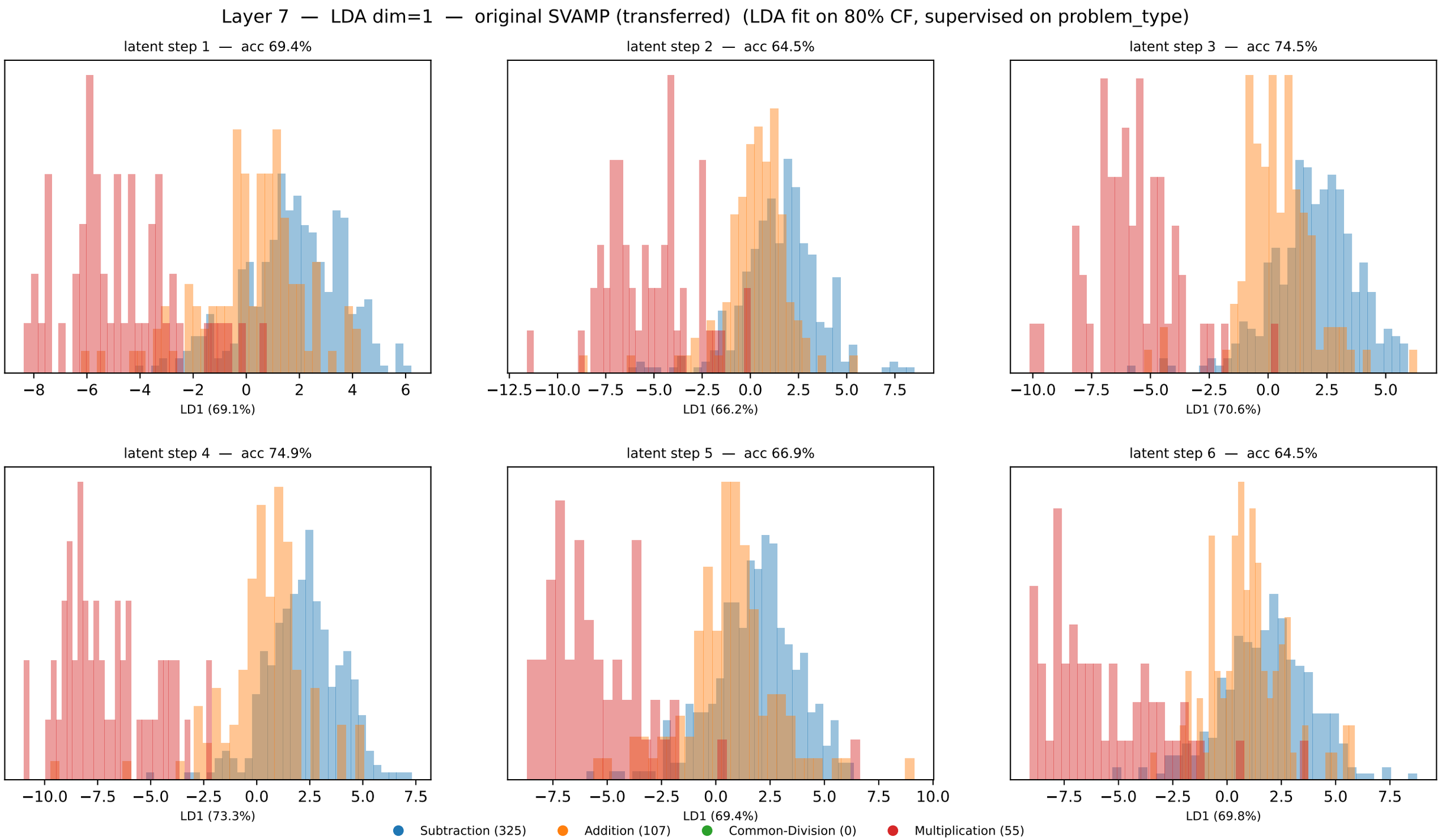

Layer 2 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 62.4%
LD1 (58.4%)
latent step 2 — acc 69.0%
LD1 (65.2%)
latent step 3 — acc 68.6%
LD1 (71.0%)
latent step 4 — acc 69.8%
LD1 (70.7%)
latent step 5 — acc 61.6%
LD1 (65.9%)
latent step 6 — acc 56.3%
LD1 (73.4%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

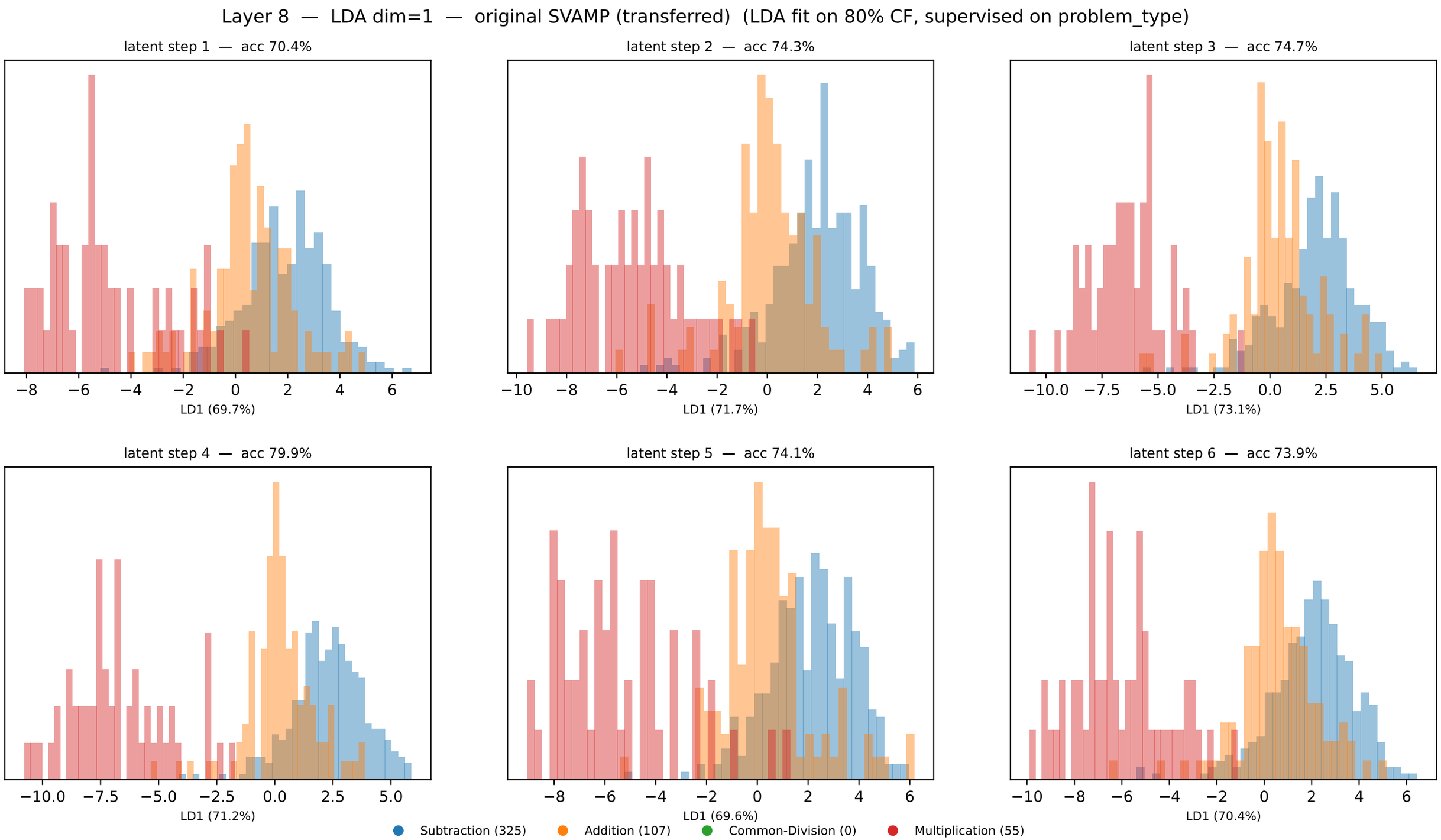

Layer 3 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 66.9%
LD1 (61.4%)
latent step 2 — acc 57.7%
LD1 (64.7%)
latent step 3 — acc 66.7%
LD1 (71.5%)
latent step 4 — acc 68.4%
LD1 (69.8%)
latent step 5 — acc 59.3%
LD1 (67.3%)
latent step 6 — acc 55.9%
LD1 (73.0%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

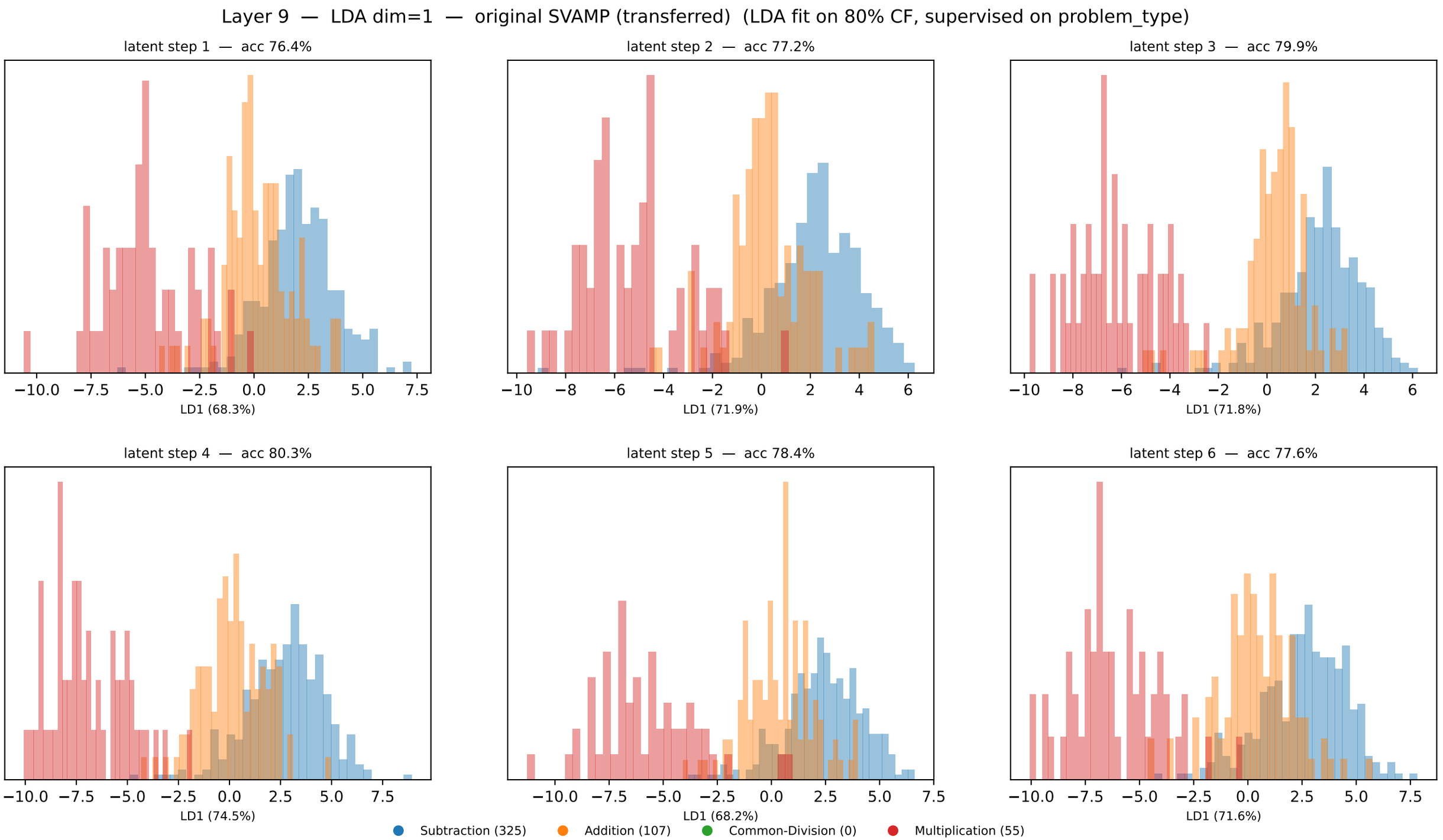

Layer 4 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 54.4%
LD1 (64.5%)
latent step 2 — acc 55.9%
LD1 (64.3%)
latent step 3 — acc 60.6%
LD1 (72.5%)
latent step 4 — acc 63.4%
LD1 (70.1%)
latent step 5 — acc 56.9%
LD1 (65.8%)
latent step 6 — acc 54.6%
LD1 (70.3%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

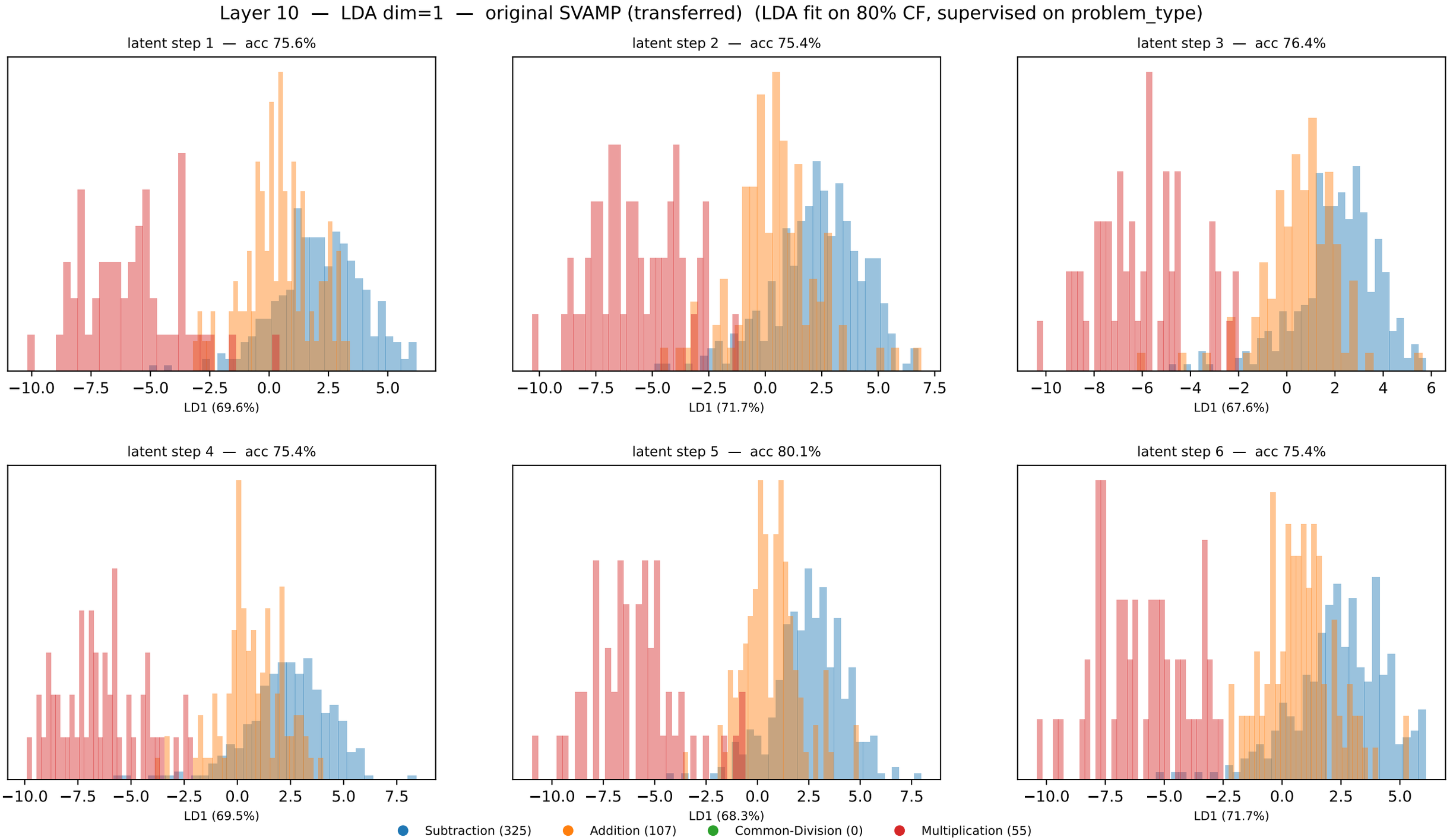

Layer 5 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 57.7%
LD1 (66.1%)
latent step 2 — acc 58.7%
LD1 (63.9%)
latent step 3 — acc 64.3%
LD1 (71.1%)
latent step 4 — acc 63.9%
LD1 (66.2%)
latent step 5 — acc 58.9%
LD1 (67.3%)
latent step 6 — acc 56.5%
LD1 (69.1%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

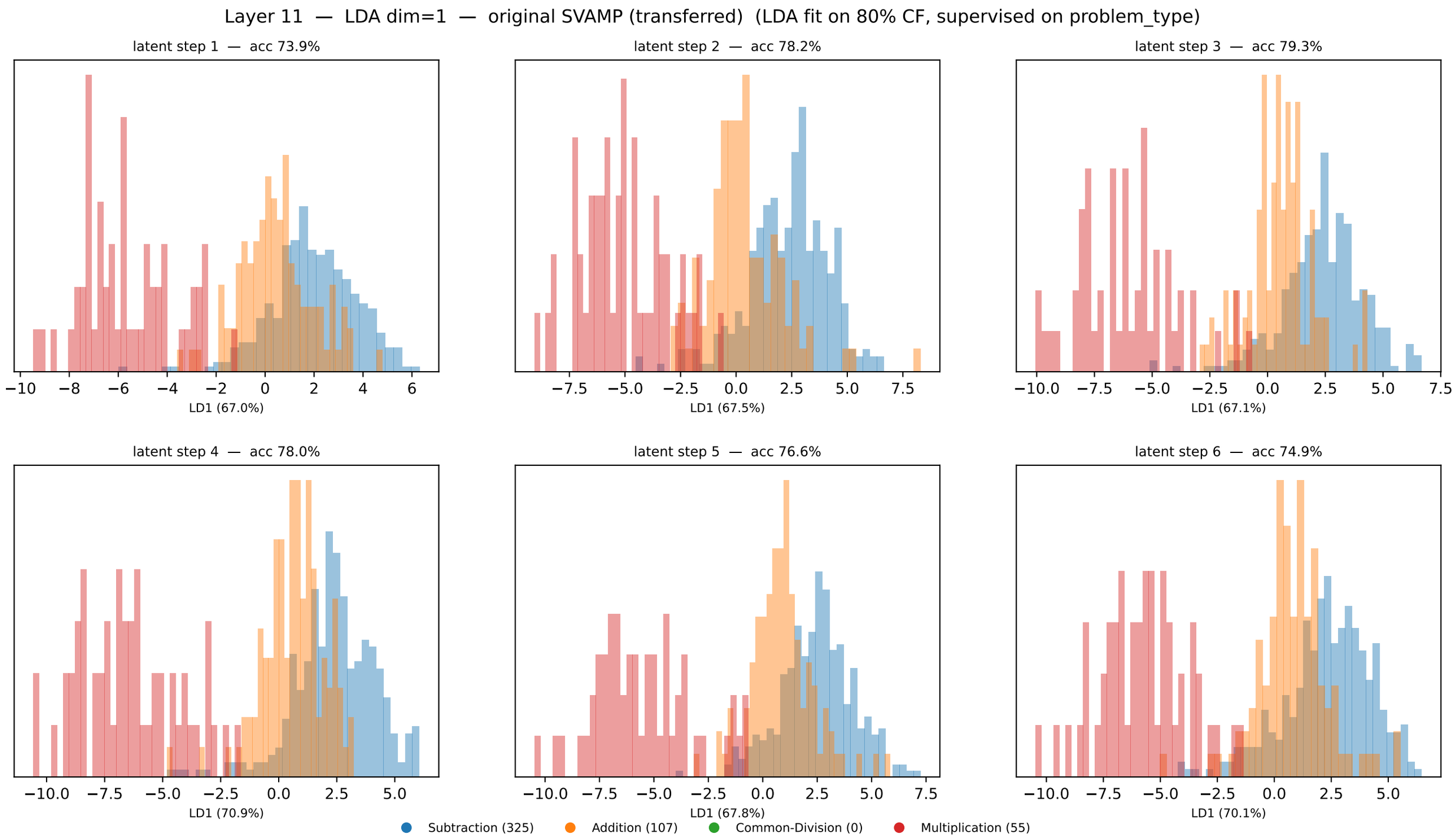

Layer 6 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 68.0%
LD1 (68.7%)
latent step 2 — acc 62.8%
LD1 (68.8%)
latent step 3 — acc 65.3%
LD1 (73.2%)
latent step 4 — acc 66.9%
LD1 (75.7%)
latent step 5 — acc 61.6%
LD1 (70.6%)
latent step 6 — acc 62.4%
LD1 (72.6%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

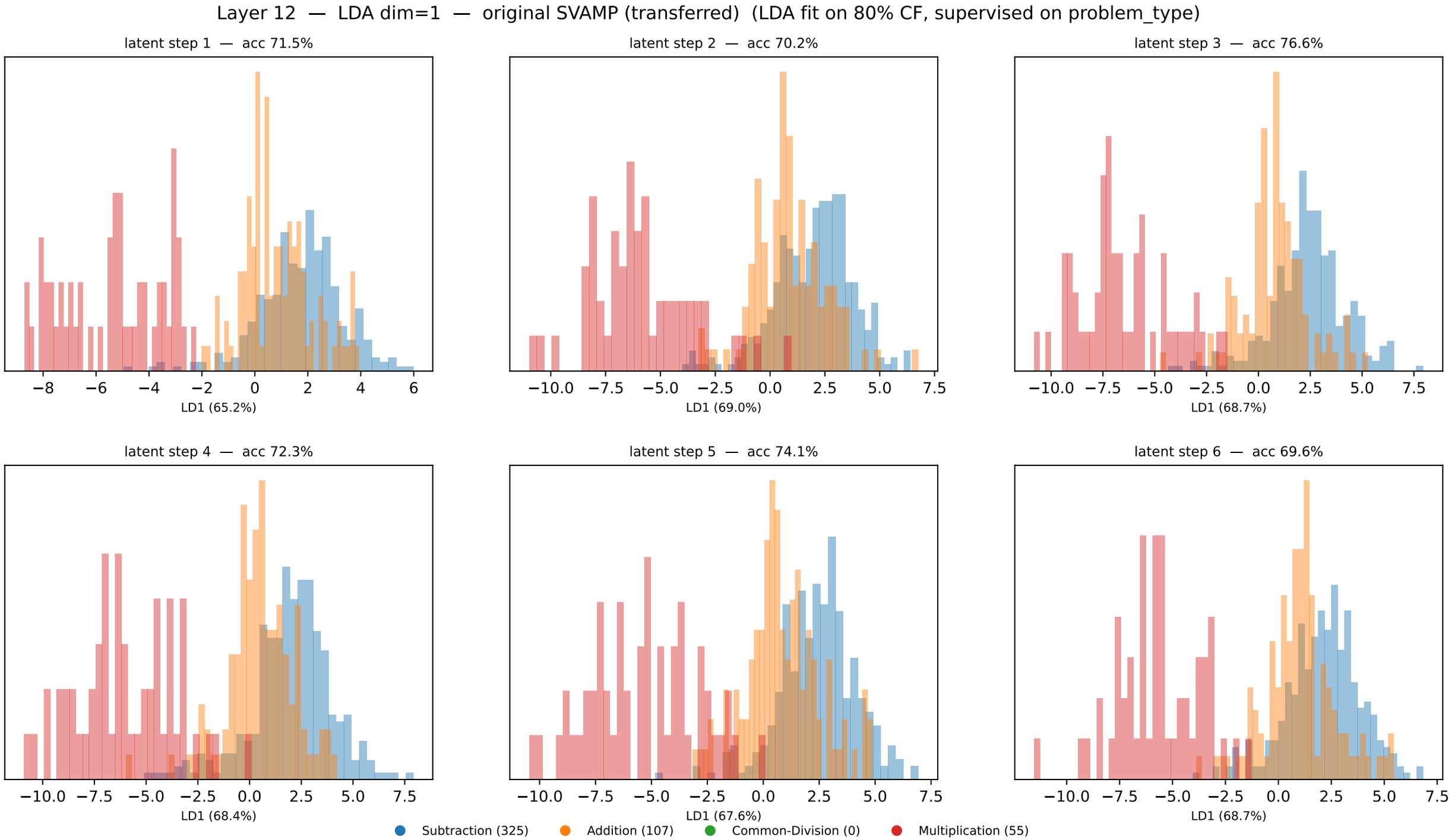

Layer 7 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 69.4%
LD1 (69.1%)
latent step 2 — acc 64.5%
LD1 (66.2%)
latent step 3 — acc 74.5%
LD1 (70.6%)
latent step 4 — acc 74.9%
LD1 (73.3%)
latent step 5 — acc 66.9%
LD1 (69.4%)
latent step 6 — acc 64.5%
LD1 (69.8%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

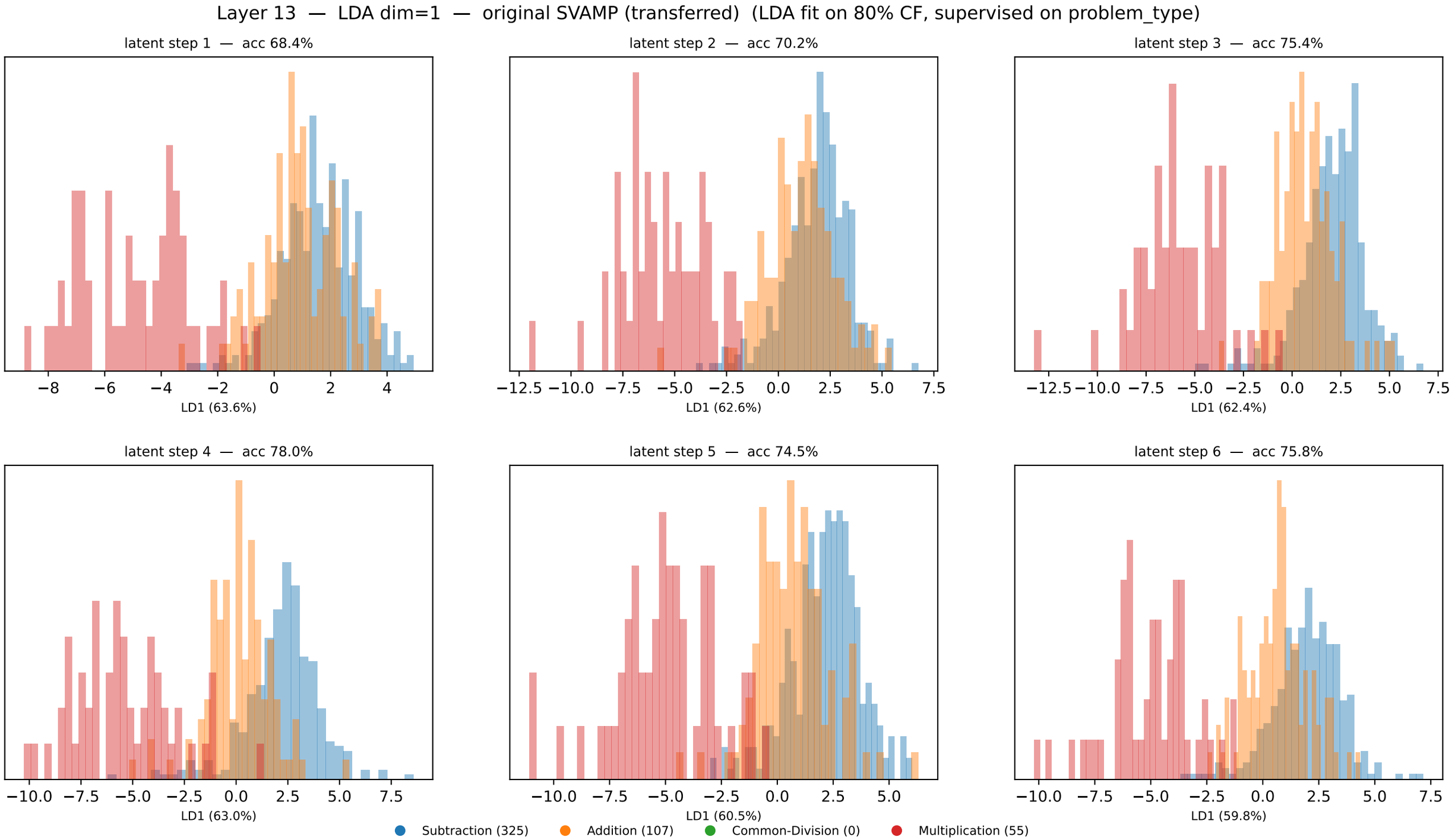

Layer 8 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 70.4%
LD1 (69.7%)
latent step 2 — acc 74.3%
LD1 (71.7%)
latent step 3 — acc 74.7%
LD1 (73.1%)
latent step 4 — acc 79.9%
LD1 (71.2%)
latent step 5 — acc 74.1%
LD1 (69.6%)
latent step 6 — acc 73.9%
LD1 (70.4%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

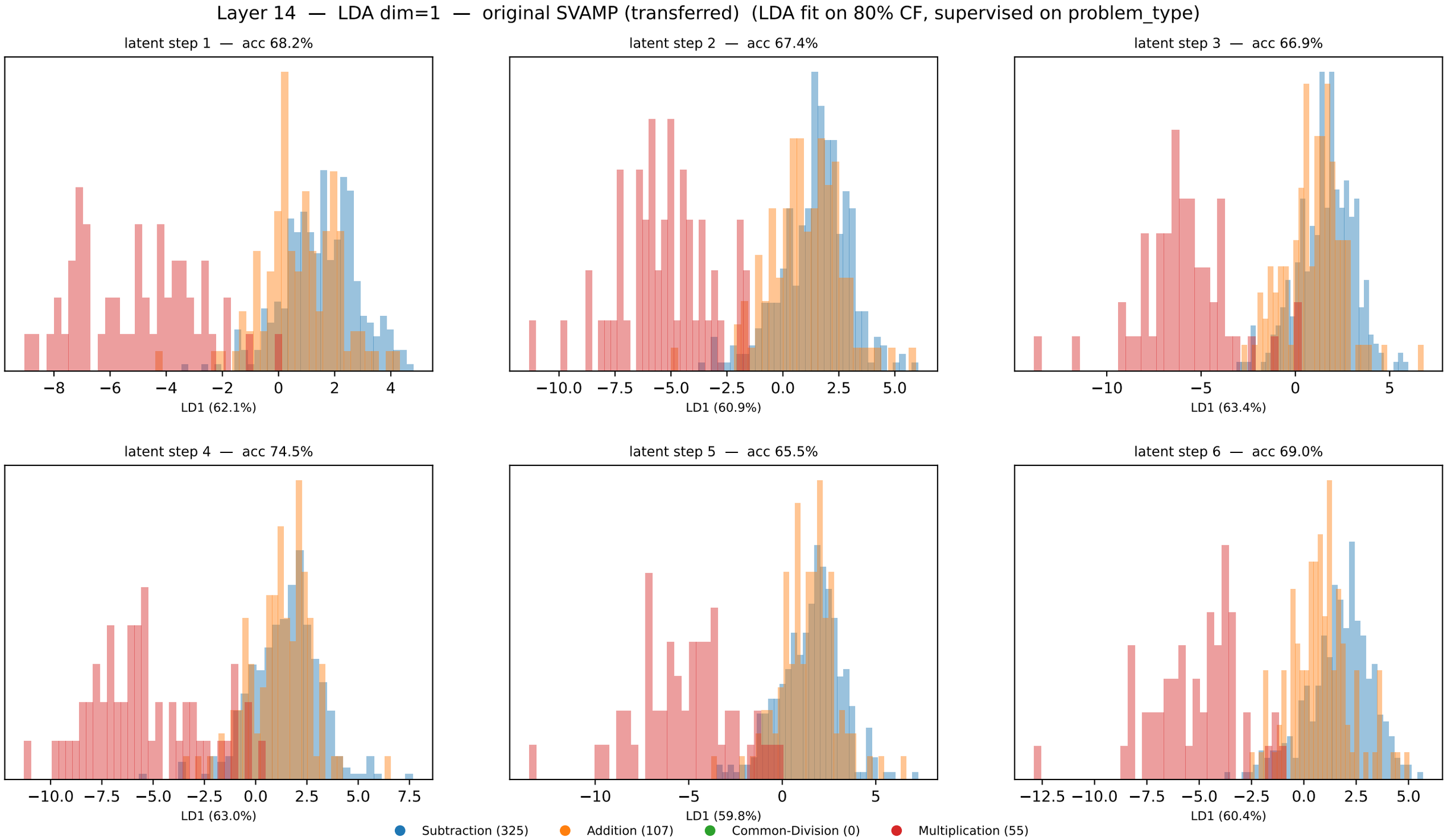

Layer 9 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 76.4%
LD1 (68.3%)
latent step 2 — acc 77.2%
LD1 (71.9%)
latent step 3 — acc 79.9%
LD1 (71.8%)
latent step 4 — acc 80.3%
LD1 (74.5%)
latent step 5 — acc 78.4%
LD1 (68.2%)
latent step 6 — acc 77.6%
LD1 (71.6%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

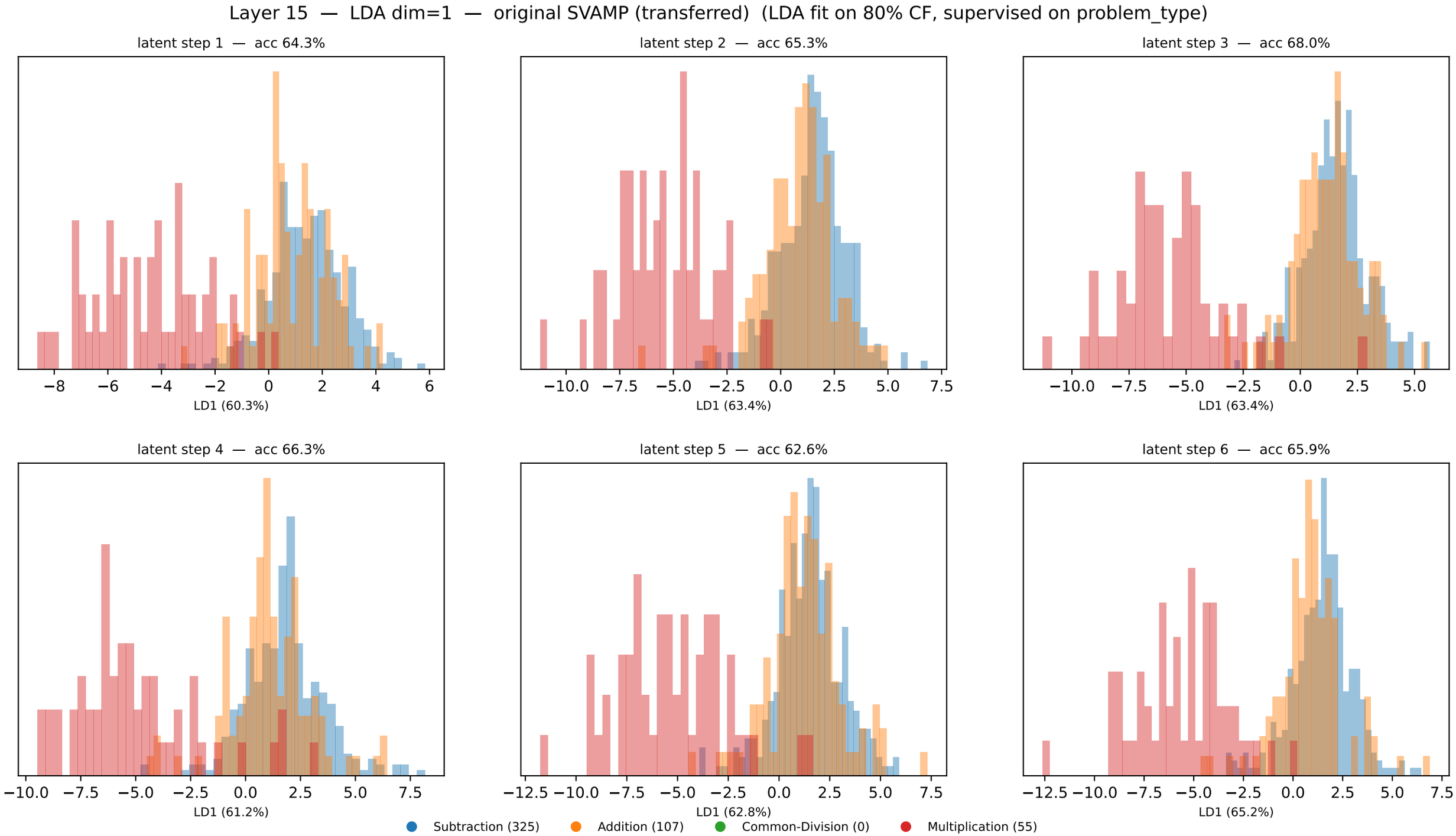

Layer 10 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 75.6%
LD1 (69.6%)
latent step 2 — acc 75.4%
LD1 (71.7%)
latent step 3 — acc 76.4%
LD1 (67.6%)
latent step 4 — acc 75.4%
LD1 (69.5%)
latent step 5 — acc 80.1%
LD1 (68.3%)
latent step 6 — acc 75.4%
LD1 (71.7%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

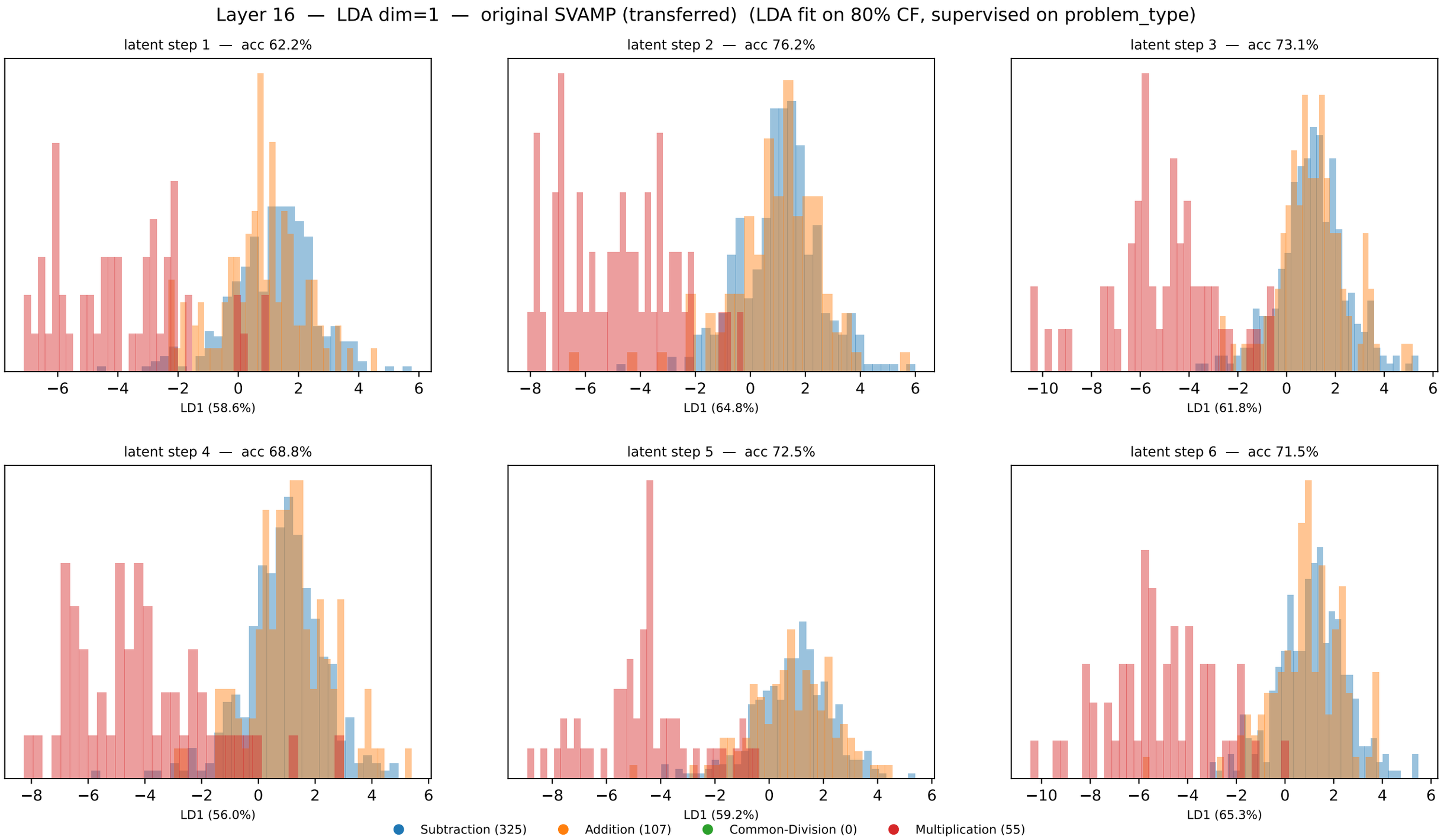

Layer 11 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 73.9%
LD1 (67.0%)
latent step 2 — acc 78.2%
LD1 (67.5%)
latent step 3 — acc 79.3%
LD1 (67.1%)
latent step 4 — acc 78.0%
LD1 (70.9%)
latent step 5 — acc 76.6%
LD1 (67.8%)
latent step 6 — acc 74.9%
LD1 (70.1%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

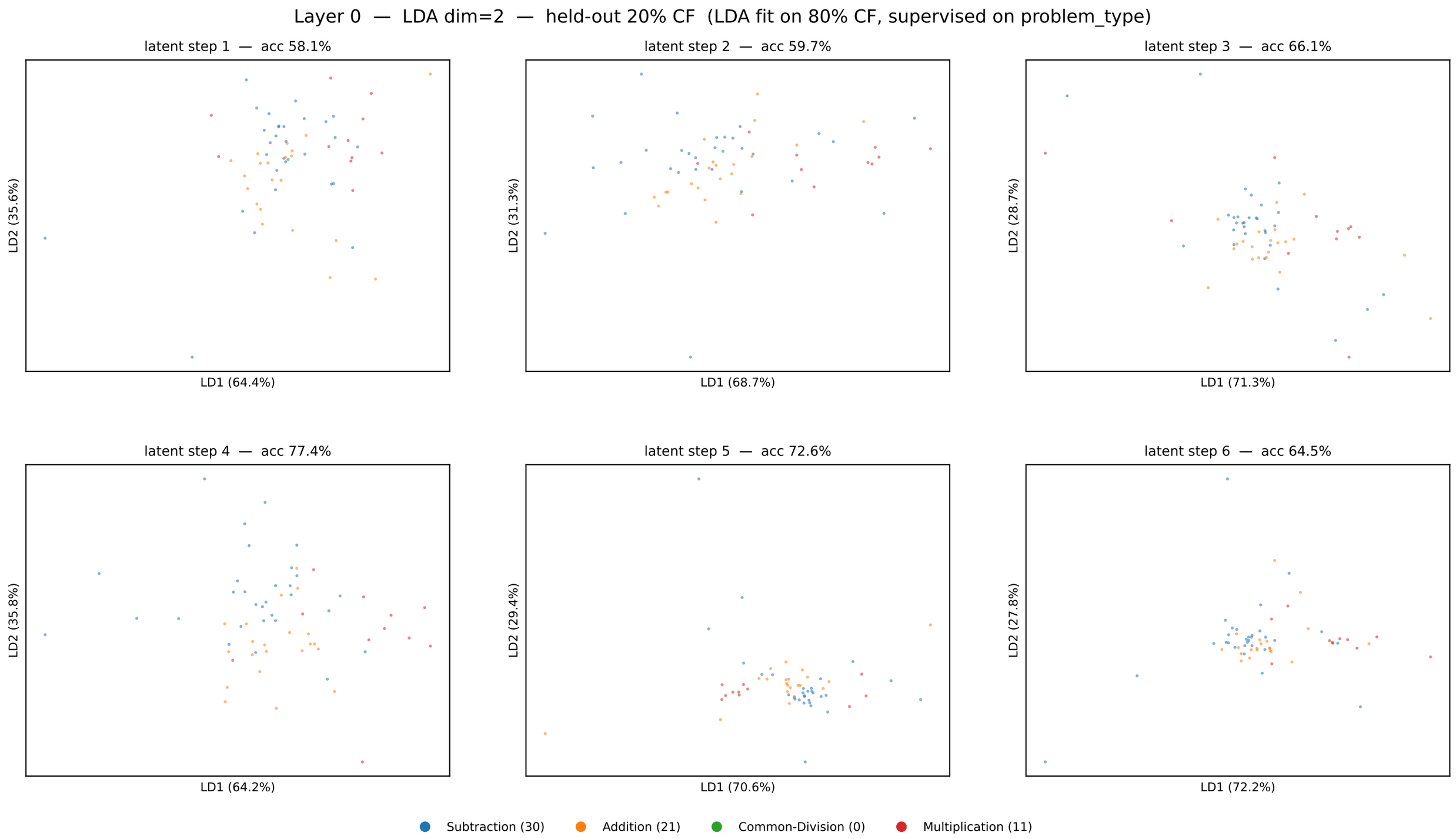

Layer 12 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 71.5%
LD1 (65.2%)
latent step 2 — acc 70.2%
LD1 (69.0%)
latent step 3 — acc 76.6%
LD1 (68.7%)
latent step 4 — acc 72.3%
LD1 (68.4%)
latent step 5 — acc 74.1%
LD1 (67.6%)
latent step 6 — acc 69.6%
LD1 (68.7%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

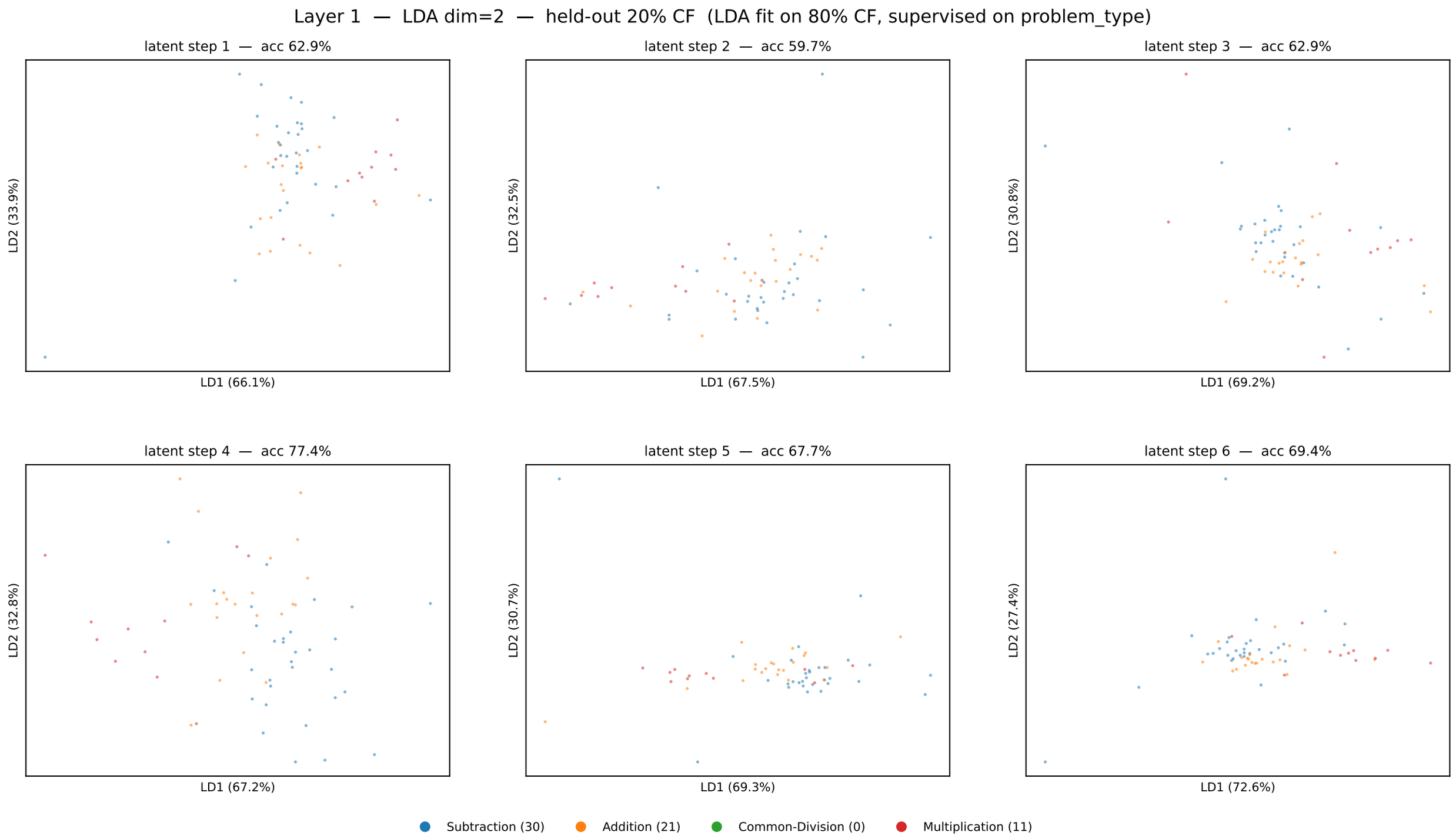

Layer 13 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 68.4%
LD1 (63.6%)
latent step 2 — acc 70.2%
LD1 (62.6%)
latent step 3 — acc 75.4%
LD1 (62.4%)
latent step 4 — acc 78.0%
LD1 (63.0%)
latent step 5 — acc 74.5%
LD1 (60.5%)
latent step 6 — acc 75.8%
LD1 (59.8%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

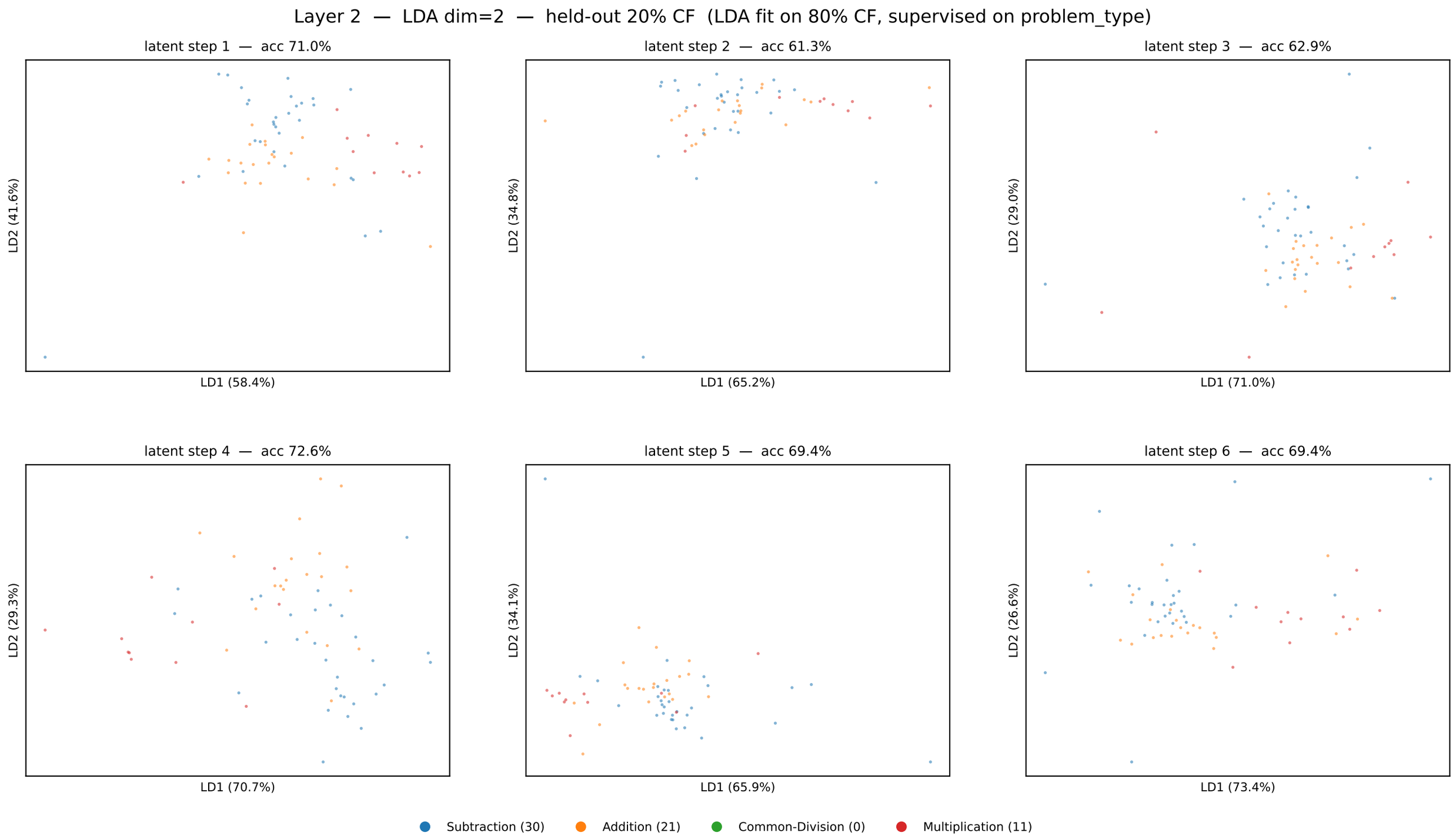

Layer 14 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 68.2%
LD1 (62.1%)
latent step 2 — acc 67.4%
LD1 (60.9%)
latent step 3 — acc 66.9%
LD1 (63.4%)
latent step 4 — acc 74.5%
LD1 (63.0%)
latent step 5 — acc 65.5%
LD1 (59.8%)
latent step 6 — acc 69.0%
LD1 (60.4%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

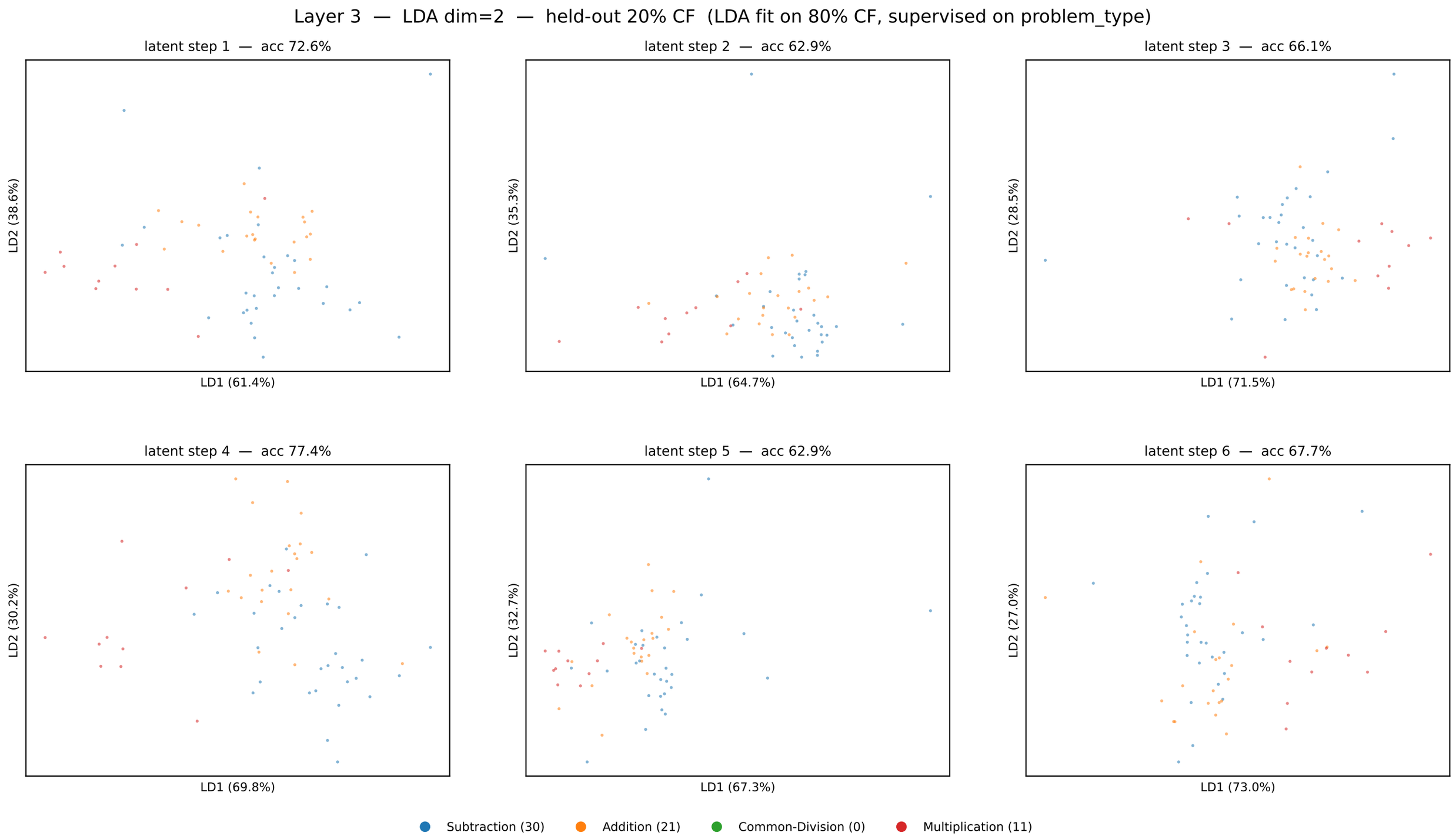

Layer 15 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 64.3%
LD1 (60.3%)
latent step 2 — acc 65.3%
LD1 (63.4%)
latent step 3 — acc 68.0%
LD1 (63.4%)
latent step 4 — acc 66.3%
LD1 (61.2%)
latent step 5 — acc 62.6%
LD1 (62.8%)
latent step 6 — acc 65.9%
LD1 (65.2%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

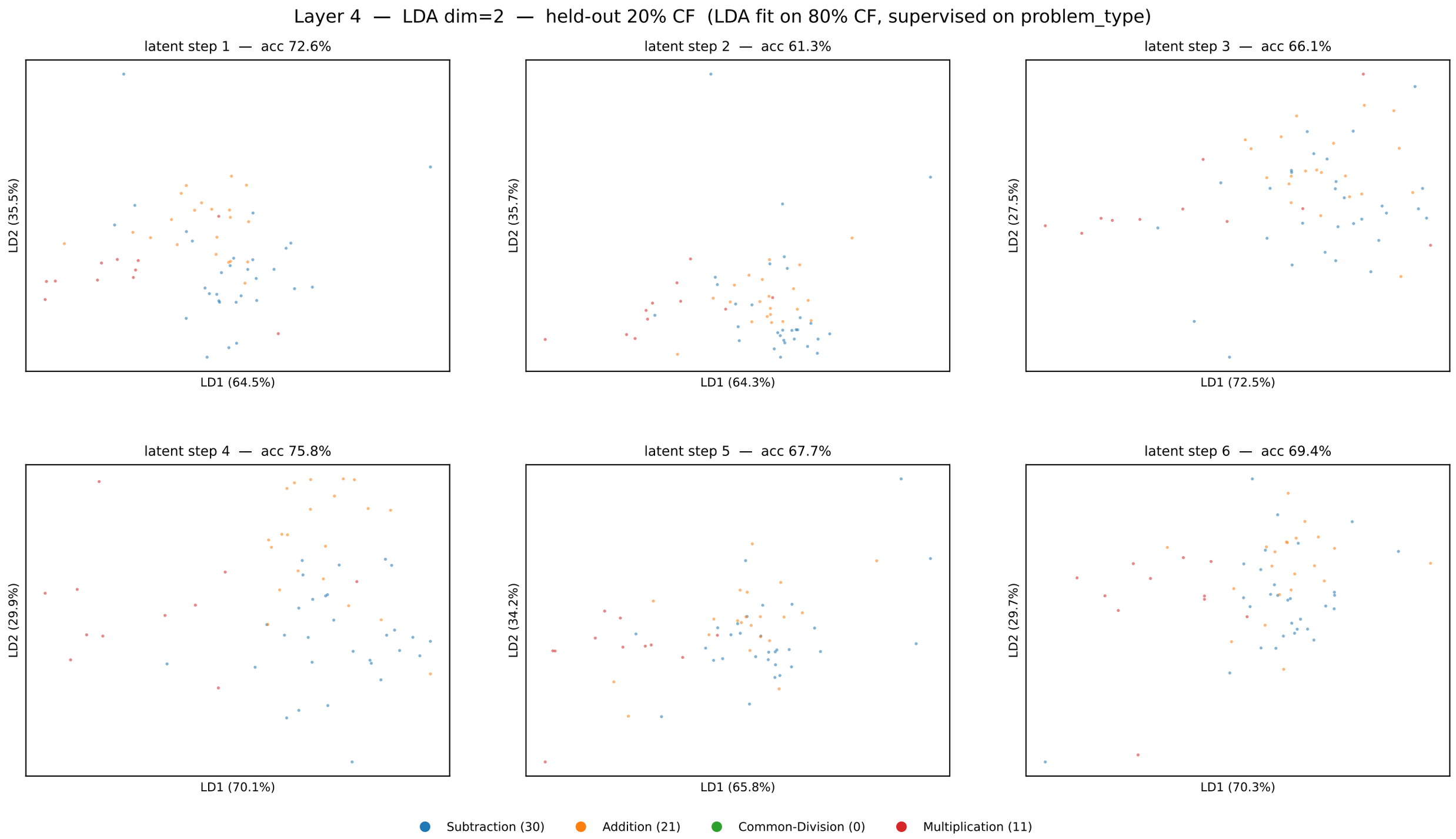

Layer 16 — LDA dim=1 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 62.2%
LD1 (58.6%)
latent step 2 — acc 76.2%
LD1 (64.8%)
latent step 3 — acc 73.1%
LD1 (61.8%)
latent step 4 — acc 68.8%
LD1 (56.0%)
latent step 5 — acc 72.5%
LD1 (59.2%)
latent step 6 — acc 71.5%
LD1 (65.3%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

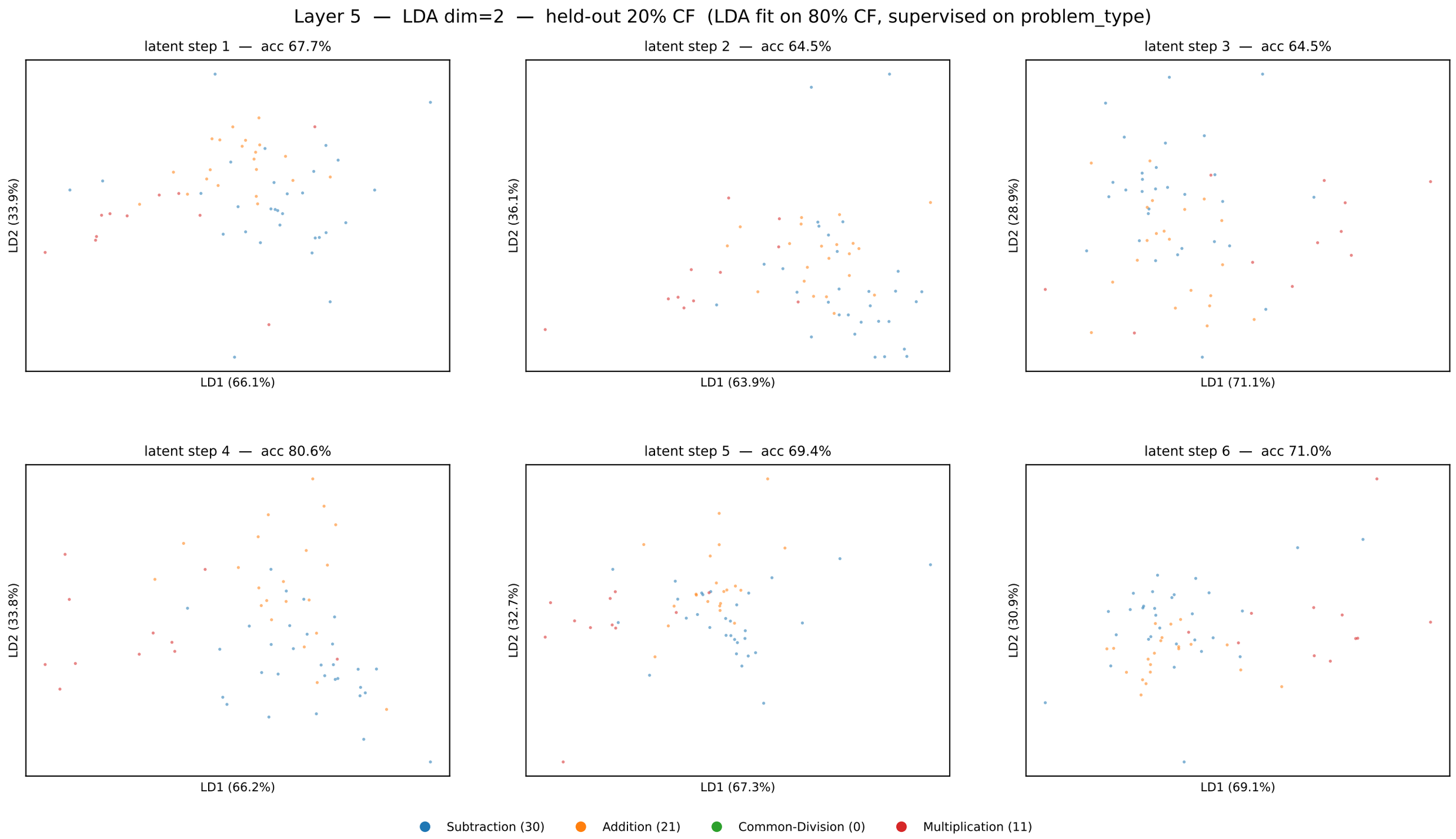

Layer 0 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 58.1%
LD2 (35.6%)
LD1 (64.4%)
latent step 2 — acc 59.7%
LD2 (31.3%)
LD1 (68.7%)
latent step 3 — acc 66.1%
LD2 (28.7%)
LD1 (71.3%)
latent step 4 — acc 77.4%
LD2 (35.8%)
LD1 (64.2%)
latent step 5 — acc 72.6%
LD2 (29.4%)
LD1 (70.6%)
latent step 6 — acc 64.5%
LD2 (27.8%)
LD1 (72.2%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

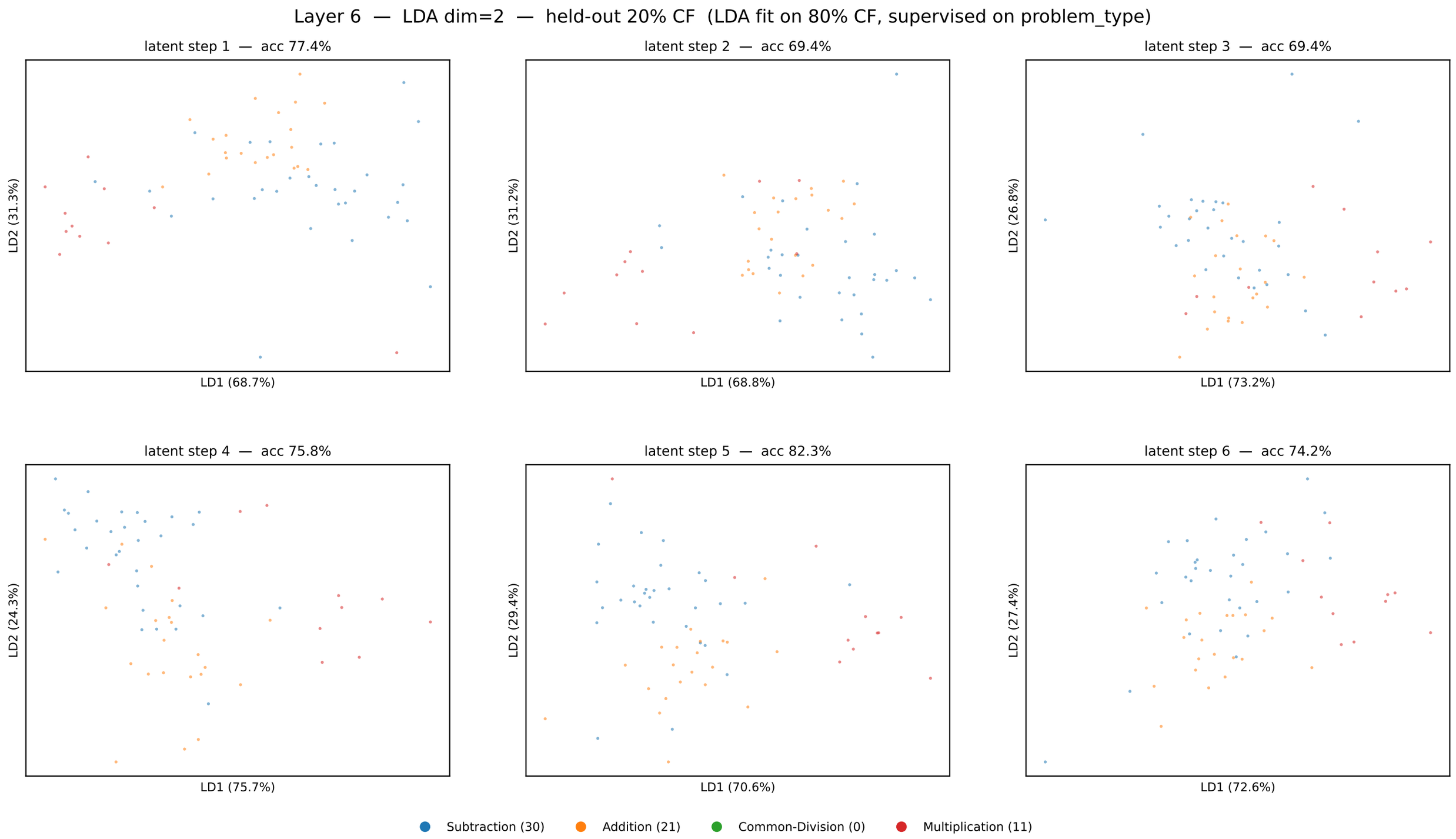

Layer 1 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 62.9%
LD2 (33.9%)
LD1 (66.1%)
latent step 2 — acc 59.7%
LD2 (32.5%)
LD1 (67.5%)
latent step 3 — acc 62.9%
LD2 (30.8%)
LD1 (69.2%)
latent step 4 — acc 77.4%
LD2 (32.8%)
LD1 (67.2%)
latent step 5 — acc 67.7%
LD2 (30.7%)
LD1 (69.3%)
latent step 6 — acc 69.4%
LD2 (27.4%)
LD1 (72.6%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

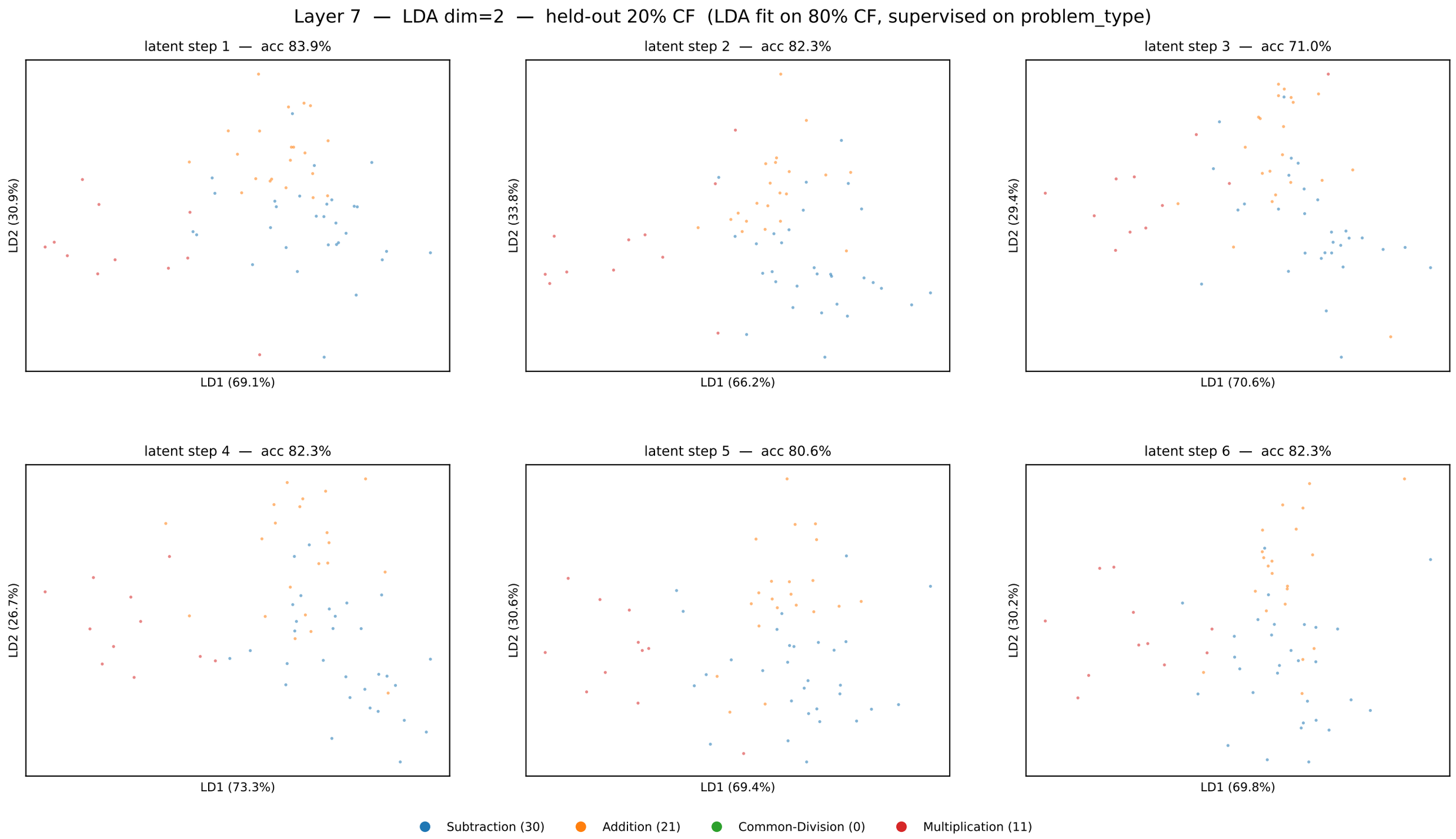

Layer 2 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 71.0%
LD2 (41.6%)
LD1 (58.4%)
latent step 2 — acc 61.3%
LD2 (34.8%)
LD1 (65.2%)
latent step 3 — acc 62.9%
LD2 (29.0%)
LD1 (71.0%)
latent step 4 — acc 72.6%
LD2 (29.3%)
LD1 (70.7%)
latent step 5 — acc 69.4%
LD2 (34.1%)
LD1 (65.9%)
latent step 6 — acc 69.4%
LD2 (26.6%)
LD1 (73.4%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

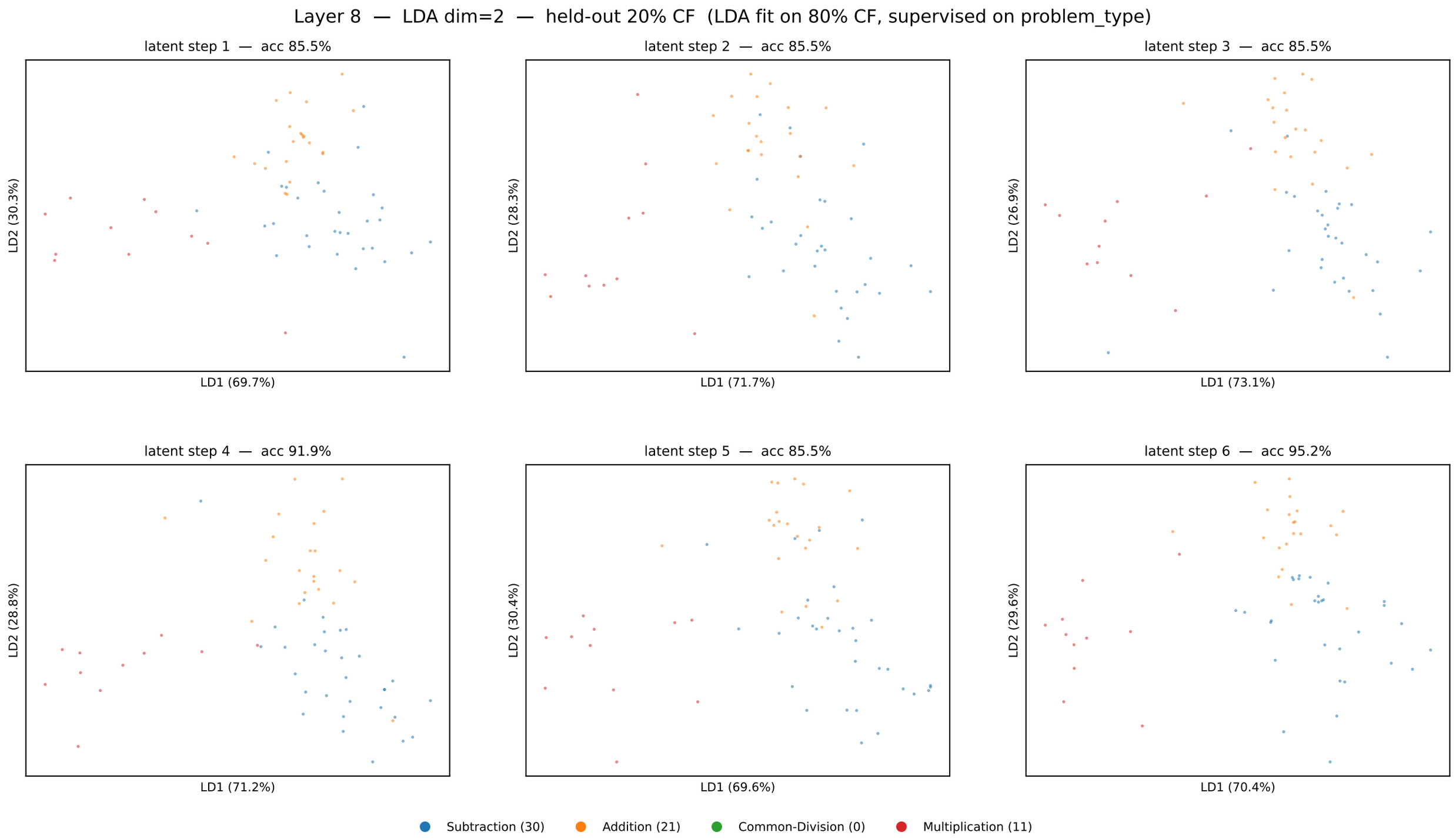

Layer 3 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 72.6%
LD2 (38.6%)
LD1 (61.4%)
latent step 2 — acc 62.9%
LD2 (35.3%)
LD1 (64.7%)
latent step 3 — acc 66.1%
LD2 (28.5%)
LD1 (71.5%)
latent step 4 — acc 77.4%
LD2 (30.2%)
LD1 (69.8%)
latent step 5 — acc 62.9%
LD2 (32.7%)
LD1 (67.3%)
latent step 6 — acc 67.7%
LD2 (27.0%)
LD1 (73.0%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

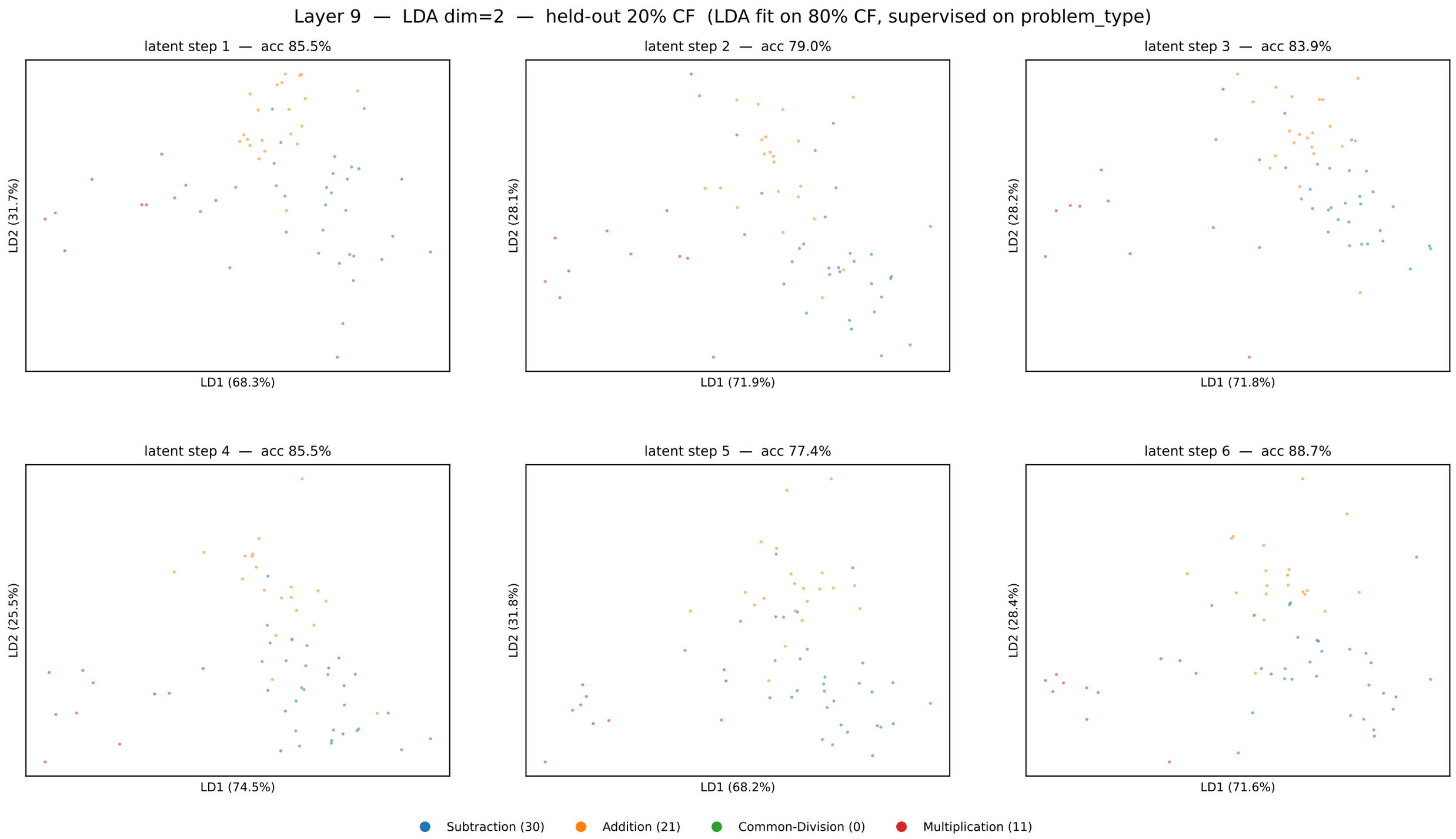

Layer 4 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 72.6%
LD2 (35.5%)
LD1 (64.5%)
latent step 2 — acc 61.3%
LD2 (35.7%)
LD1 (64.3%)
latent step 3 — acc 66.1%
LD2 (27.5%)
LD1 (72.5%)
latent step 4 — acc 75.8%
LD2 (29.9%)
LD1 (70.1%)
latent step 5 — acc 67.7%
LD2 (34.2%)
LD1 (65.8%)
latent step 6 — acc 69.4%
LD2 (29.7%)
LD1 (70.3%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

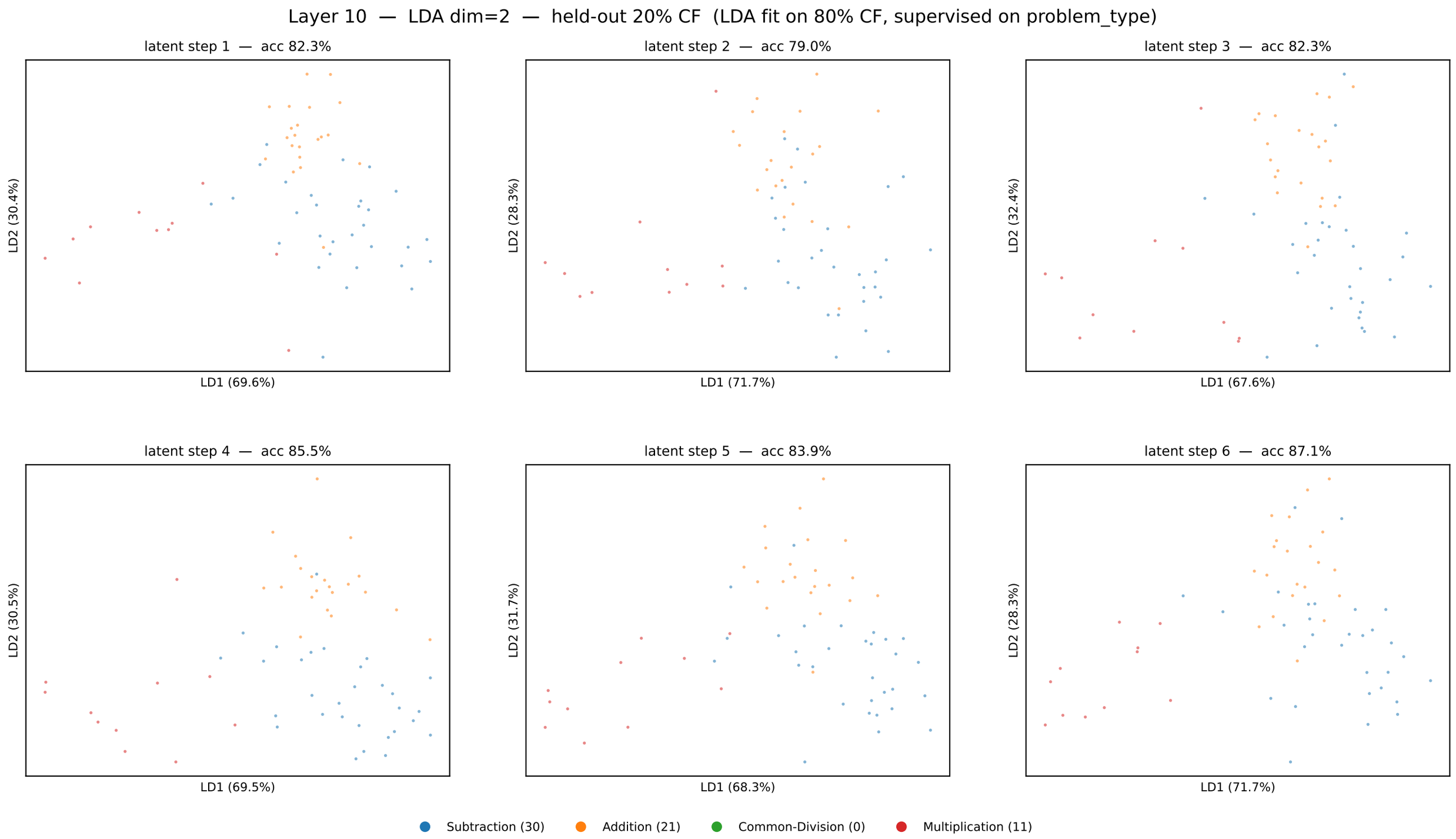

Layer 5 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 67.7%
LD2 (33.9%)
LD1 (66.1%)
latent step 2 — acc 64.5%
LD2 (36.1%)
LD1 (63.9%)
latent step 3 — acc 64.5%
LD2 (28.9%)
LD1 (71.1%)
latent step 4 — acc 80.6%
LD2 (33.8%)
LD1 (66.2%)
latent step 5 — acc 69.4%
LD2 (32.7%)
LD1 (67.3%)
latent step 6 — acc 71.0%
LD2 (30.9%)
LD1 (69.1%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

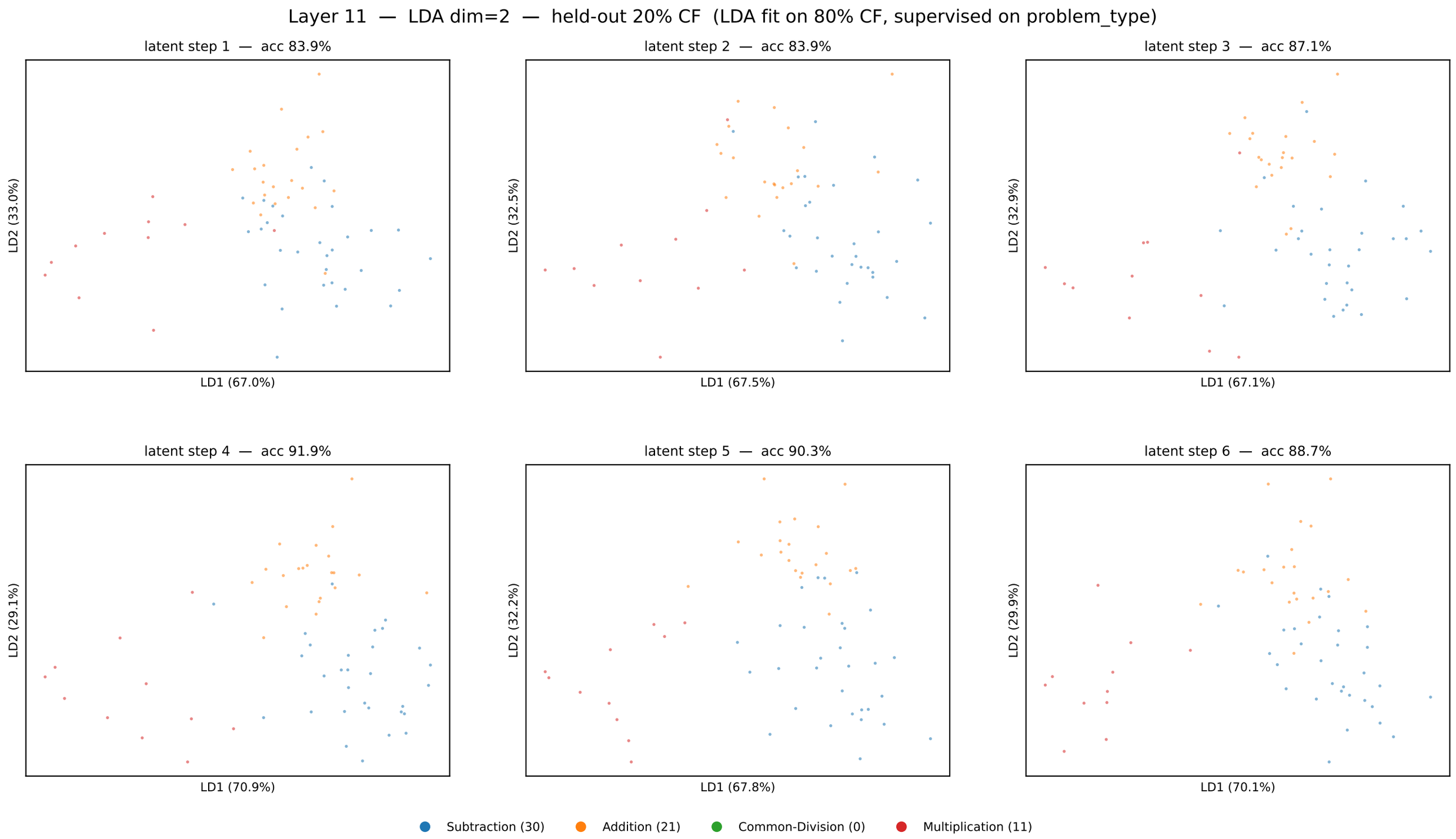

Layer 6 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 77.4%
LD2 (31.3%)
LD1 (68.7%)
latent step 2 — acc 69.4%
LD2 (31.2%)
LD1 (68.8%)
latent step 3 — acc 69.4%
LD2 (26.8%)
LD1 (73.2%)
latent step 4 — acc 75.8%
LD2 (24.3%)
LD1 (75.7%)
latent step 5 — acc 82.3%
LD2 (29.4%)
LD1 (70.6%)
latent step 6 — acc 74.2%
LD2 (27.4%)
LD1 (72.6%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

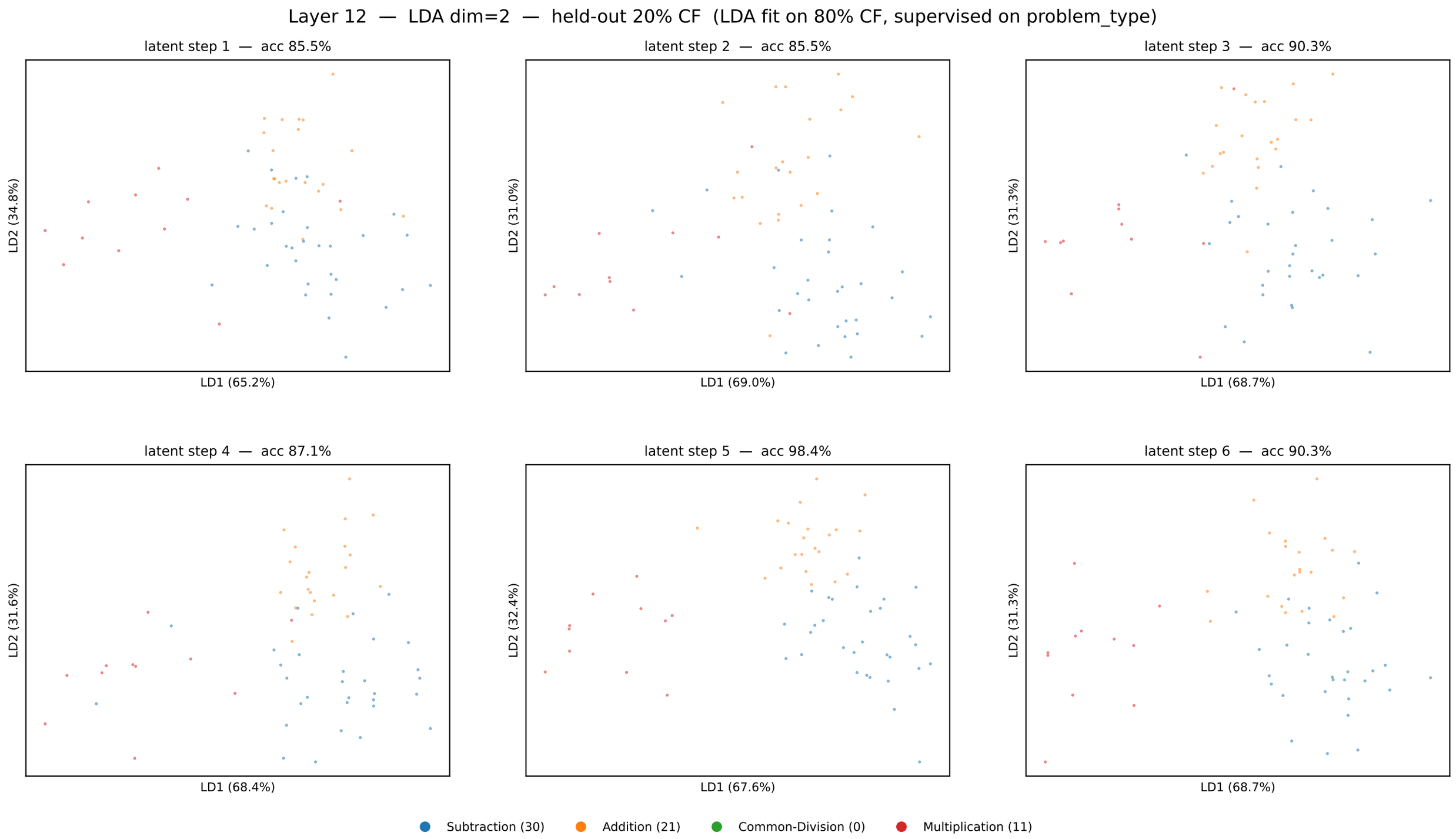

Layer 7 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 83.9%
LD2 (30.9%)
LD1 (69.1%)
latent step 2 — acc 82.3%
LD2 (33.8%)
LD1 (66.2%)
latent step 3 — acc 71.0%
LD2 (29.4%)
LD1 (70.6%)
latent step 4 — acc 82.3%
LD2 (26.7%)
LD1 (73.3%)
latent step 5 — acc 80.6%
LD2 (30.6%)
LD1 (69.4%)
latent step 6 — acc 82.3%
LD2 (30.2%)
LD1 (69.8%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

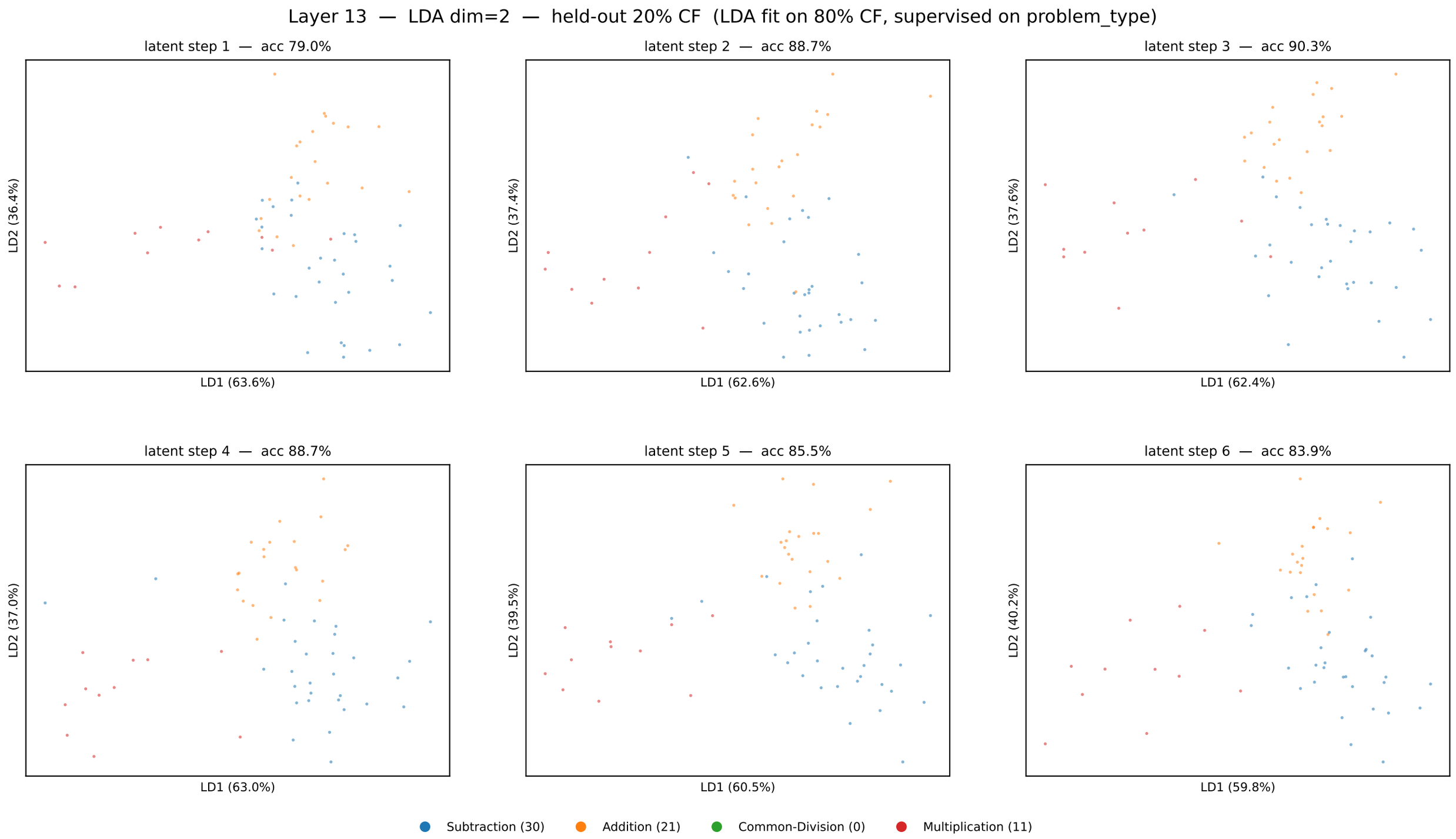

Layer 8 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 85.5%
LD2 (30.3%)
LD1 (69.7%)
latent step 2 — acc 85.5%
LD2 (28.3%)
LD1 (71.7%)
latent step 3 — acc 85.5%
LD2 (26.9%)
LD1 (73.1%)
latent step 4 — acc 91.9%
LD2 (28.8%)
LD1 (71.2%)
latent step 5 — acc 85.5%
LD2 (30.4%)
LD1 (69.6%)
latent step 6 — acc 95.2%
LD2 (29.6%)
LD1 (70.4%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

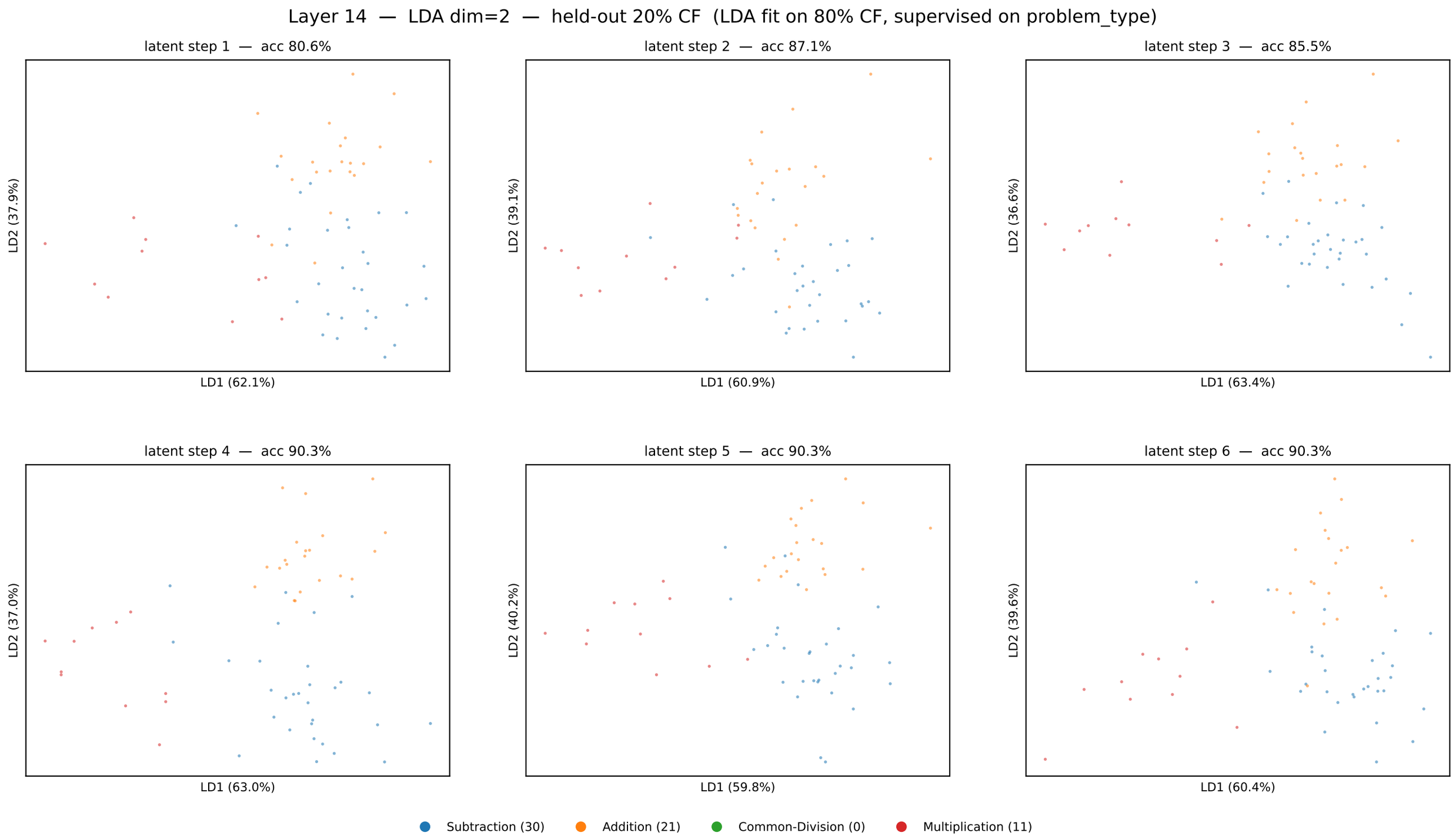

Layer 9 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 85.5%
LD2 (31.7%)
LD1 (68.3%)
latent step 2 — acc 79.0%
LD2 (28.1%)
LD1 (71.9%)
latent step 3 — acc 83.9%
LD2 (28.2%)
LD1 (71.8%)
latent step 4 — acc 85.5%
LD2 (25.5%)
LD1 (74.5%)
latent step 5 — acc 77.4%
LD2 (31.8%)
LD1 (68.2%)
latent step 6 — acc 88.7%
LD2 (28.4%)
LD1 (71.6%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

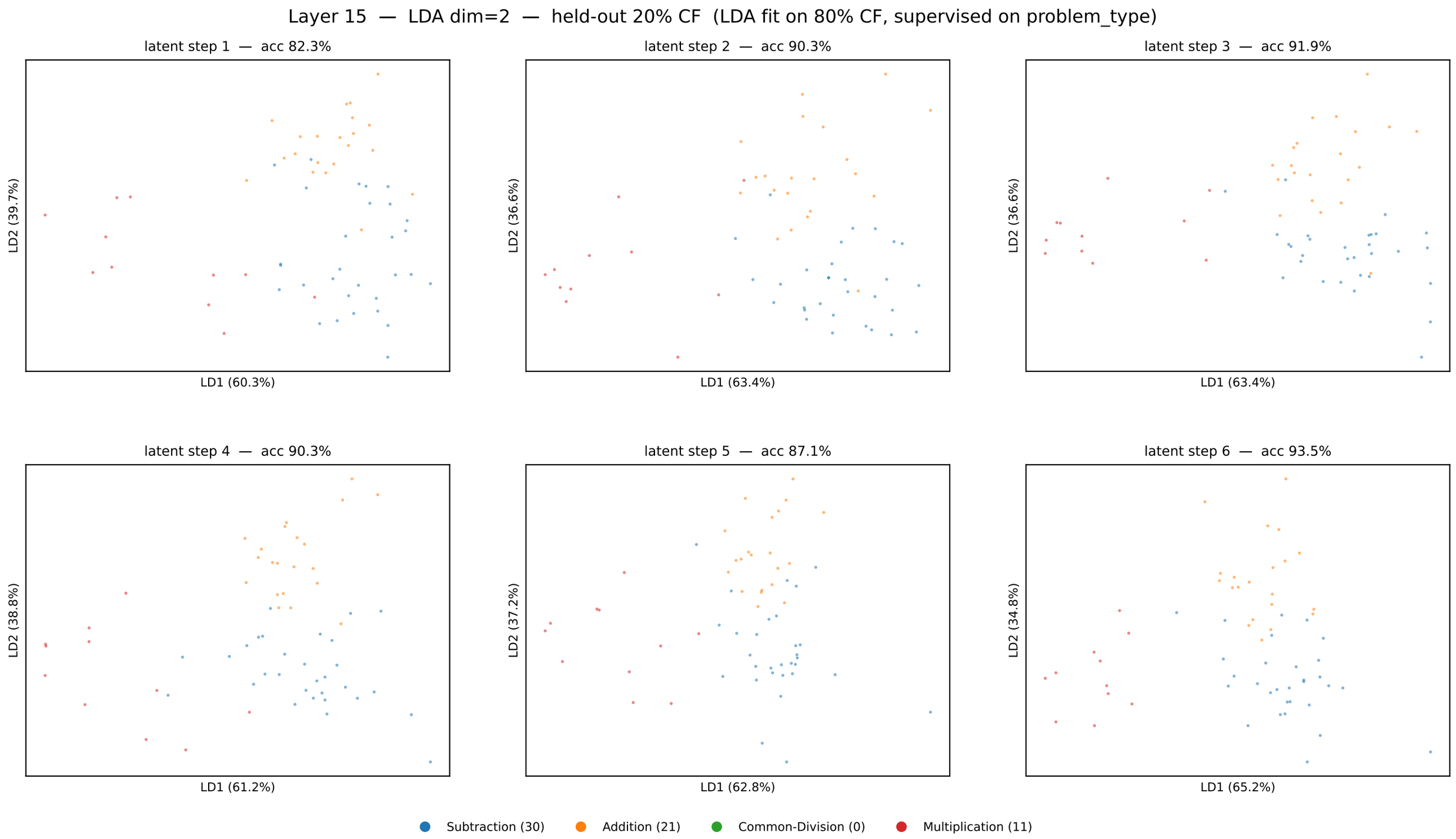

Layer 10 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 82.3%
LD2 (30.4%)
LD1 (69.6%)
latent step 2 — acc 79.0%
LD2 (28.3%)
LD1 (71.7%)
latent step 3 — acc 82.3%
LD2 (32.4%)
LD1 (67.6%)
latent step 4 — acc 85.5%
LD2 (30.5%)
LD1 (69.5%)
latent step 5 — acc 83.9%
LD2 (31.7%)
LD1 (68.3%)
latent step 6 — acc 87.1%
LD2 (28.3%)
LD1 (71.7%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

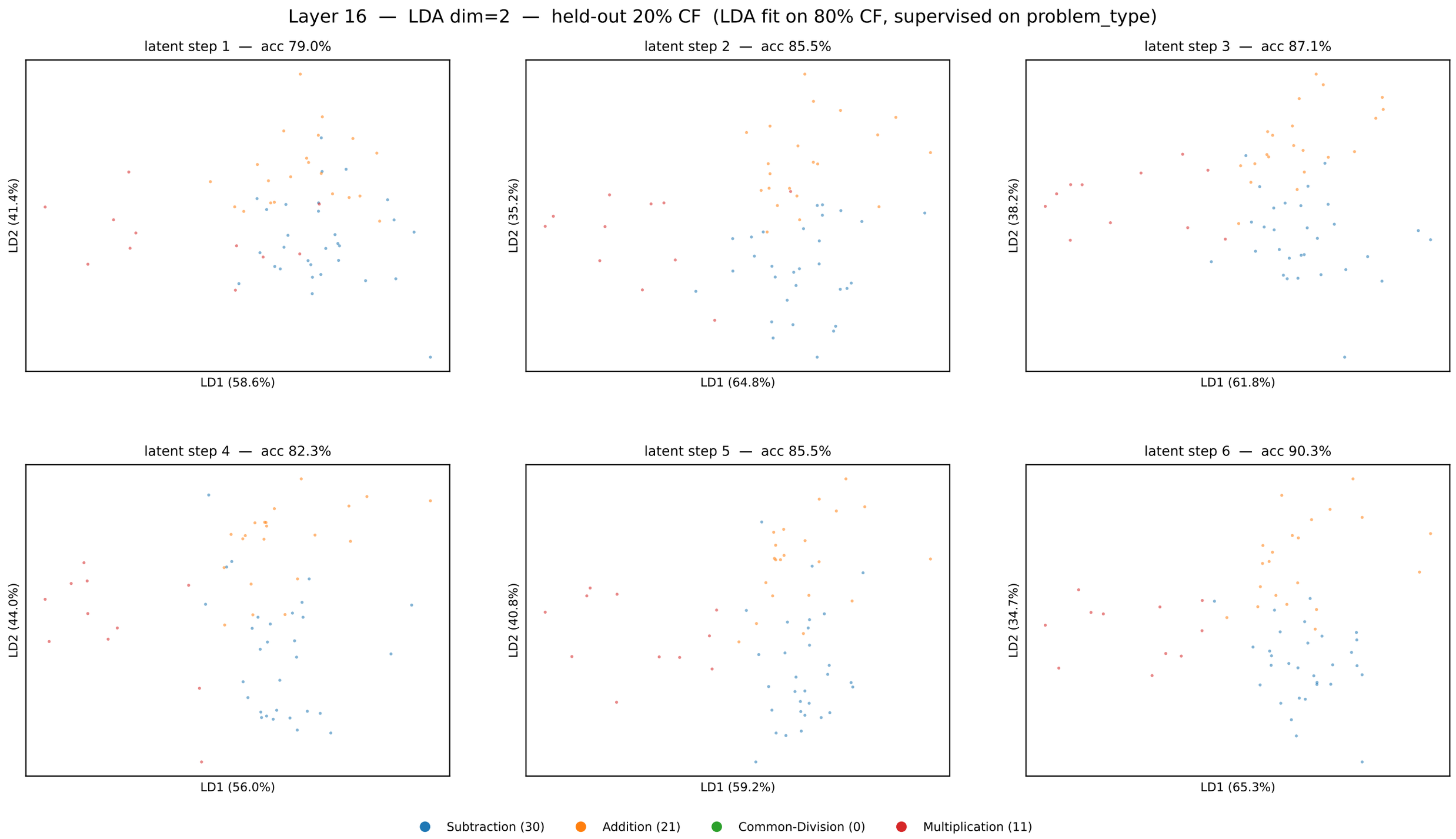

Layer 11 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 83.9%
LD2 (33.0%)
LD1 (67.0%)
latent step 2 — acc 83.9%
LD2 (32.5%)
LD1 (67.5%)
latent step 3 — acc 87.1%
LD2 (32.9%)
LD1 (67.1%)
latent step 4 — acc 91.9%
LD2 (29.1%)
LD1 (70.9%)
latent step 5 — acc 90.3%
LD2 (32.2%)
LD1 (67.8%)
latent step 6 — acc 88.7%
LD2 (29.9%)
LD1 (70.1%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

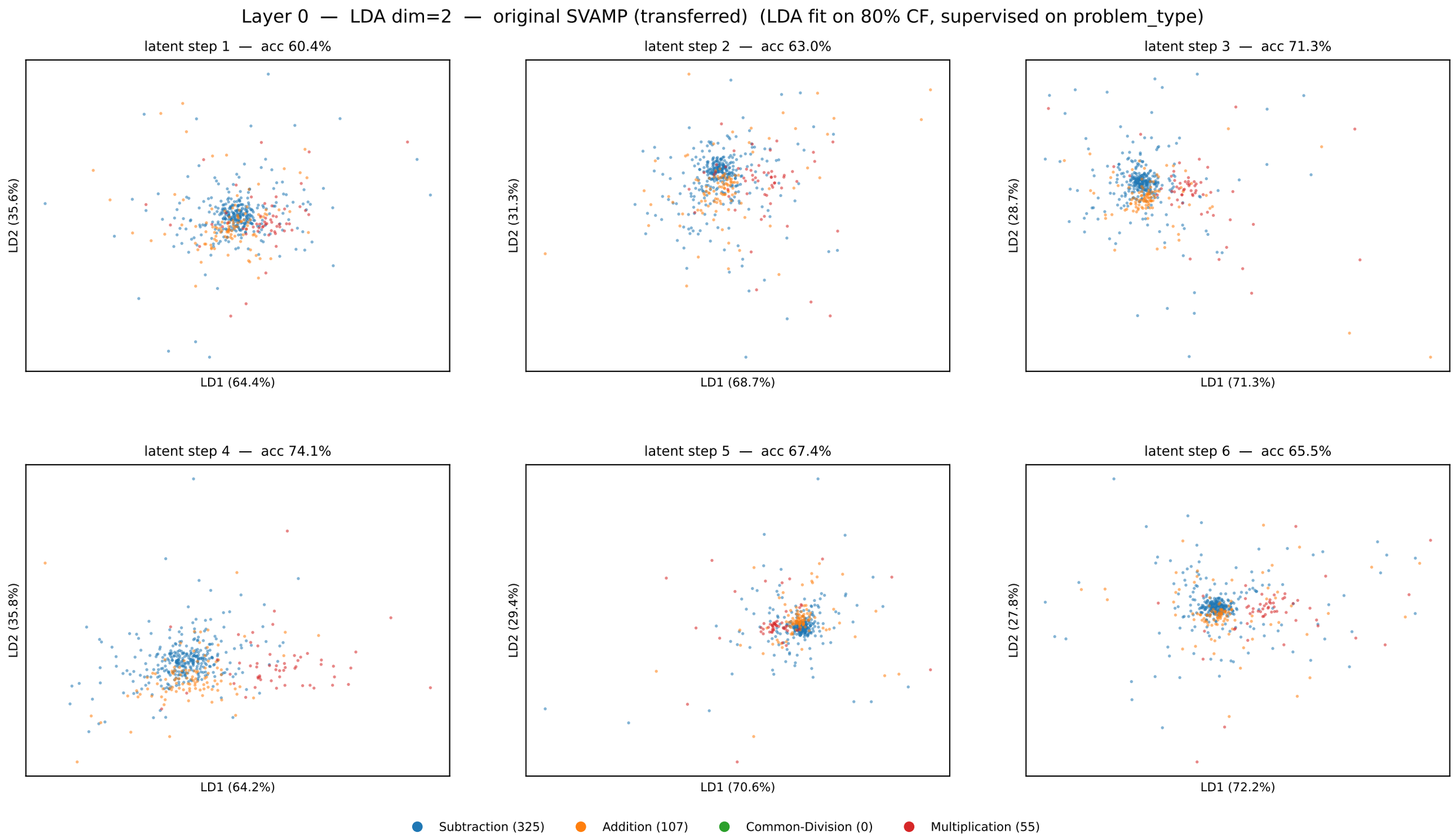

Layer 12 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 85.5%
LD2 (34.8%)
LD1 (65.2%)
latent step 2 — acc 85.5%
LD2 (31.0%)
LD1 (69.0%)
latent step 3 — acc 90.3%
LD2 (31.3%)
LD1 (68.7%)
latent step 4 — acc 87.1%
LD2 (31.6%)
LD1 (68.4%)
latent step 5 — acc 98.4%
LD2 (32.4%)
LD1 (67.6%)
latent step 6 — acc 90.3%
LD2 (31.3%)
LD1 (68.7%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

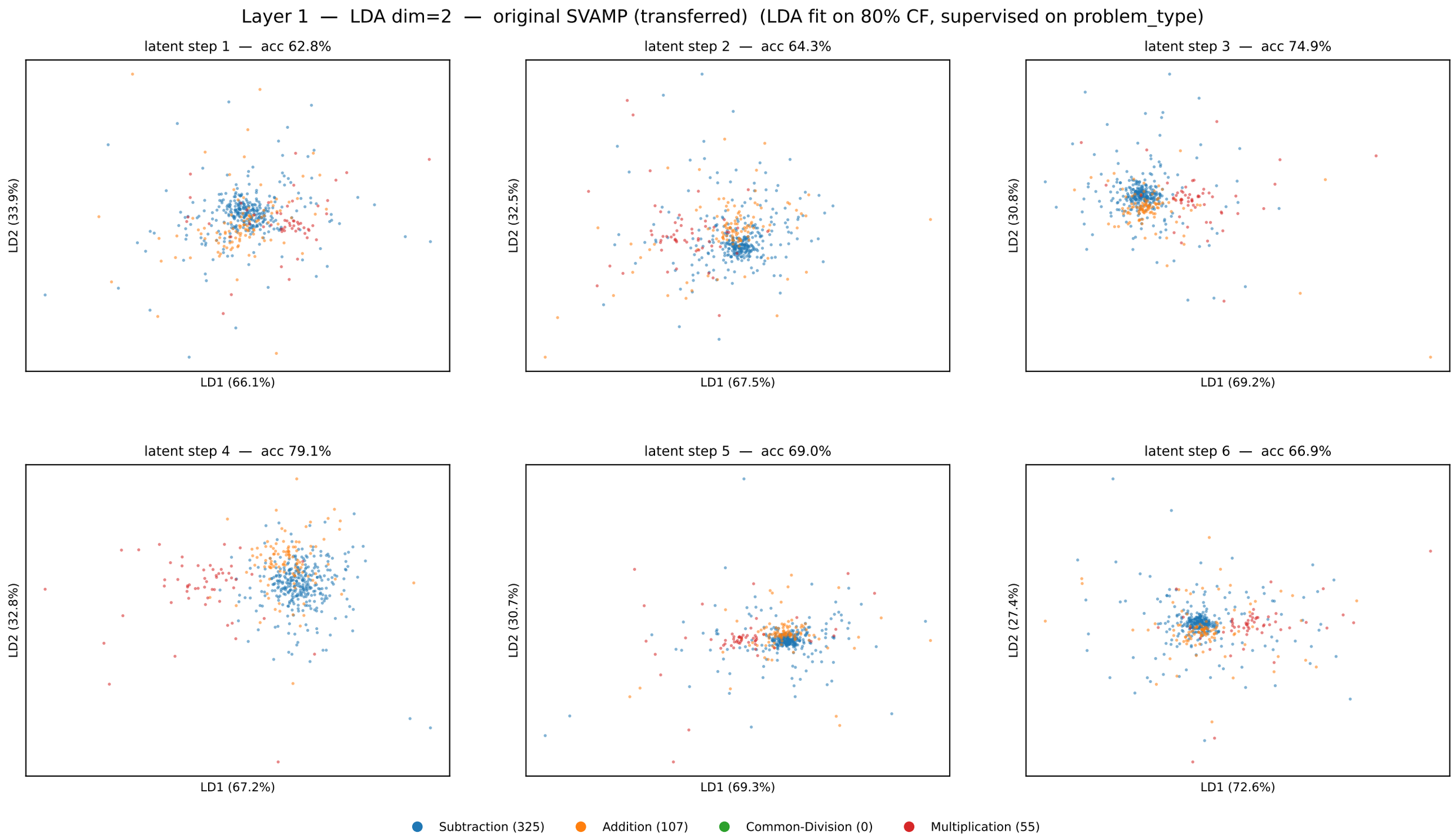

Layer 13 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 79.0%
LD2 (36.4%)
LD1 (63.6%)
latent step 2 — acc 88.7%
LD2 (37.4%)
LD1 (62.6%)
latent step 3 — acc 90.3%
LD2 (37.6%)
LD1 (62.4%)
latent step 4 — acc 88.7%
LD2 (37.0%)
LD1 (63.0%)
latent step 5 — acc 85.5%
LD2 (39.5%)
LD1 (60.5%)
latent step 6 — acc 83.9%
LD2 (40.2%)
LD1 (59.8%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

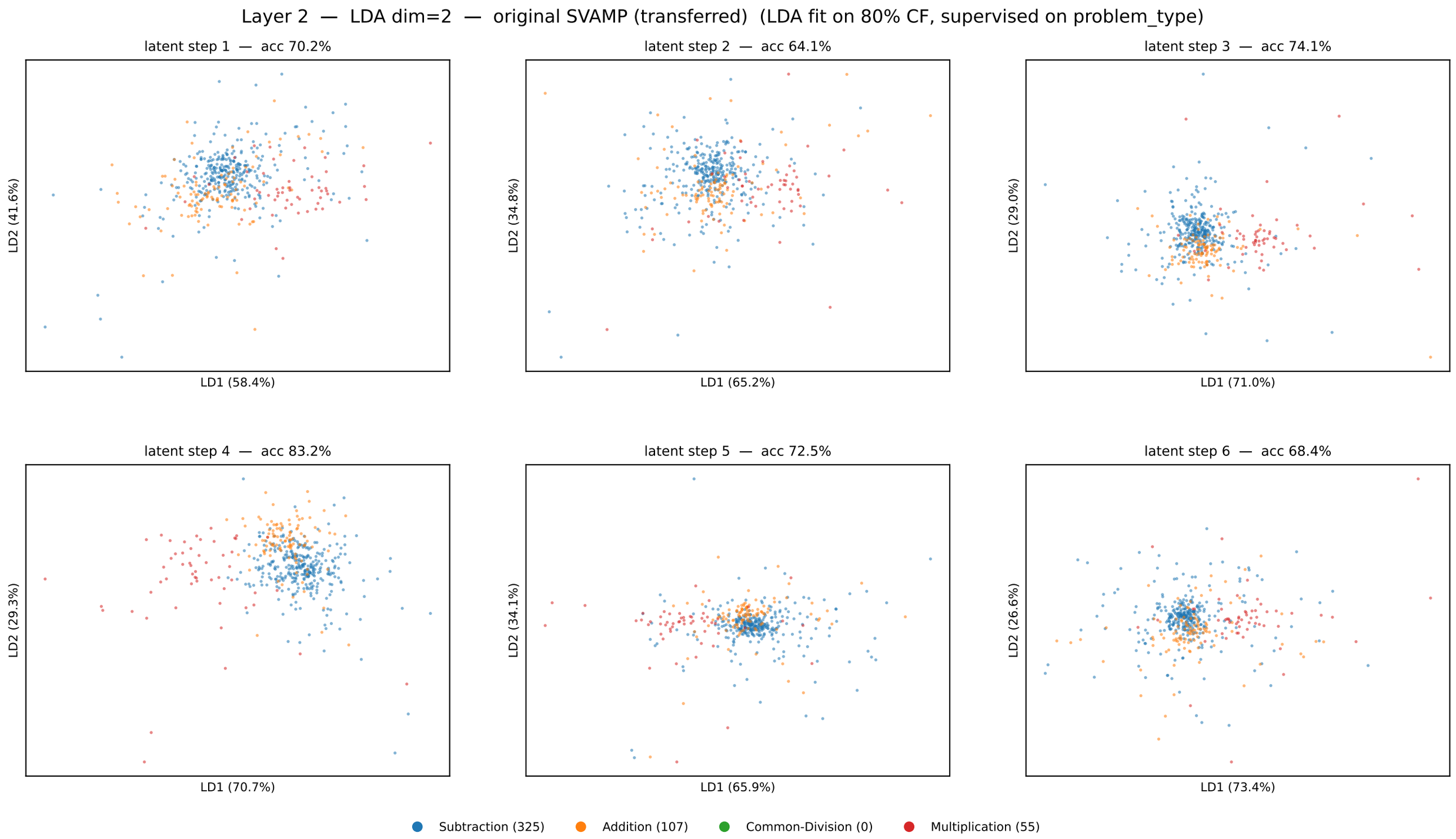

Layer 14 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 80.6%
LD2 (37.9%)
LD1 (62.1%)
latent step 2 — acc 87.1%
LD2 (39.1%)
LD1 (60.9%)
latent step 3 — acc 85.5%
LD2 (36.6%)
LD1 (63.4%)
latent step 4 — acc 90.3%
LD2 (37.0%)
LD1 (63.0%)
latent step 5 — acc 90.3%
LD2 (40.2%)
LD1 (59.8%)
latent step 6 — acc 90.3%
LD2 (39.6%)
LD1 (60.4%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

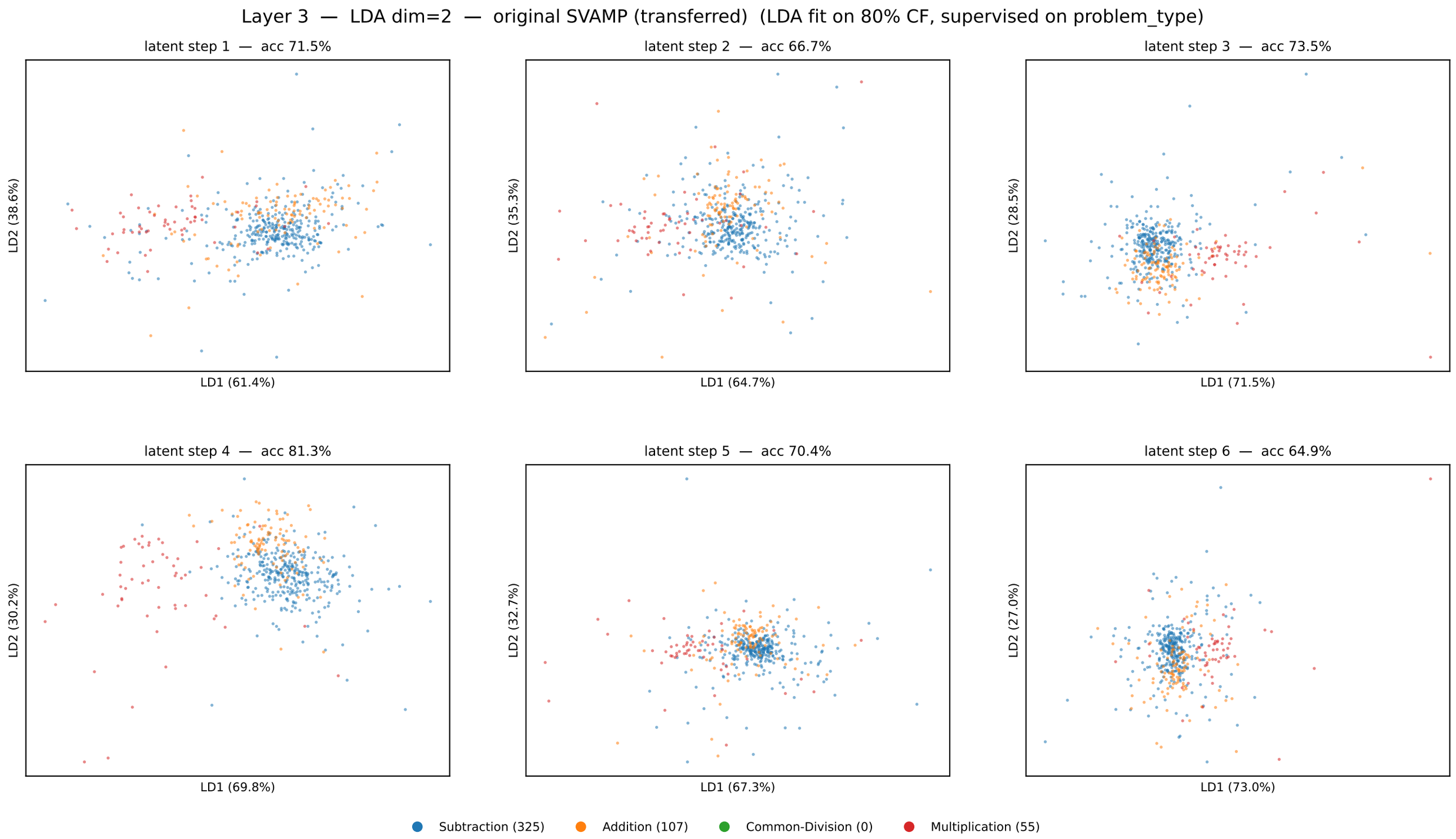

Layer 15 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 82.3%
LD2 (39.7%)
LD1 (60.3%)
latent step 2 — acc 90.3%
LD2 (36.6%)
LD1 (63.4%)
latent step 3 — acc 91.9%
LD2 (36.6%)
LD1 (63.4%)
latent step 4 — acc 90.3%
LD2 (38.8%)
LD1 (61.2%)
latent step 5 — acc 87.1%
LD2 (37.2%)
LD1 (62.8%)
latent step 6 — acc 93.5%
LD2 (34.8%)
LD1 (65.2%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

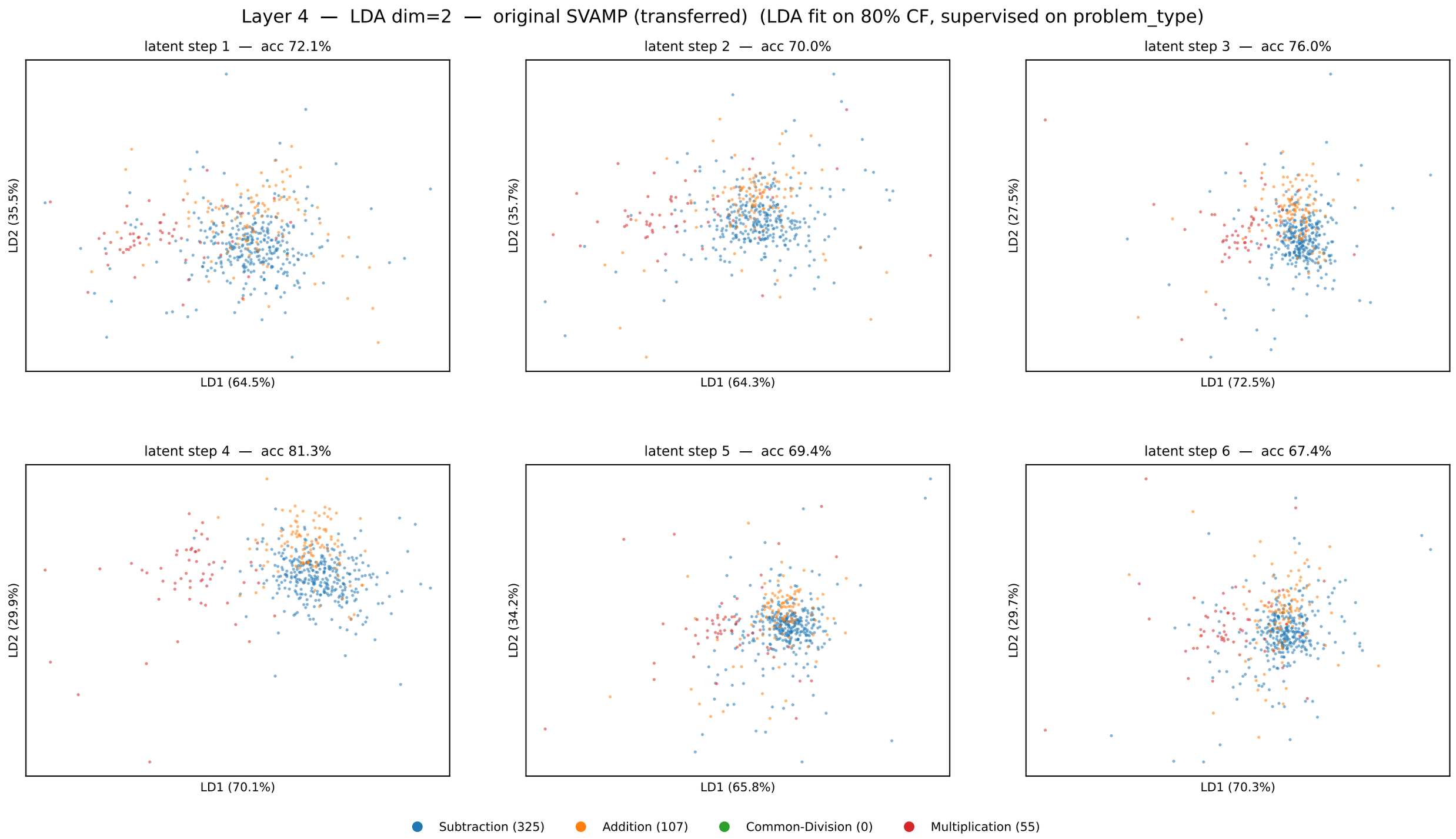

Layer 16 — LDA dim=2 — held-out 20% CF (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 79.0%
LD2 (41.4%)
LD1 (58.6%)
latent step 2 — acc 85.5%
LD2 (35.2%)
LD1 (64.8%)
latent step 3 — acc 87.1%
LD2 (38.2%)
LD1 (61.8%)
latent step 4 — acc 82.3%
LD2 (44.0%)
LD1 (56.0%)
latent step 5 — acc 85.5%
LD2 (40.8%)
LD1 (59.2%)
latent step 6 — acc 90.3%
LD2 (34.7%)
LD1 (65.3%)
Subtraction (30)
Addition (21)
Common-Division (0)
Multiplication (11)

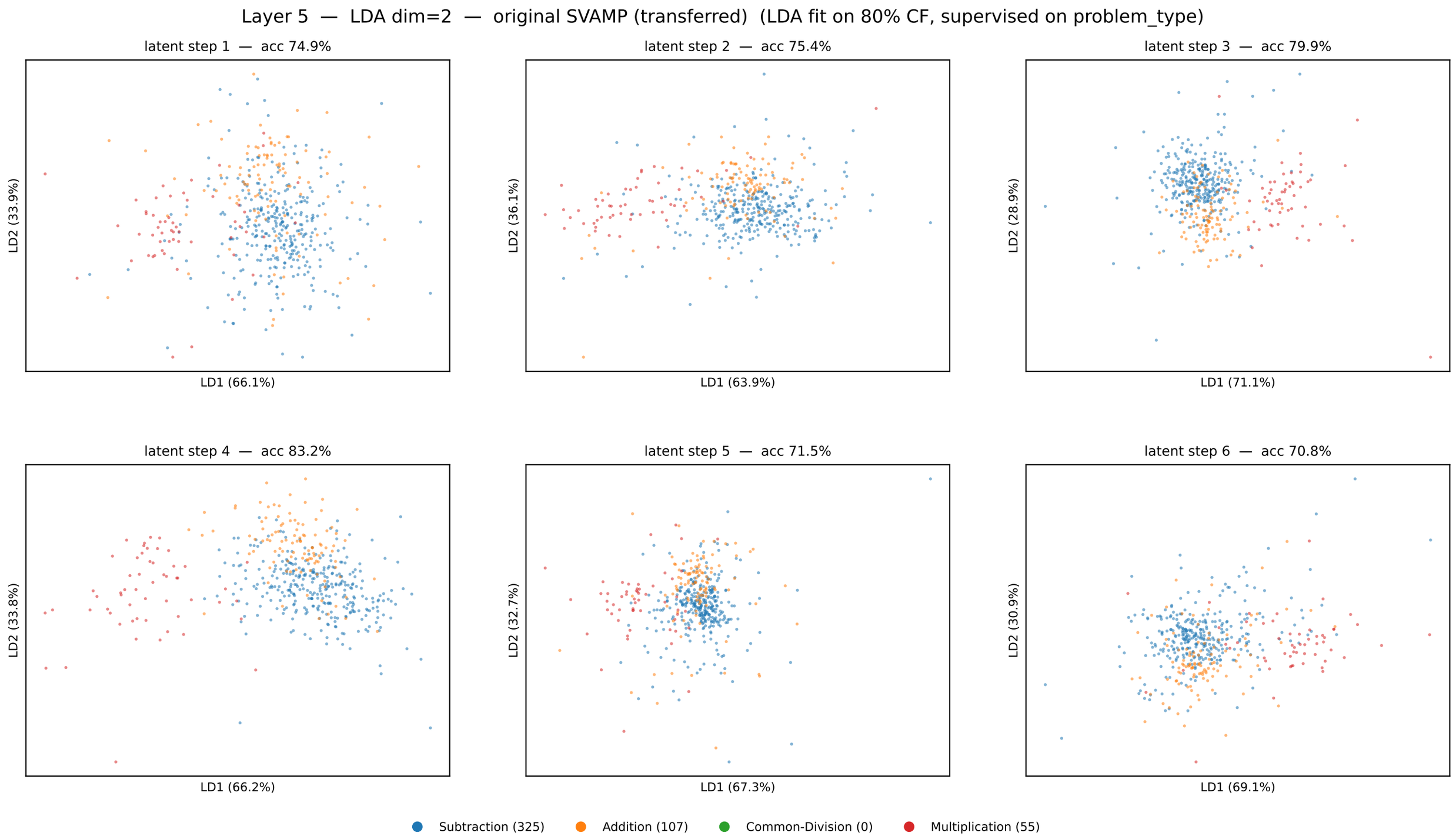

Layer 0 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 60.4%
LD2 (35.6%)
LD1 (64.4%)
latent step 2 — acc 63.0%
LD2 (31.3%)
LD1 (68.7%)
latent step 3 — acc 71.3%
LD2 (28.7%)
LD1 (71.3%)
latent step 4 — acc 74.1%
LD2 (35.8%)
LD1 (64.2%)
latent step 5 — acc 67.4%
LD2 (29.4%)
LD1 (70.6%)
latent step 6 — acc 65.5%
LD2 (27.8%)
LD1 (72.2%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

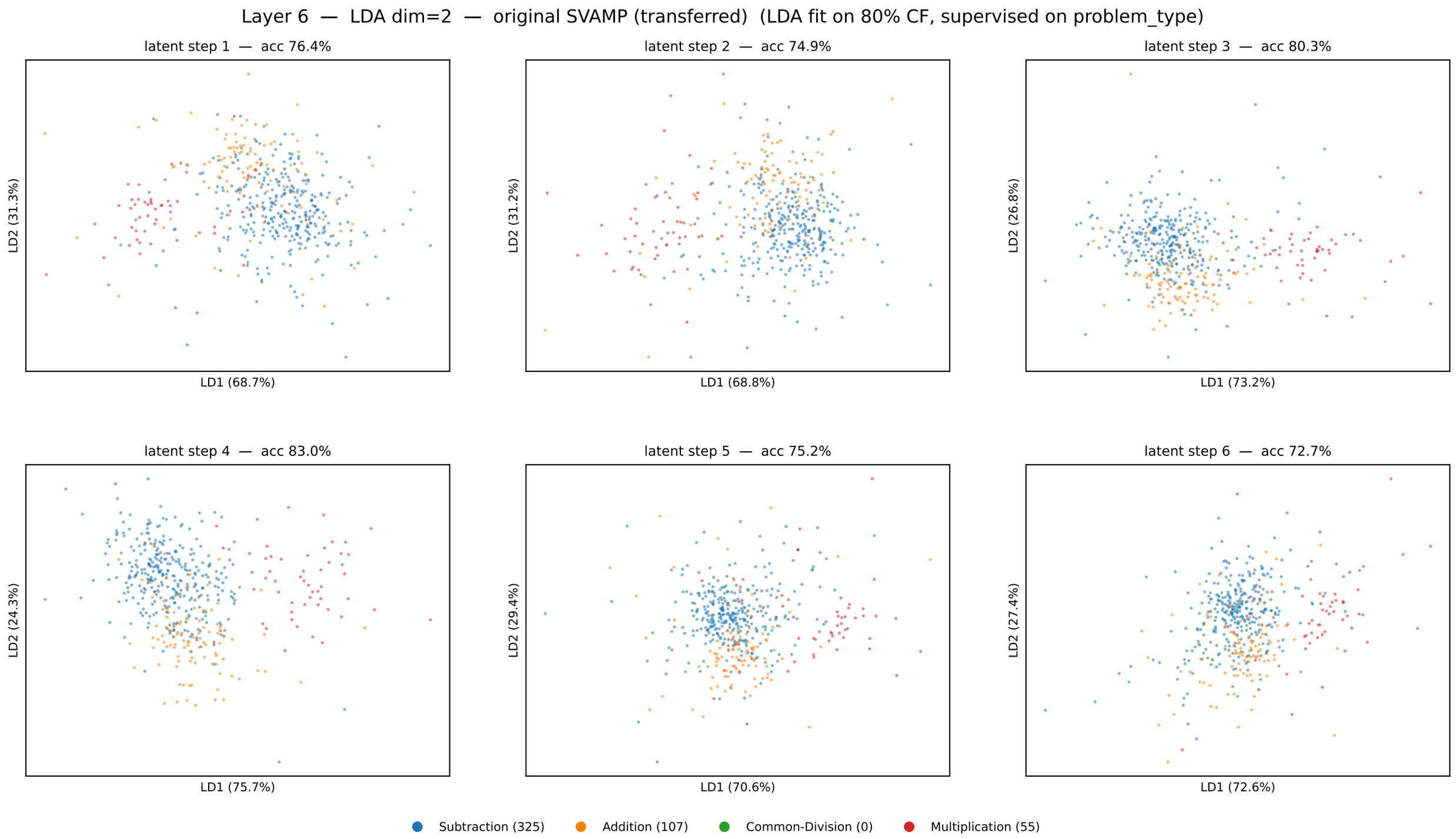

Layer 1 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 62.8%
LD2 (33.9%)
LD1 (66.1%)
latent step 2 — acc 64.3%
LD2 (32.5%)
LD1 (67.5%)
latent step 3 — acc 74.9%
LD2 (30.8%)
LD1 (69.2%)
latent step 4 — acc 79.1%
LD2 (32.8%)
LD1 (67.2%)
latent step 5 — acc 69.0%
LD2 (30.7%)
LD1 (69.3%)
latent step 6 — acc 66.9%
LD2 (27.4%)
LD1 (72.6%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

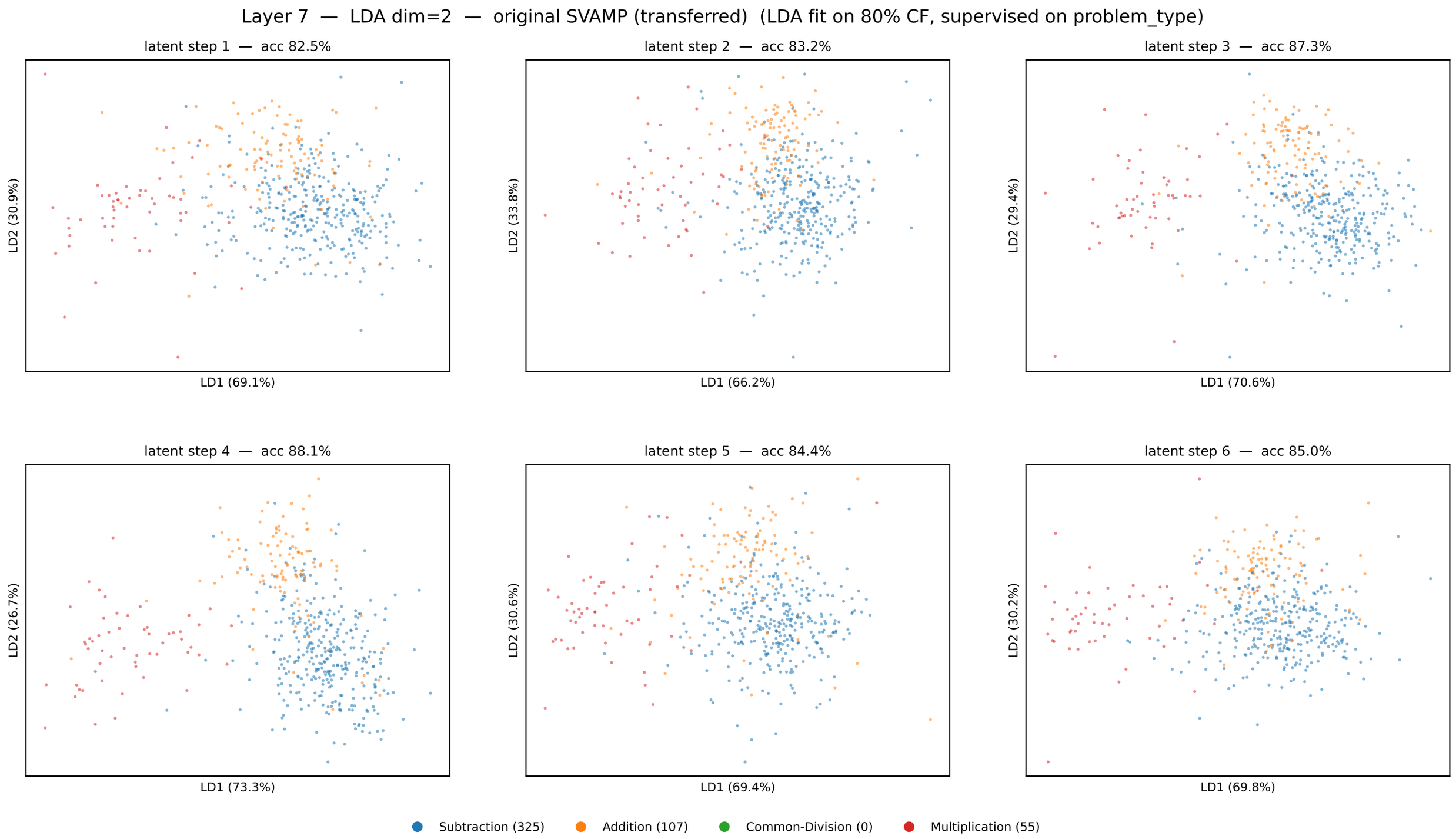

Layer 2 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 70.2%
LD2 (41.6%)
LD1 (58.4%)
latent step 2 — acc 64.1%
LD2 (34.8%)
LD1 (65.2%)
latent step 3 — acc 74.1%
LD2 (29.0%)
LD1 (71.0%)
latent step 4 — acc 83.2%
LD2 (29.3%)
LD1 (70.7%)
latent step 5 — acc 72.5%
LD2 (34.1%)
LD1 (65.9%)
latent step 6 — acc 68.4%
LD2 (26.6%)
LD1 (73.4%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

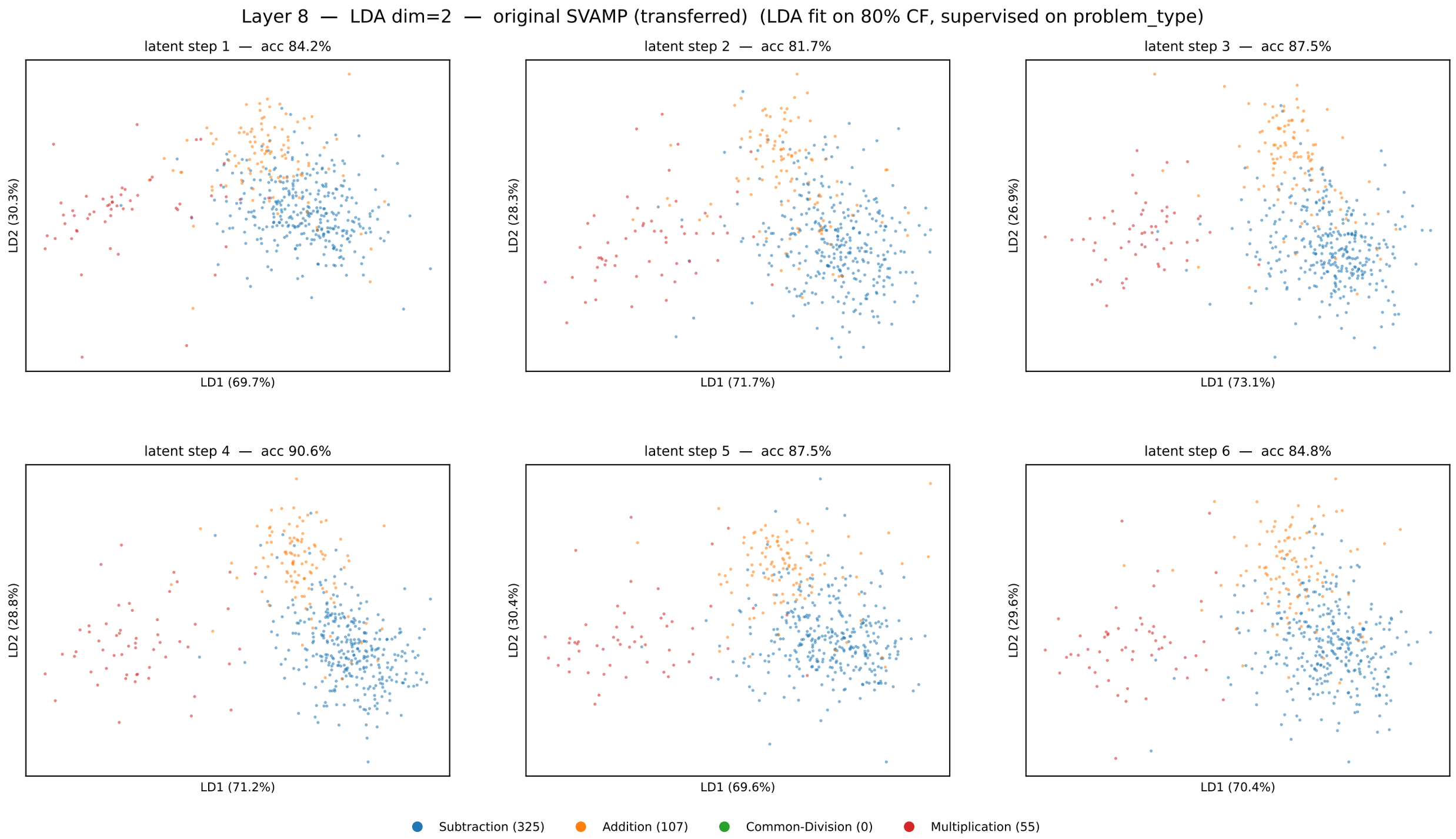

Layer 3 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 71.5%
LD2 (38.6%)
LD1 (61.4%)
latent step 2 — acc 66.7%
LD2 (35.3%)
LD1 (64.7%)
latent step 3 — acc 73.5%
LD2 (28.5%)
LD1 (71.5%)
latent step 4 — acc 81.3%
LD2 (30.2%)
LD1 (69.8%)
latent step 5 — acc 70.4%
LD2 (32.7%)
LD1 (67.3%)
latent step 6 — acc 64.9%
LD2 (27.0%)
LD1 (73.0%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)


Layer 4 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 72.1%
LD2 (35.5%)
LD1 (64.5%)
latent step 2 — acc 70.0%
LD2 (35.7%)
LD1 (64.3%)
latent step 3 — acc 76.0%
LD2 (27.5%)
LD1 (72.5%)
latent step 4 — acc 81.3%
LD2 (29.9%)
LD1 (70.1%)
latent step 5 — acc 69.4%
LD2 (34.2%)
LD1 (65.8%)
latent step 6 — acc 67.4%
LD2 (29.7%)
LD1 (70.3%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

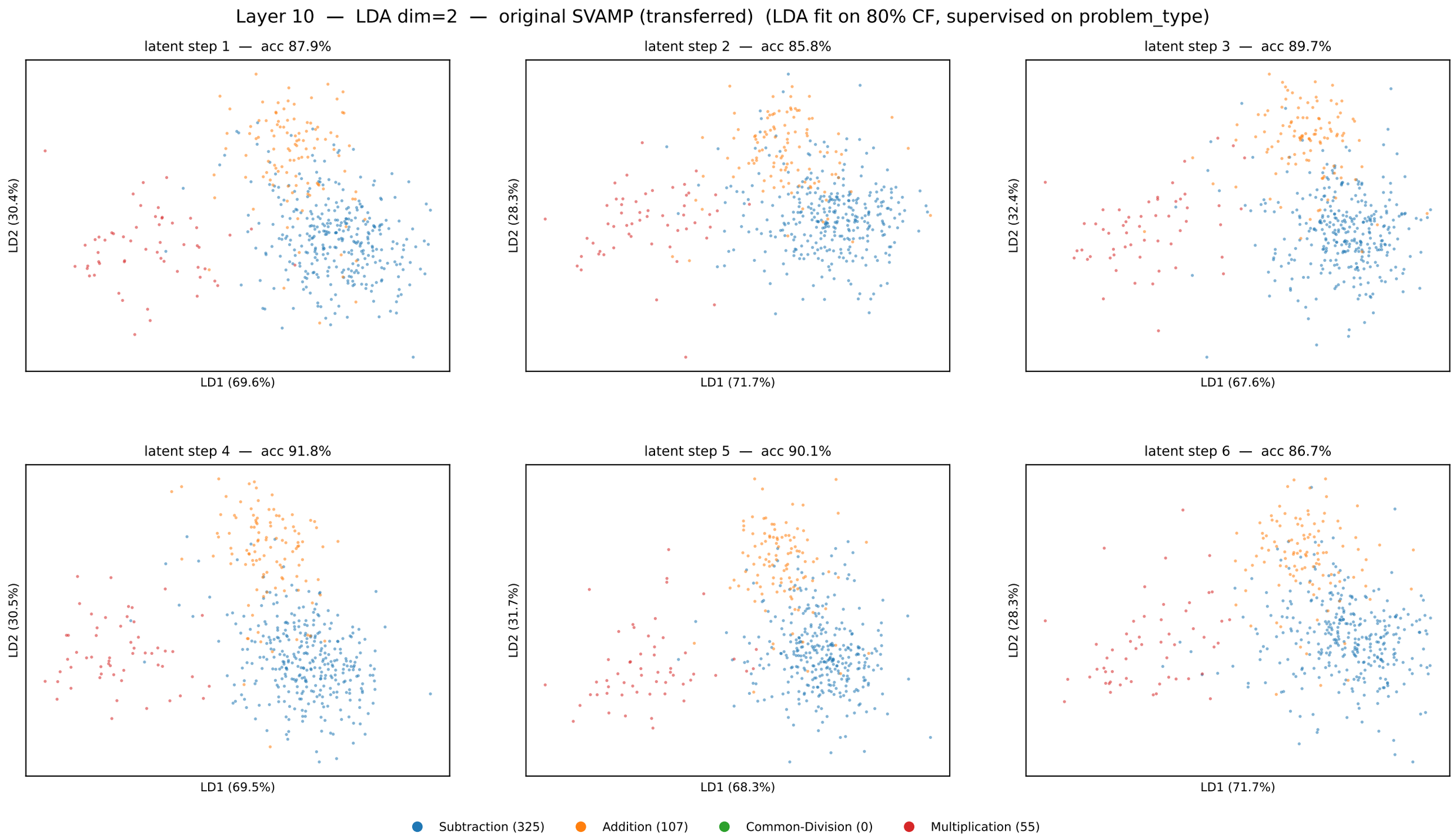

Layer 5 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 74.9%
LD2 (33.9%)
LD1 (66.1%)
latent step 2 — acc 75.4%
LD2 (36.1%)
LD1 (63.9%)
latent step 3 — acc 79.9%
LD2 (28.9%)
LD1 (71.1%)
latent step 4 — acc 83.2%
LD2 (33.8%)
LD1 (66.2%)
latent step 5 — acc 71.5%
LD2 (32.7%)
LD1 (67.3%)
latent step 6 — acc 70.8%
LD2 (30.9%)
LD1 (69.1%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

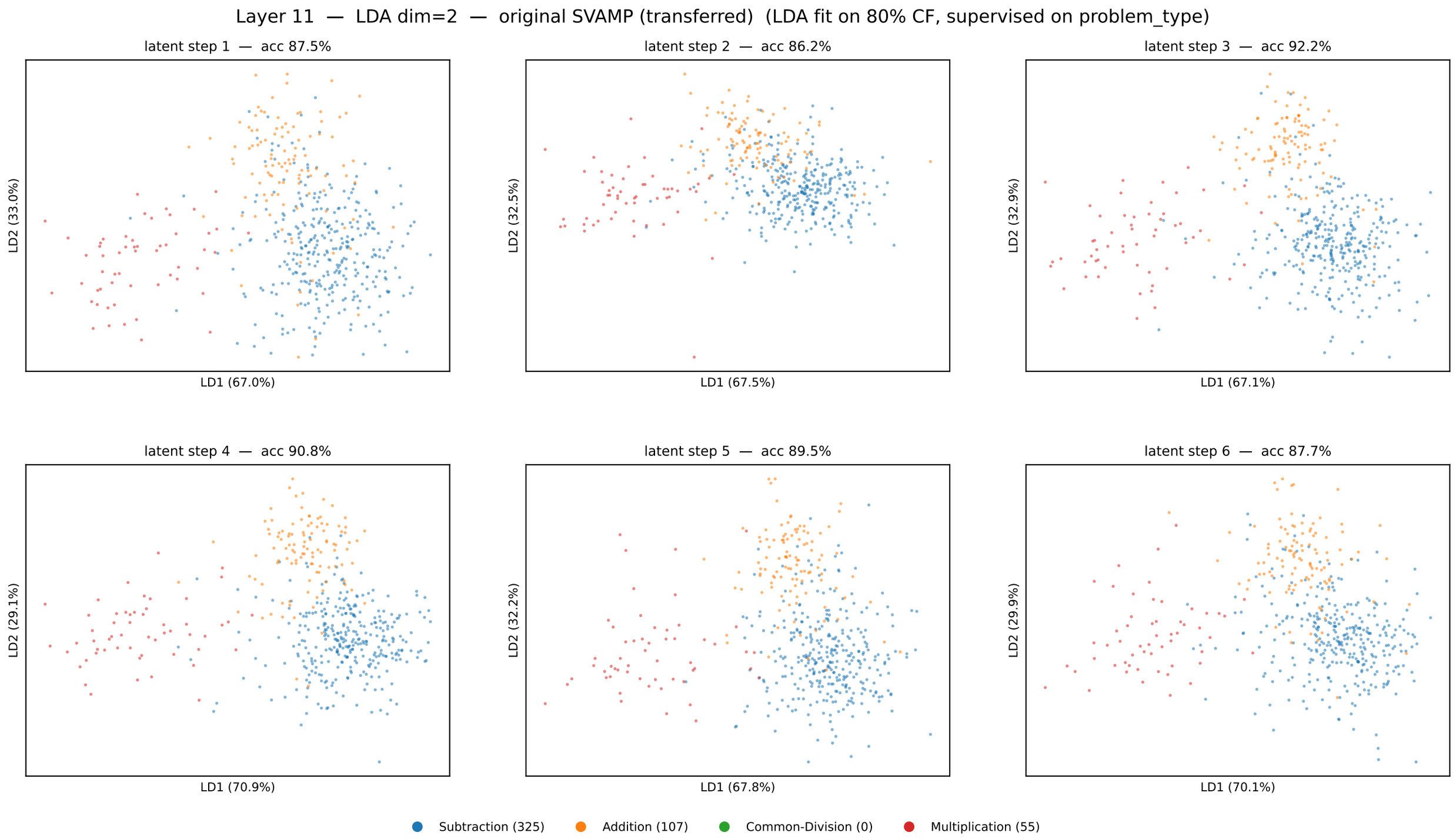

Layer 6 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 76.4%
LD2 (31.3%)
LD1 (68.7%)
latent step 2 — acc 74.9%
LD2 (31.2%)
LD1 (68.8%)
latent step 3 — acc 80.3%
LD2 (26.8%)
LD1 (73.2%)
latent step 4 — acc 83.0%
LD2 (24.3%)
LD1 (75.7%)
latent step 5 — acc 75.2%
LD2 (29.4%)
LD1 (70.6%)
latent step 6 — acc 72.7%
LD2 (27.4%)
LD1 (72.6%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

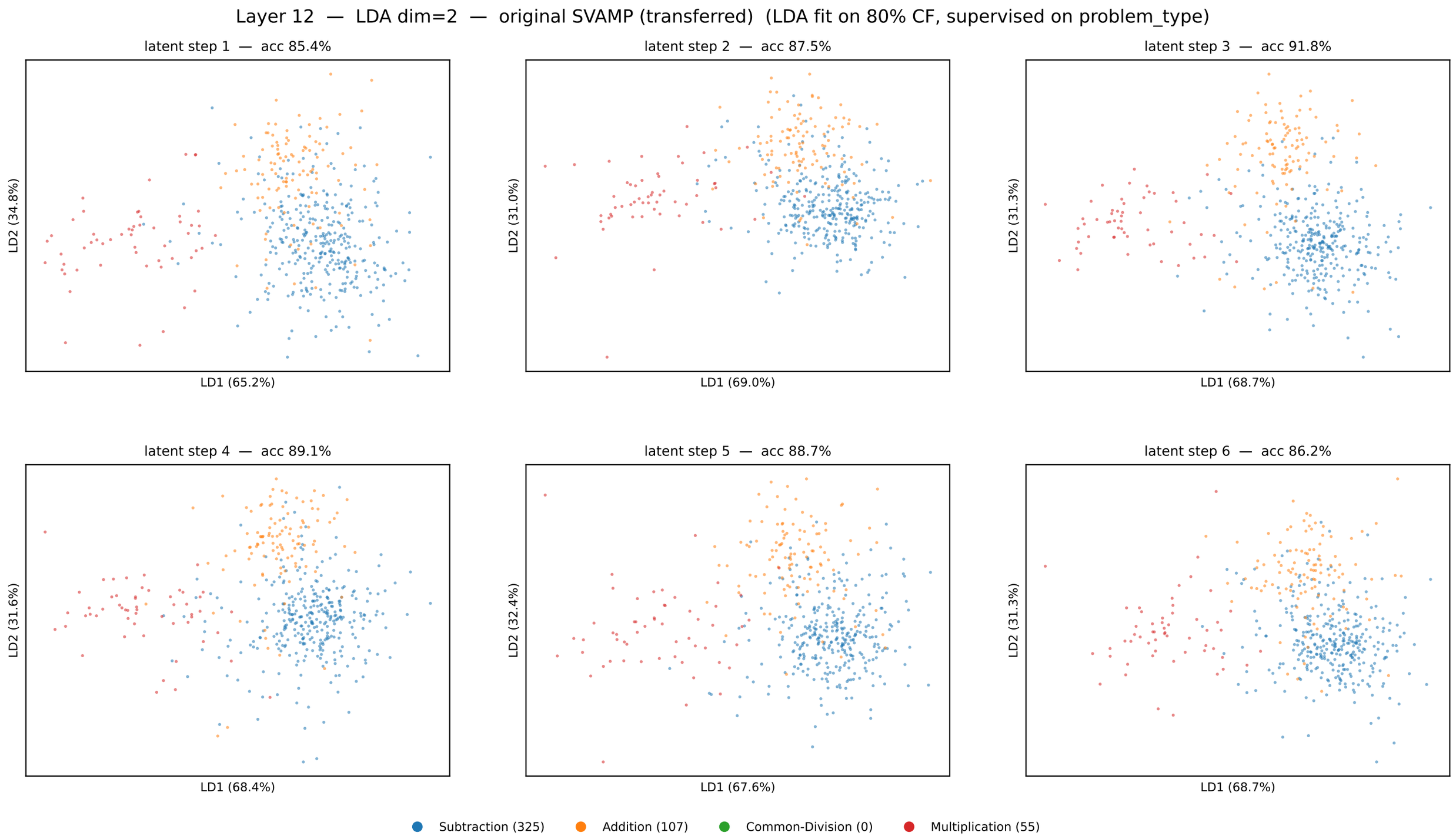

Layer 7 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 82.5%
LD2 (30.9%)
LD1 (69.1%)
latent step 2 — acc 83.2%
LD2 (33.8%)
LD1 (66.2%)
latent step 3 — acc 87.3%
LD2 (29.4%)
LD1 (70.6%)
latent step 4 — acc 88.1%
LD2 (26.7%)
LD1 (73.3%)
latent step 5 — acc 84.4%
LD2 (30.6%)
LD1 (69.4%)
latent step 6 — acc 85.0%
LD2 (30.2%)
LD1 (69.8%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

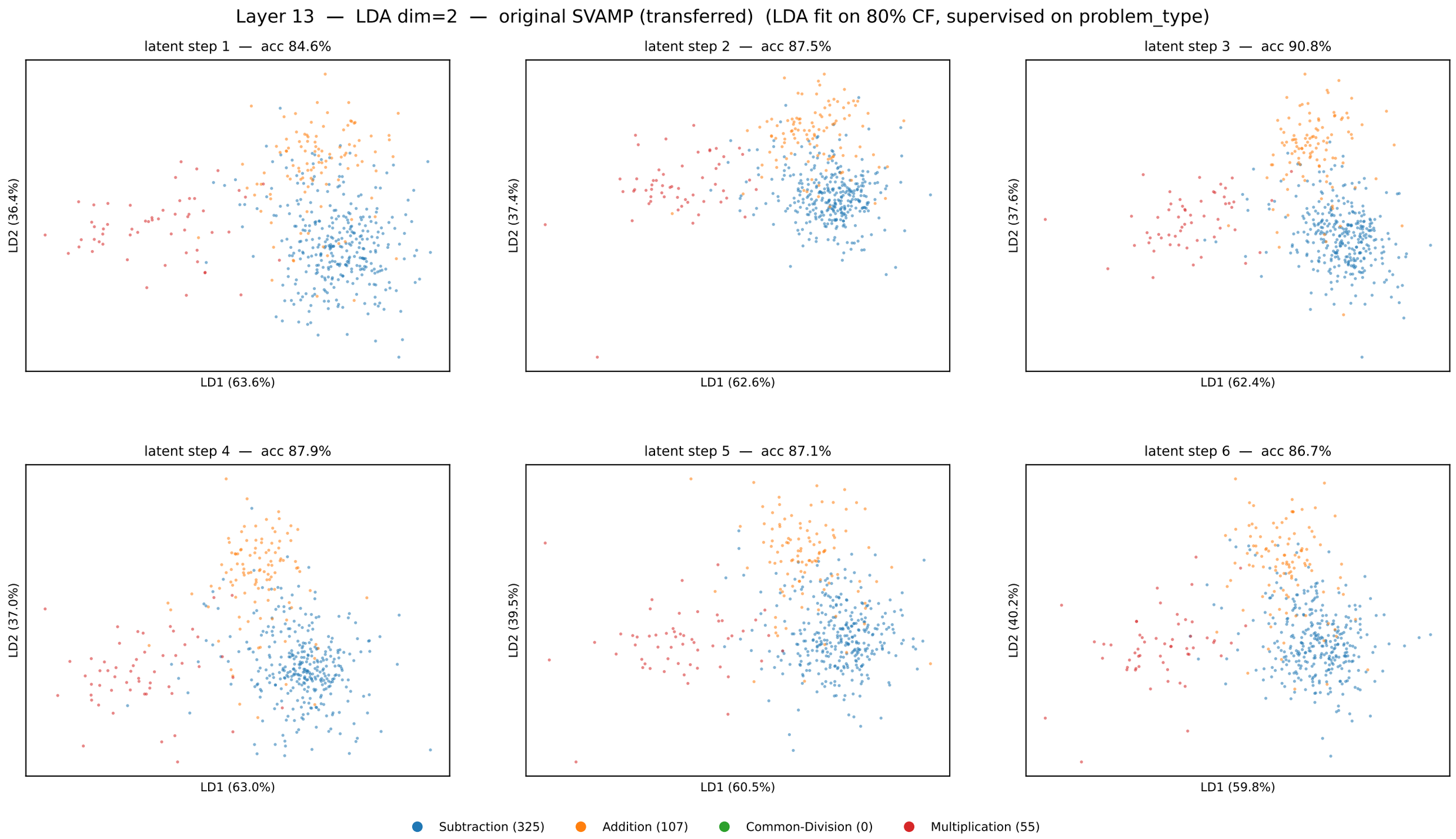

Layer 8 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 84.2%
LD2 (30.3%)
LD1 (69.7%)
latent step 2 — acc 81.7%
LD2 (28.3%)
LD1 (71.7%)
latent step 3 — acc 87.5%
LD2 (26.9%)
LD1 (73.1%)
latent step 4 — acc 90.6%
LD2 (28.8%)
LD1 (71.2%)
latent step 5 — acc 87.5%
LD2 (30.4%)
LD1 (69.6%)
latent step 6 — acc 84.8%
LD2 (29.6%)
LD1 (70.4%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

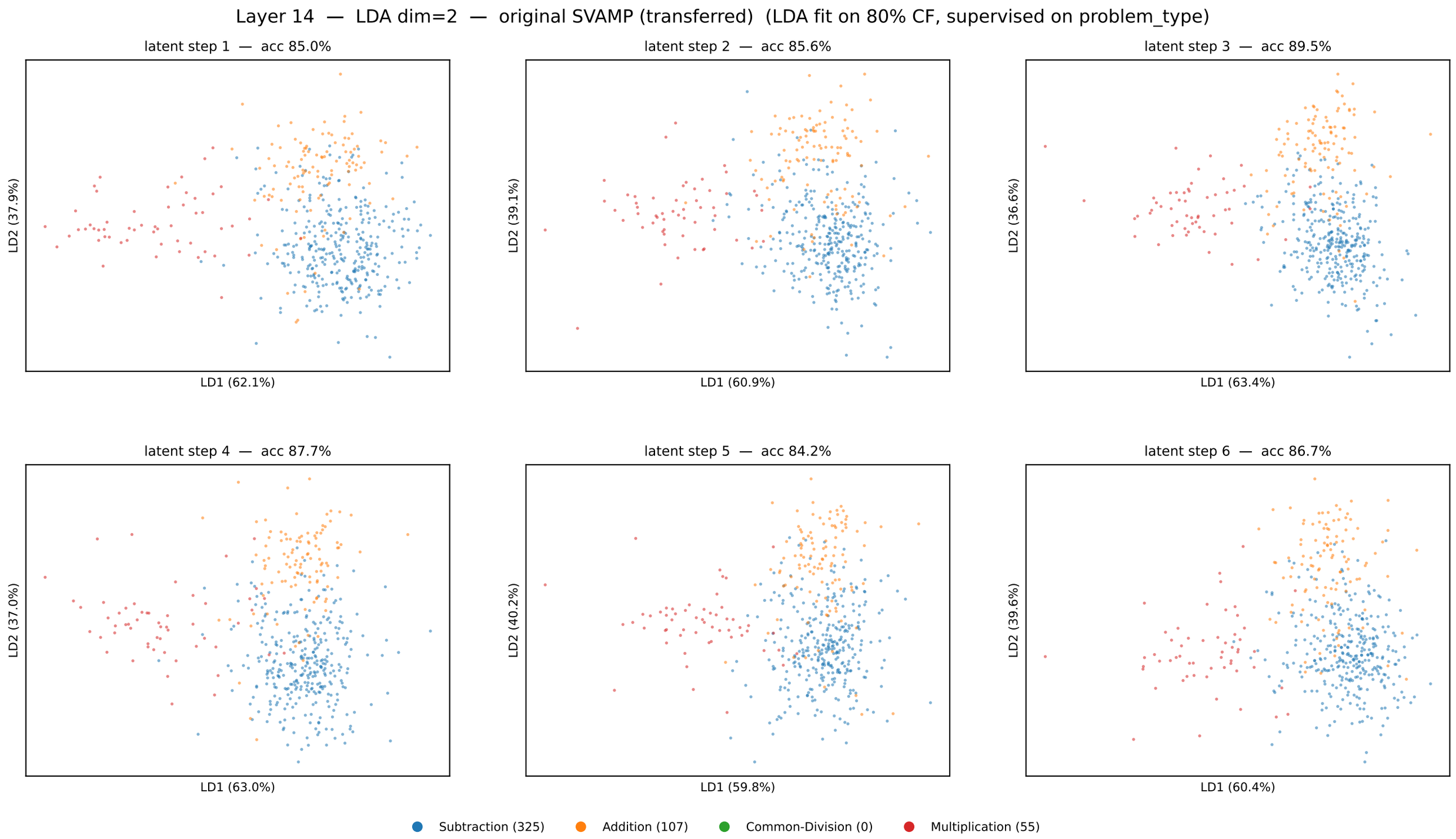
Layer 9 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 86.7%
LD2 (31.7%)
LD1 (68.3%)
latent step 2 — acc 83.4%
LD2 (28.1%)
LD1 (71.9%)
latent step 3 — acc 88.5%
LD2 (28.2%)
LD1 (71.8%)
latent step 4 — acc 89.1%
LD2 (25.5%)
LD1 (74.5%)
latent step 5 — acc 88.3%
LD2 (31.8%)
LD1 (68.2%)
latent step 6 — acc 87.5%
LD2 (28.4%)
LD1 (71.6%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

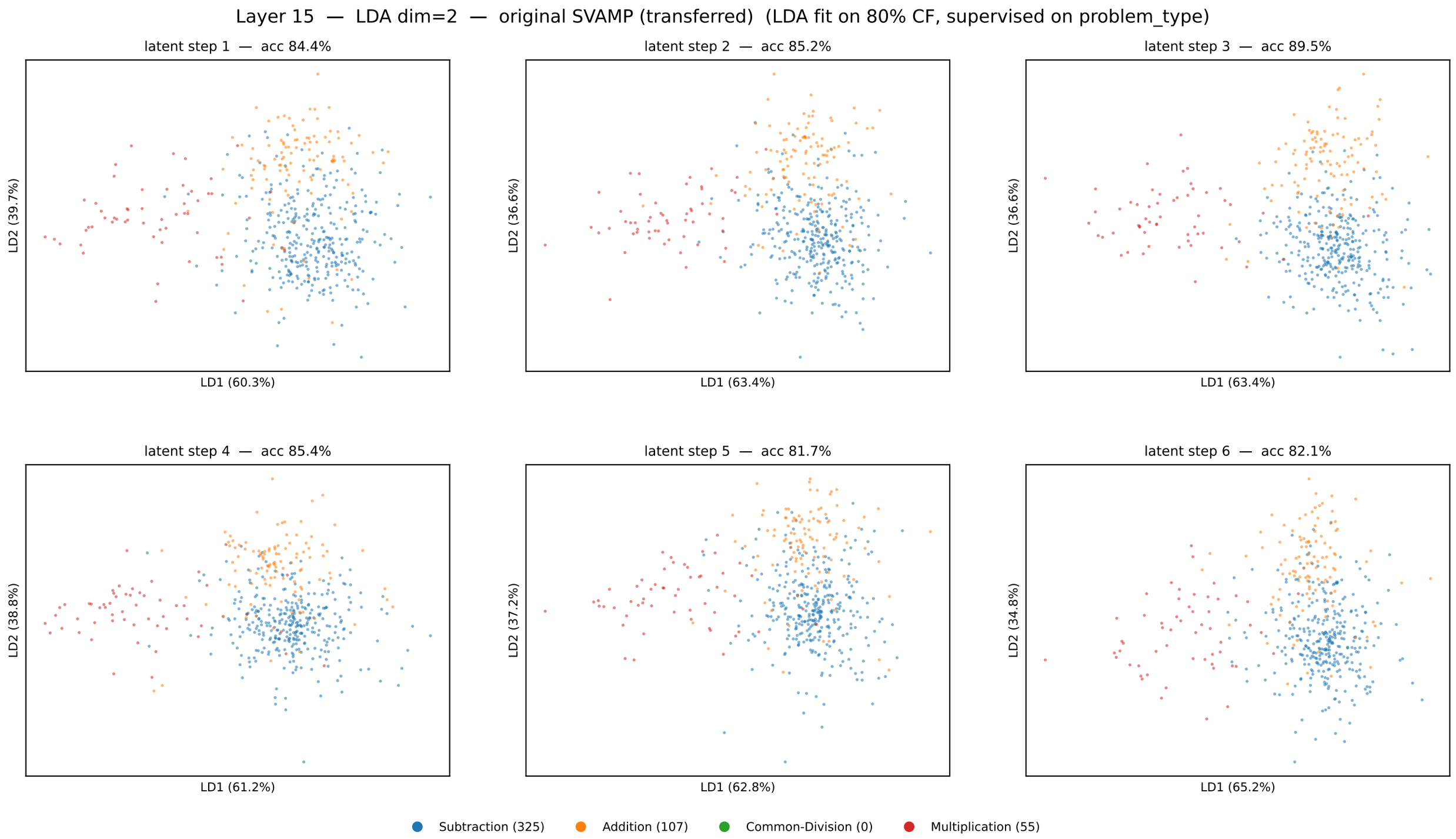

Layer 10 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 87.9%
LD2 (30.4%)
LD1 (69.6%)
latent step 2 — acc 85.8%
LD2 (28.3%)
LD1 (71.7%)
latent step 3 — acc 89.7%
LD2 (32.4%)
LD1 (67.6%)
latent step 4 — acc 91.8%
LD2 (30.5%)
LD1 (69.5%)
latent step 5 — acc 90.1%
LD2 (31.7%)
LD1 (68.3%)
latent step 6 — acc 86.7%
LD2 (28.3%)
LD1 (71.7%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)

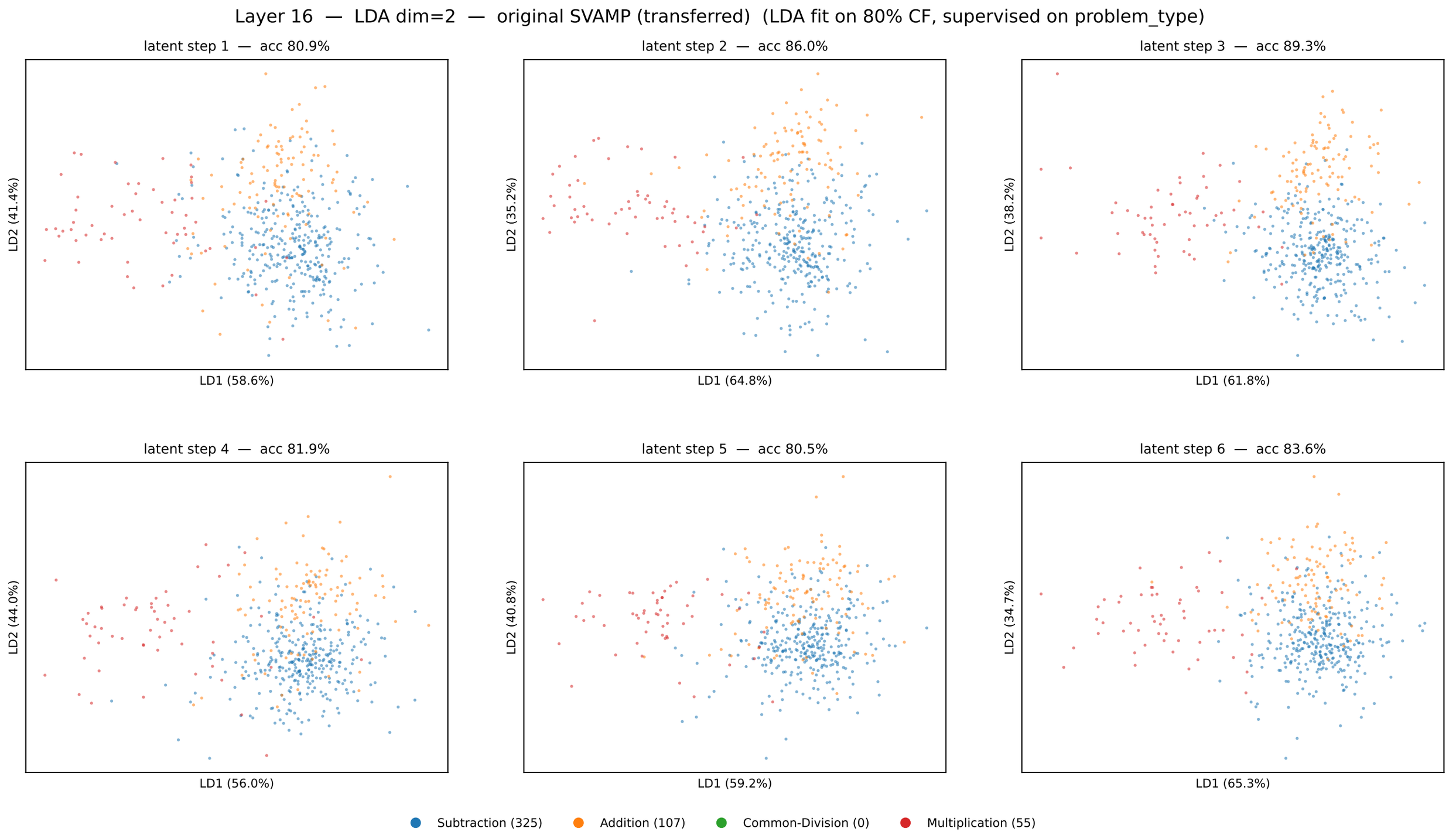

Layer 11 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 87.5%
LD2 (33.0%)
LD1 (67.0%)
latent step 2 — acc 86.2%
LD2 (32.5%)
LD1 (67.5%)
latent step 3 — acc 92.2%
LD2 (32.9%)
LD1 (67.1%)
latent step 4 — acc 90.8%
LD2 (29.1%)
LD1 (70.9%)
latent step 5 — acc 89.5%
LD2 (32.2%)
LD1 (67.8%)
latent step 6 — acc 87.7%
LD2 (29.9%)
LD1 (70.1%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)


Layer 12 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 85.4%
LD2 (34.8%)
LD1 (65.2%)
latent step 2 — acc 87.5%
LD2 (31.0%)
LD1 (69.0%)
latent step 3 — acc 91.8%
LD2 (31.3%)
LD1 (68.7%)
latent step 4 — acc 89.1%
LD2 (31.6%)
LD1 (68.4%)
latent step 5 — acc 88.7%
LD2 (32.4%)
LD1 (67.6%)
latent step 6 — acc 86.2%
LD2 (31.3%)
LD1 (68.7%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)


Layer 13 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 84.6%
LD2 (36.4%)
LD1 (63.6%)
latent step 2 — acc 87.5%
LD2 (37.4%)
LD1 (62.6%)
latent step 3 — acc 90.8%
LD2 (37.6%)
LD1 (62.4%)
latent step 4 — acc 87.9%
LD2 (37.0%)
LD1 (63.0%)
latent step 5 — acc 87.1%
LD2 (39.5%)
LD1 (60.5%)
latent step 6 — acc 86.7%
LD2 (40.2%)
LD1 (59.8%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)


Layer 14 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 85.0%
LD2 (37.9%)
LD1 (62.1%)
latent step 2 — acc 85.6%
LD2 (39.1%)
LD1 (60.9%)
latent step 3 — acc 89.5%
LD2 (36.6%)
LD1 (63.4%)
latent step 4 — acc 87.7%
LD2 (37.0%)
LD1 (63.0%)
latent step 5 — acc 84.2%
LD2 (40.2%)
LD1 (59.8%)
latent step 6 — acc 86.7%
LD2 (39.6%)
LD1 (60.4%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)


Layer 15 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 84.4%
LD2 (39.7%)
LD1 (60.3%)
latent step 2 — acc 85.2%
LD2 (36.6%)
LD1 (63.4%)
latent step 3 — acc 89.5%
LD2 (36.6%)
LD1 (63.4%)
latent step 4 — acc 85.4%
LD2 (38.8%)
LD1 (61.2%)
latent step 5 — acc 81.7%
LD2 (37.2%)
LD1 (62.8%)
latent step 6 — acc 82.1%
LD2 (34.8%)
LD1 (65.2%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)


Layer 16 — LDA dim=2 — original SVAMP (transferred) (LDA fit on 80% CF, supervised on problem_type)
latent step 1 — acc 80.9%
LD2 (41.4%)
LD1 (58.6%)
latent step 2 — acc 86.0%
LD2 (35.2%)
LD1 (64.8%)
latent step 3 — acc 89.3%
LD2 (38.2%)
LD1 (61.8%)
latent step 4 — acc 81.9%
LD2 (44.0%)
LD1 (56.0%)
latent step 5 — acc 80.5%
LD2 (40.8%)
LD1 (59.2%)
latent step 6 — acc 83.6%
LD2 (34.7%)
LD1 (65.3%)
Subtraction (325)
Addition (107)
Common-Division (0)
Multiplication (55)